# Contents

**AIME Number Theory Worked Solutions** **7**
1983 AIME Problem 6: Powers one away from seven . . . 7
1983 AIME Problem 8: A prime divisor of a central binomial coefficient . . . 8
1983 AIME Problem 12: A diameter with reversed chord digits . . . 8
1984 AIME Problem 2: The smallest zero-eight multiple . . . 9
1984 AIME Problem 14: The last even number without two odd composites . . . 9
1985 AIME Problem 3: An integer complex cube . . . 10
1985 AIME Problem 7: Two perfect powers nineteen apart . . . 10
1985 AIME Problem 13: The largest gcd of adjacent quadratic values . . . 11
1986 AIME Problem 5: A variable divisor and a constant remainder . . . 11
1986 AIME Problem 6: A page counted twice . . . 12
1986 AIME Problem 7: Ranking subsets by ternary value . . . 12
1986 AIME Problem 8: Logarithms of proper divisors . . . 13
1987 AIME Problem 1: Addition without carrying . . . 13
1987 AIME Problem 3: Numbers equal to their proper-divisor product . . . 14
1987 AIME Problem 5: A factored equation in two integer squares . . . 14
1987 AIME Problem 7: LCM triples . . . 15
1987 AIME Problem 8: Exactly one integer in a moving interval . . . 15
1987 AIME Problem 11: The longest consecutive-integer representation . . . 16
1987 AIME Problem 12: An integer just above a cube . . . 16
1988 AIME Problem 2: A digit-sum iteration . . . 17
1988 AIME Problem 5: Divisor probability . . . 17
1988 AIME Problem 8: A functional equation hiding the least common multiple . . . 18
1988 AIME Problem 9: The smallest cube ending in eight-eight-eight . . . 18
1989 AIME Problem 3: A repeated decimal digit forced by divisibility . . . 19
1989 AIME Problem 4: One middle term, two perfect powers . . . 19
1989 AIME Problem 9: Recovering a fifth root with congruences . . . 20
1989 AIME Problem 14: Real integers written in a complex base . . . 20
1990 AIME Problem 1: Ranking nonsquares and noncubes . . . 21
1990 AIME Problem 3: A largest regular-polygon side count . . . 21
1990 AIME Problem 5: The smallest multiple with seventy-five divisors . . . 22
1990 AIME Problem 10: Products of two root-of-unity groups . . . 23
1990 AIME Problem 11: A factorial as a shorter consecutive product . . . 23
1990 AIME Problem 13: Leading nines in powers . . . 24
1991 AIME Problem 1: Symmetric sums with an integer restriction . . . 24
1991 AIME Problem 5: Coprime factor allocations . . . 25
1991 AIME Problem 6: Recovering a number from a floor sum . . . 25
1991 AIME Problem 8: Integer roots via a shifted factorization . . . 26
1991 AIME Problem 13: Sock counts from a probability . . . 26
1991 AIME Problem 15: An attainable vector minimum that is integral . . . 27
1992 AIME Problem 1: Pairing reduced fractions around five . . . 27
1992 AIME Problem 5: Numerators of repeating decimals . . . 28
1992 AIME Problem 6: Consecutive integers without carries . . . 28
1992 AIME Problem 11: The order of two successive reflections . . . 29
1992 AIME Problem 15: Missing factorial trailing-zero counts . . . 29
1993 AIME Problem 4: Equal outer and inner sums . . . 30
1993 AIME Problem 6: One integer as three consecutive sums . . . 31

1993 AIME Problem 9: Colliding labels in a circular walk . . . 31
1994 AIME Problem 1: Indexing squares one above multiples of three . . . 32
1994 AIME Problem 4: A sum of binary logarithm floors . . . 32
1994 AIME Problem 5: A weighted sum over digit strings . . . 33
1994 AIME Problem 7: Lines with only integral circle intersections . . . 33
1994 AIME Problem 10: A prime-cube projection in an integer right triangle . . . 34
1994 AIME Problem 11: Distinct heights of a brick tower . . . 34
1994 AIME Problem 12: The finest square grid within a fence budget . . . 35
1995 AIME Problem 2: A product of roots without solving them . . . 35
1995 AIME Problem 6: Divisors below a square root . . . 36
1995 AIME Problem 8: Two simultaneous integer ratios . . . 36
1995 AIME Problem 10: A composite term in five consecutive residue steps . . . 37
1995 AIME Problem 11: A self-similar cut prism . . . 38
1995 AIME Problem 13: Grouping a sum by rounded fourth roots . . . 38
1996 AIME Problem 2: Even logarithm floors . . . 39
1996 AIME Problem 8: Integer pairs of prescribed harmonic mean . . . 39
1996 AIME Problem 10: An integer angle from a tangent identity . . . 40
1996 AIME Problem 14: Cubes crossed by a space diagonal . . . 40
1997 AIME Problem 1: Differences of squares . . . 41
1997 AIME Problem 3: Concatenation mistaken for multiplication . . . 41
1997 AIME Problem 5: Decimal numbers closest to two sevenths . . . 42
1997 AIME Problem 6: An equilateral attachment and another regular polygon . . . 42
1998 AIME Problem 1: Counting possible LCM partners . . . 43
1998 AIME Problem 5: A four-term cycle of signed triangular numbers . . . 43
1998 AIME Problem 8: The longest subtractive recurrence . . . 44
1998 AIME Problem 14: A sharp bound for a box volume ratio . . . 44
1999 AIME Problem 1: Five primes in arithmetic progression . . . 45
1999 AIME Problem 3: A quadratic square condition with four solutions . . . 46
1999 AIME Problem 5: Possible digit-sum changes . . . 46
1999 AIME Problem 7: Four-position divisor switches . . . 47
1999 AIME Problem 13: Distinct tournament win totals . . . 47
2000 AIME I Problem 1: Zero-free factors of powers of ten . . . 48
2000 AIME I Problem 2: A reflected pentagon . . . 48
2000 AIME I Problem 5: Two urns with a fixed total . . . 49
2000 AIME I Problem 6: A gap between means . . . 49
2000 AIME I Problem 11: Independent prime exponents . . . 50
2000 AIME II Problem 2: Lattice points on a difference of squares . . . 50
2000 AIME II Problem 4: Prescribed odd and even divisor counts . . . 51
2000 AIME II Problem 11: Integral equal legs restrict base directions . . . 51
2000 AIME II Problem 14: Borrowing in factorial base . . . 52
2001 AIME I Problem 1: Two digits dividing their number . . . 52
2001 AIME I Problem 8: Largest number with doubled reinterpreted digits . . . 53
2001 AIME I Problem 11: Two numberings of a rectangular array . . . 53
2001 AIME II Problem 1: Digit adjacency as a directed graph . . . 54
2001 AIME II Problem 5: Triangle-free ordered sets . . . 55
2001 AIME II Problem 10: Counting exponent gaps . . . 55
2001 AIME II Problem 14: Unit roots of a sparse polynomial . . . 56
2002 AIME I Problem 3: Reversed two-digit ages . . . 56
2002 AIME I Problem 4: A telescope followed by a prime-square factorization . . . 57

2002 AIME I Problem 7: Three fractional digits of an enormous power . . . 57
2002 AIME I Problem 8: The first repeated Fibonacci-type endpoint . . . 58
2002 AIME I Problem 9: Three arithmetic progressions partition the integers . . . 58
2002 AIME I Problem 11: Unfolding a reflected ray in a cube . . . 59
2002 AIME I Problem 14: Integer means after deletion . . . 59
2002 AIME II Problem 1: Differences of reversed three-digit integers . . . 60
2002 AIME II Problem 3: An integral geometric progression with a square gap . . . 60
2002 AIME II Problem 5: When a power fails to divide another . . . 61
2002 AIME II Problem 7: A square sum divisible by 200 . . . 61
2002 AIME II Problem 8: The first missing integer quotient . . . 62
2003 AIME I Problem 1: The largest factorial factor . . . 62
2003 AIME I Problem 7: Integral distances in an isosceles triangle . . . 63
2003 AIME I Problem 8: An arithmetic triple followed by a geometric triple . . . 63
2003 AIME I Problem 14: The smallest denominator containing a decimal block . . . 64
2003 AIME II Problem 1: A product equal to six times a sum . . . 65
2003 AIME II Problem 2: Largest distinct-digit multiple of eight . . . 65
2003 AIME II Problem 10: A nonsquare radical sum with fixed difference . . . 66
2003 AIME II Problem 12: Minimum occupancy with percentage constraints . . . 66
2004 AIME I Problem 1: Four descending consecutive digits . . . 67
2004 AIME I Problem 5: Strict daily rates with a larger aggregate score . . . 67
2004 AIME I Problem 8: Regular star polygons . . . 68
2004 AIME I Problem 15: Reverse dynamics and forbidden operation runs . . . 68
2004 AIME II Problem 3: Hidden cubes . . . 69
2004 AIME II Problem 6: Integer shares in a three-stage distribution . . . 69
2004 AIME II Problem 8: Divisor counts as exponent allocations . . . 70
2004 AIME II Problem 10: Two binary ones modulo nine . . . 71
2004 AIME II Problem 14: Distinct digit lengths of a sum . . . 71
2005 AIME I Problem 2: Arithmetic progressions reaching 2005 . . . 72
2005 AIME I Problem 3: Exactly three small proper divisors . . . 72
2005 AIME I Problem 4: A rectangular formation five above a square . . . 73
2005 AIME I Problem 12: Parity changes only at squares . . . 73
2005 AIME II Problem 4: Union of divisor sets . . . 74
2005 AIME II Problem 5: Reciprocal logarithms force powers . . . 74
2005 AIME II Problem 6: Fixed points of a shuffle . . . 75
2005 AIME II Problem 9: A swapped sine-cosine power identity . . . 75
2005 AIME II Problem 13: Integer polynomial values constrain integer roots . . . 76
2006 AIME I Problem 3: Deleting the leading digit . . . 76
2006 AIME I Problem 4: Valuation of a product of factorials . . . 77
2006 AIME I Problem 9: Integral geometric progressions and logarithms . . . 77
2006 AIME I Problem 13: Valuation layers . . . 78
2006 AIME II Problem 3: Three-adic valuation of an odd factorial . . . 79
2006 AIME II Problem 7: Complementary decimal numbers . . . 79
2006 AIME II Problem 13: Five representations by consecutive odd integers . . . 80
2006 AIME II Problem 14: Reciprocal digit frequency . . . 80
2007 AIME I Problem 1: Square multiples of twenty-four . . . 81
2007 AIME I Problem 4: The next planetary alignment . . . 81
2007 AIME I Problem 5: A periodic rounding map . . . 82
2007 AIME I Problem 7: A floor-ceiling logarithm indicator . . . 82
2007 AIME I Problem 11: Rounding square roots in blocks . . . 83

2007 AIME II Problem 2: Divisibility becomes a composition . . . 83
2007 AIME II Problem 5: Grid cells below a rational line . . . 84
2007 AIME II Problem 7: Multiples inside a cube-root interval . . . 84
2007 AIME II Problem 12: The largest term controls a geometric sum . . . 85
2007 AIME II Problem 13: Pascal propagation modulo three . . . 85
2008 AIME I Problem 3: Integer speeds from two distance totals . . . 86
2008 AIME I Problem 4: A shifted difference of squares . . . 86
2008 AIME I Problem 6: Summing an arithmetic row repeatedly . . . 87
2008 AIME I Problem 7: Occupied hundred-blocks . . . 87
2008 AIME II Problem 4: Balanced ternary support . . . 88
2008 AIME II Problem 8: A trigonometric telescope and divisibility . . . 88
2008 AIME II Problem 15: Consecutive cubes and a second square . . . 89
2009 AIME I Problem 1: Extremal three-digit geometric progressions . . . 89
2009 AIME I Problem 6: Integer values of a stepped power function . . . 90
2009 AIME I Problem 7: An exponential recurrence that telescopes . . . 90
2009 AIME I Problem 11: Integral triangle areas on a line . . . 91
2009 AIME I Problem 13: Infinite integrality forces a two-cycle . . . 91
2009 AIME I Problem 14: Minimizing a second moment with fixed first and fourth moments 92
2009 AIME II Problem 4: The smallest winning score . . . 93
2009 AIME II Problem 7: A sum with strictly increasing dyadic denominators . . . 93
2009 AIME II Problem 9: An injection leaves two boundary slices . . . 94
2009 AIME II Problem 11: Exactly fifty logarithmic neighbors . . . 94
2010 AIME I Problem 1: One square divisor in a pair . . . 95
2010 AIME I Problem 2: A product of repeated nines . . . 95
2010 AIME I Problem 5: Equality in a difference-of-squares bound . . . 96
2010 AIME I Problem 8: The smallest disk enclosing floor-defined squares . . . 96
2010 AIME I Problem 10: Digits allowed to carry . . . 97
2010 AIME I Problem 12: A multiplicative two-color threshold . . . 97
2010 AIME I Problem 14: A monotone sum of decimal digit counts . . . 98
2010 AIME II Problem 1: The largest multiple with distinct even digits . . . 98
2010 AIME II Problem 3: Valuation of all pair differences . . . 99
2010 AIME II Problem 6: The first reducible quartic . . . 99
2010 AIME II Problem 10: Quadratics modulo interchange of roots . . . 100
2010 AIME II Problem 12: Two isosceles triangles with equal area and perimeter . . . 100
2011 AIME I Problem 5: Sliding sums force periodic residues . . . 101
2011 AIME I Problem 7: Splitting a power into equal smaller powers . . . 101
2011 AIME I Problem 10: Obtuse triangles in a regular polygon . . . 102
2011 AIME I Problem 11: Summing a power-residue orbit without its order . . . 103
2011 AIME I Problem 15: Bounding integer roots of a depressed cubic . . . 103
2011 AIME II Problem 3: Integer angles in arithmetic progression . . . 104
2011 AIME II Problem 14: Permutations respecting residue partitions . . . 104
2011 AIME II Problem 15: Floor equality on three intervals . . . 105
2012 AIME I Problem 1: Divisibility in both digit directions . . . 105
2012 AIME I Problem 4: A shared horse and the first integer meeting point . . . 106
2012 AIME I Problem 5: Consecutive binary numbers with equal weight . . . 107
2012 AIME I Problem 6: A positive imaginary part selects a root-of-unity denominator . . . 107
2012 AIME I Problem 10: Order the squares with a prescribed three-digit suffix . . . 108
2012 AIME I Problem 11: Reachability in a lattice semigroup . . . 109
2012 AIME I Problem 15: Multiplication changes every circular separation . . . 109

2012 AIME II Problem 1: Count lattice points on a positive Diophantine segment . . . . . 110
2012 AIME II Problem 12: Independent forbidden residues . . . . . 110
2013 AIME I Problem 2: Uniform digit sums modulo five . . . . . 111
2013 AIME I Problem 8: A function domain imposes an integer divisor . . . . . 111
2013 AIME I Problem 10: Conjugate integer roots and distinct cubics . . . . . 112
2013 AIME I Problem 11: Three remainder conditions after an lcm reduction . . . . . 113
2013 AIME I Problem 15: Two arithmetic progressions with reordered residues . . . . . 113
2013 AIME II Problem 2: Nested logarithms impose a power-of-two divisor . . . . . 114
2013 AIME II Problem 6: The first complete thousand-block between squares . . . . . 114
2013 AIME II Problem 7: Staff reductions force a unique feasible integer solution . . . . . 115
2013 AIME II Problem 14: Maximize a remainder by separating quotient ranges . . . . . 116
2014 AIME I Problem 3: Reduced fractions with fixed numerator-plus-denominator . . . . . 116
2014 AIME I Problem 6: Two parabolas with a shared axis . . . . . 117
2014 AIME I Problem 8: A four-digit idempotent . . . . . 117
2014 AIME II Problem 4: Compatible repeating blocks . . . . . 118
2014 AIME II Problem 15: Prime multiplication as binary carrying . . . . . 118
2015 AIME I Problem 3: A prime concealed in a cube . . . . . 119
2015 AIME I Problem 8: Digit sums determine the carries . . . . . 120
2015 AIME I Problem 11: Integral isosceles triangles with a fixed incenter distance . . . . . 120
2015 AIME II Problem 1: Two percentage relations force a least common multiple . . . . . 121
2015 AIME II Problem 3: A multiple with a prescribed digit sum . . . . . 121
2015 AIME II Problem 6: Two possible cubics with positive integer roots . . . . . 122
2015 AIME II Problem 8: An integer inequality bounds a cubic quotient . . . . . 122
2015 AIME II Problem 13: Negative partial sums of successive sines . . . . . 123
2016 AIME I Problem 5: Two arithmetic totals share a day count . . . . . 123
2016 AIME I Problem 7: Real values of a complex radical expression . . . . . 124
2016 AIME I Problem 10: Alternating geometric and arithmetic triples . . . . . 124
2016 AIME I Problem 12: The first value with four prime factors . . . . . 125
2016 AIME II Problem 5: An infinite chain of right-triangle altitudes . . . . . 125
2016 AIME II Problem 7: Three nested squares with two integer areas . . . . . 126
2016 AIME II Problem 8: Three distinct factors of a product . . . . . 126
2016 AIME II Problem 9: An arithmetic sequence plus a geometric sequence . . . . . 127
2016 AIME II Problem 11: Divisor counts of perfect powers . . . . . 127
2017 AIME I Problem 2: Equal remainders determine two divisors . . . . . 128
2017 AIME I Problem 3: A periodic sequence of triangular-number digits . . . . . 128
2017 AIME I Problem 5: One rational number in two bases . . . . . 129
2017 AIME I Problem 9: A concatenation-type recurrence modulo99 . . . . . 129
2017 AIME I Problem 12: Product-free subsets . . . . . 130
2017 AIME I Problem 13: The last gap between consecutive cubes . . . . . 130
2017 AIME I Problem 14: Nested logarithms become a monotone power equation . . . . . 131
2017 AIME II Problem 4: Ternary digit avoidance . . . . . 132
2017 AIME II Problem 6: A square squeezed between nearby shifted squares . . . . . 132
2017 AIME II Problem 8: Integrality of a truncated exponential series . . . . . 133
2018 AIME I Problem 2: A digit permutation in three bases . . . . . 133
2018 AIME I Problem 6: Two disjoint root-of-unity families . . . . . 134
2018 AIME I Problem 11: A two-digit ending requires prime-square orders . . . . . 134
2018 AIME I Problem 12: Subset sums modulo three . . . . . 135
2018 AIME II Problem 2: A recurring triple proves periodicity . . . . . 135
2018 AIME II Problem 3: A square and a cube in the same base . . . . . 136

2018 AIME II Problem 7: Rational lengths among equal-area parallel slices . . . 136
2019 AIME I Problem 1: A sum of strings of nines . . . 137
2019 AIME I Problem 7: Prime exponents hidden in gcd and lcm logarithms . . . 137
2019 AIME I Problem 9: Consecutive integers with seven divisors in total . . . 138
2019 AIME I Problem 11: A circle tangent to all three excircles . . . 138
2019 AIME I Problem 14: An eighth power forces order sixteen . . . 139
2019 AIME II Problem 4: A square product of four die rolls . . . 139
2019 AIME II Problem 9: Multiples of twenty having twenty divisors . . . 140
2019 AIME II Problem 10: A tangent sign itinerary fixes every binary digit . . . 140
2019 AIME II Problem 12: Divisibility chains with prescribed sum . . . 141
2019 AIME II Problem 14: A three-denomination Frobenius problem . . . 141
2020 AIME I Problem 3: A shared set of digits in bases eleven and eight . . . 142
2020 AIME I Problem 4: A truncated prefix must divide the suffix . . . 142
2020 AIME I Problem 9: A random chain of divisors . . . 143
2020 AIME I Problem 10: A missing prime power constrains exponential divisibility . . . 143
2020 AIME I Problem 12: Prime-power divisibility after selecting an exponent class . . . 144
2020 AIME II Problem 1: Square divisors of a prime-power product . . . 145
2020 AIME II Problem 5: The smallest integer with nested digit-sum restrictions . . . 145
2020 AIME II Problem 10: Clear denominators before reducing a cube sum . . . 146
2021 AIME I Problem 3: Differences of powers of two . . . 146
2021 AIME I Problem 5: Integer arithmetic progressions satisfying a square identity . . . 147
2021 AIME I Problem 10: A rational recurrence reduces at its first shared prime . . . 147
2021 AIME I Problem 14: A universal divisor-sum congruence without advanced prime distribution . . . 148
2021 AIME II Problem 3: Cyclic products modulo three . . . 148
2021 AIME II Problem 9: Counting exponent pairs by valuations . . . 149
2021 AIME II Problem 13: Lift an exponential fixed point through powers of five . . . 149
2021 AIME II Problem 15: Walking to the next square with two step sizes . . . 150
2022 AIME I Problem 2: A cyclic digit shift in two bases . . . 151
2022 AIME I Problem 7: A minimal positive ratio of digit products . . . 151
2022 AIME I Problem 13: Distinct reduced numerators . . . 152
2022 AIME I Problem 14: Perimeter-bisecting cleavers determine a120-degree angle . . . 152
2022 AIME II Problem 1: A population ratio constrains the total size . . . 153
2022 AIME II Problem 5: Prime-difference triangles . . . 153
2022 AIME II Problem 8: Identifiability from three floor values . . . 154
2022 AIME II Problem 13: A coefficient reduced to stars and bars . . . 154
2022 AIME II Problem 14: Complete coverage by a finite stamp collection . . . 155
2023 AIME I Problem 4: Sum all divisors with prescribed exponent parity . . . 156
2023 AIME I Problem 7: Pairwise-distinct remainders . . . 156
2023 AIME I Problem 10: Bound the floor error before summing residues . . . 157
2023 AIME I Problem 15: A bounded Gaussian-integer search with all sign cases . . . 157
2023 AIME II Problem 2: Enumerate the four-digit octal palindromes . . . 158
2023 AIME II Problem 5: The cancellation divisor determines a rational number . . . 158
2023 AIME II Problem 10: Balanced proper colorings of a ladder . . . 159
2023 AIME II Problem 13: Prove the integrality class before counting last digits . . . 159
2023 AIME II Problem 15: A finite carry state detects repeated modular inverses . . . 160
2024 AIME I Problem 13: Lift the four roots of minus one . . . 161
2024 AIME II Problem 2: A mode and a missing median constrain an integer partition . . . 161
2024 AIME II Problem 6: Encoding a set by maxima of subsets . . . 162

2024 AIME II Problem 7: Sum digit replacements before solving the digits . . . . . 162
2024 AIME II Problem 13: Evaluate a root product at two conjugate constants . . . . . 163
2024 AIME II Problem 14: Digit-square conditions count CRT idempotents . . . . . 163
2025 AIME I Problem 1: Reduce a base-divisibility condition to one fixed integer . . . . . 164
2025 AIME I Problem 4: Count a union of two primitive lattice lines . . . . . 164
2025 AIME I Problem 5: Digit permutations divisible by twenty-two . . . . . 165
2025 AIME I Problem 15: Cubic residues and three-variable counting . . . . . 165
2025 AIME II Problem 2: Evaluate a polynomial at the modulus root . . . . . 166
2025 AIME II Problem 7: The least common multiple of a random divisor subset . . . . . 167
2025 AIME II Problem 8: When greedy coin change succeeds . . . . . 167
2025 AIME II Problem 13: Track reduction before simplifying a rational recurrence . . . . . 168
2026 AIME I Problem 4: Products of two unequal factors . . . . . 168
2026 AIME I Problem 6: A logarithmic quadratic determines a product . . . . . 169
2026 AIME I Problem 8: Divisor residues determined by exponent parity . . . . . 169
2026 AIME I Problem 13: Binomial residue sums over a finite field . . . . . 170
2026 AIME II Problem 1: Arithmetic progressions indexed by common divisors . . . . . 170
2026 AIME II Problem 3: An indented pentagon gives one residue class . . . . . 171
2026 AIME II Problem 4: A numeral equal to its value in a smaller base . . . . . 172
2026 AIME II Problem 5: Equal hypergeometric probabilities . . . . . 172
2026 AIME II Problem 8: An integer isosceles triangle with a tiny incenter triangle . . . . . 173
2026 AIME II Problem 9: A divisor-generating decimal with controlled carries . . . . . 173
2026 AIME II Problem 10: An angle bisector and tangent secants . . . . . 174
2026 AIME II Problem 15: A Fano-plane residue count . . . . . 174

# AIME Number Theory Worked Solutions

New exposition after source consultation. Each entry records its source review and verification; see the coverage and provenance reports for their precise scope.

## 1983 AIME Problem 6: Powers one away from seven

Find the remainder of 6^83+8^83 upon division by 49.

Published problem and solutions · MAA AMC.

### Solution

Modulo $7^2$, terms containing $7^2$ vanish in the binomial expansions. Because 83 is odd, (7-1)^83≡-1+83 · 7 and (7+1)^83≡1+83 · 7. Their sum is 14 · 83=1162≡35 (mod 49).

**Answer:** 035.

**Methods:** modular arithmetic, binomial expansion.

**Verification:** 1162=23 · 49+35; an independent modular-power check is readily reproducible. Independent exact check passed: Independent modular exponentiation.

**Source comparison:** Five entries inspected. Binomial, Euler-inverse, and factorization methods are valid. Index 2 asserts a false constant-residue pattern for odd exponents; index 4 supplies no actual modular computation.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Expansion around 7 determines two large powers modulo 49.

### 1983 AIME Problem 8: A prime divisor of a central binomial coefficient

Find the largest two-digit prime dividing C(200,100).

Published problem and solutions · MAA AMC.

#### Solution

For a prime p>50, $p^2$>200 and the exponent in C(200,100) is floor(200/p)-2 floor(100/p). Every two-digit prime p≥67 has exponent 2-2=0. For p=61 the exponent is 3-2=1. Since 61 is the largest prime below 67, it is the largest two-digit prime divisor.

**Answer:** 061.

**Methods:** prime valuations, factorials.

**Verification:** The valuation is 1 at 61 and 0 for each possible larger two-digit prime 67,71,73,79,83,89,97. Independent exact check passed: Form the exact binomial integer and test all two-digit primes.

**Source comparison:** Both alternatives use factorial valuations. Index 1 incorrectly displays floor(200/p)>3; the required condition in the relevant range is at least 3. Index 0 uses a strict inequality 3p<200, harmless here because 200 is not three times a prime.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Prime valuations of a factorial ratio identify the largest two-digit prime divisor.

### 1983 AIME Problem 12: A diameter with reversed chord digits

A circle has a two-digit integer diameter AB. A perpendicular chord CD has length obtained by reversing those digits. Their intersection H has positive rational distance from the center O. Find AB.

Published problem and solutions · MAA AMC.

#### Solution

Write AB=10a+b and CD=10b+a. The perpendicular diameter bisects the chord, so $(2\,OH)^2=(10a+b)^2-(10b+a)^2=99(a+b)(a-b)$. Positivity gives a>b. A rational square root of an integer is integral; therefore this product is a square. Its factor 11 forces 11|(a+b)(a-b). Since 0<a-b≤9 and a+b≤18, we get a+b=11. Now $99\cdot 11(a-b)=33^2(a-b)$, so a-b is a positive square. With a+b=11 and digits at most 9, the odd difference is at most 7; its only square possibility is 1. Thus a=6,b=5 and AB=65.

**Answer:** 065.

**Methods:** digit constraints, perfect squares, circle chords.

**Verification:** $65^2-56^2=1089=33^2$, so OH=33/2 is positive rational. Independent exact check passed: Exhaust all90 digit pairs and test the squared chord-distance difference.

**Source comparison:** Both alternatives use the right-triangle or power-of-point identity followed by the prime 11. The bound on the digit difference is needed to rule out square 9 after a+b=11.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory. The reversed-digit chord condition imposes a perfect-square constraint from a right triangle.

### 1984 AIME Problem 2: The smallest zero-eight multiple

Let n be the smallest positive multiple of 15 whose decimal digits all belong to{0,8}. Find n/15.

Published problem and solutions · MAA AMC.

#### Solution

Divisibility by 5 forces the last digit to be 0. Divisibility by 3 forces the number of 8s to be a positive multiple of 3. Thus at least three 8s and a final 0 are necessary. The smallest such numeral is 8880, which is divisible by 15. Therefore n/15=592.

**Answer:** 592.

**Methods:** divisibility, decimal digits.

**Verification:** 8880=15 · 592; no positive numeral with fewer than four digits can have three 8s and a final 0. Independent exact check passed: Exhaust all admissible numerals through four digits.

**Source comparison:** The sole alternative correctly combines divisibility by 3 and 5 with the minimal-digit-length argument.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Divisibility by3 and5 constrains an integer containing only0 and8.

### 1984 AIME Problem 14: The last even number without two odd composites

Find the largest even integer that cannot be written as the sum of two odd composite positive integers.

Published problem and solutions · MAA AMC.

#### Solution

The odd composites below 38 are 9,15,21,25,27,33. Subtracting these from 38 gives 29,23,17,13,11,5, all prime, so 38 has no representation. Every even n>38 has one: if n≡0(mod 6), use 9+(n-9); if n≡4(mod 6), use 25+(n-25); if n≡2(mod 6), use 35+(n-35). In the respective ranges n≥42,40,44, the second summand is an odd multiple of 3 at least 9. Both summands are therefore odd composites, proving the maximum 38.

**Answer:** 038.

**Methods:** residue classes, constructive number theory.

**Verification:** The six possible odd-composite addends below 38 are exhausted, and the three residue classes cover every larger even integer. Independent exact check passed: Enumerate all odd-composite sums through1000; the proof handles the infinite tail.

**Source comparison:** All seven entries read. The modulo 6 and modulo 5 constructions are sound. Indices 2 and 5 invoke a Frobenius bound without proving that its arbitrary linear combination splits into two odd composites. Indices 5 and 6 also falsely claim every candidate must be twice a prime; this omits even numbers whose halves are even. The direct residue construction is complete.

**Archived entries inspected:** 7 of 7. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Residue classes give explicit odd-composite decompositions of every sufficiently large even integer.

### 1985 AIME Problem 3: An integer complex cube

Positive integers a,b,c satisfy $c=(a+bi)^3-107i$, where $i^2=-1$. Find c.

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#### Solution

Comparing imaginary parts gives $b(3a^2-b^2)=107$. Since 107 is prime and $b>0$, b is 1 or 107. The second choice would give $3a^2=107^2+1\equiv 2 \pmod 3$, impossible. Thus $b=1$ and $3a^2=108$, so $a=6$. The real part is $c=a^3-3ab^2=216-18=198$.

**Answer:** 198.

**Methods:** complex expansion, prime divisors, congruences.

**Verification:** $(6+i)^3=198+107i$, so the candidate satisfies the full complex equation. Independent exact check passed: Enumerate the prime-divisor cases and all possible positive a in their bounds.

**Source comparison:** The sole alternative correctly combines real/imaginary parts with the primality of 107 and a modulo 3 exclusion.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. The imaginary part of a complex cube yields a prime-divisor restriction on a positive integer.

### 1985 AIME Problem 7: Two perfect powers nineteen apart

Positive integers a,b,c,d satisfy $a^5=b^4$, $c^3=d^2$, and $c-a=19$. Find $d-b$.

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#### Solution

Prime exponents in $a^5=b^4$ show $a=t^4, b=t^5$ for a positive integer t; similarly $c=s^2, d=s^3$. Thus $(s-t^2)(s+t^2)=19$. Both factors are positive integers and 19 is prime, so they are 1 and 19. Consequently $s=10, t^2=9, t=3$. Therefore $d-b=10^3-3^5=757$.

**Answer:** 757.

**Methods:** perfect powers, prime valuations, difference of squares.

**Verification:** $a=81, b=243, c=100, d=1000$ satisfy the two power equations and $c-a=19$.

**Source comparison:** The sole alternative gives the same complete prime-exponent and factorization argument; positivity rules out negative factor pairs.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Coprime exponent equations force perfect powers and a difference-of-squares factorization.

## 1985 AIME Problem 13: The largest gcd of adjacent quadratic values

Let $a_n=100+n^2$ for positive integers n. Find the largest possible $\gcd(a_n,a_{n+1})$.

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### Solution

The gcd is $d=\gcd(n^2+100,2n+1)$. Since $2n+1$ is odd, multiplying the first entry by 4 does not change d. Modulo $2n+1$, the identity $4(n^2+100)=(2n+1)^2-2(2n+1)+401$ gives $d=\gcd(401,2n+1)\leq 401$. At $n=200$, both entries are divisible by 401: $n^2+100=100\cdot 401$ and $2n+1=401$. Thus the maximum is 401.

**Answer:** 401.

**Methods:** Euclidean algorithm, polynomial gcd, attainment.

**Verification:** At $n=200$, $a_n=40100$ and $a_{n+1}=40501$ have gcd 401; the identity proves the bound for every n. Independent exact check passed: Direct gcd values across two complete401-residue periods.

**Source comparison:** All four alternatives inspected. Indices 0 and 3 establish a constant gcd bound. Index 1 merely chooses $n=200$ after an intermediate factorization, without proving this maximizes the gcd; index 2 supplies the missing Euclidean reduction.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. The Euclidean algorithm reduces neighboring quadratic values to a fixed gcd bound.

## 1986 AIME Problem 5: A variable divisor and a constant remainder

Find the largest positive integer n such that $n+10$ divides $n^3+100$.

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### Solution

Modulo $n+10$, we have $n\equiv -10$ and therefore $n^3+100\equiv -900$. The condition is exactly $n+10\mid 900$. Since $n+10$ is positive, its largest possible value is 900, attained when $n=890$. Thus the maximum is 890.

**Answer:** 890.

**Methods:** polynomial remainders, divisibility.

**Verification:** The polynomial identity $n^3+100=(n+10)(n^2-10n+100)-900$ proves both necessity and sufficiency. Independent exact check passed: Direct division for every n up to the proven divisor bound.

**Source comparison:** All five alternatives correctly reduce the condition to n+10 dividing 900, using Euclidean division, substitution, or a difference of cubes.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Polynomial reduction modulo n+10 turns divisibility into a divisor condition on900.

## 1986 AIME Problem 6: A page counted twice

The page numbers 1 through n were summed, but one page number was added twice. The resulting total was 1986. Find the repeated page.

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### Solution

Write 1986=n(n+1)/2+k with 1≤k≤n. If n≤61, the total is at most $61 \cdot 62/2+61=1952$. If n≥63, the triangular sum alone is at least $63 \cdot 64/2=2016$. Thus n=62, and k=1986-$62 \cdot 63/2$=33, which lies between 1 and 62.

**Answer:** 033.

**Methods:** triangular numbers, integer bounds.

**Verification:** 1+···+62=1953 and 1953+33=1986. Independent exact check passed: Exhaust all feasible page counts below the triangular-sum bound.

**Source comparison:** Both alternatives obtain 33. Index 0 uses an approximate root and says k<n, though k=n is permitted; the exact adjacent bounds remove both issues.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. An integral triangular-number interval identifies the page count and repeated page.

## 1986 AIME Problem 7: Ranking subsets by ternary value

Arrange all positive sums of distinct nonnegative powers of 3 in increasing order. Find the hundredth term.

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### Solution

These are exactly positive ternary numerals using only 0 and 1. Reading the same digit strings in binary preserves order: the first differing digit dominates all following digits in either base. Thus the hundredth ternary string is the binary representation of 100, namely 1100100. Its ternary value is $3^6 + 3^5 + 3^2 = 981$.

**Answer:** 981.

**Methods:** order-preserving bijection, subsets.

**Verification:** Independent exact check passed: Enumerated and sorted every nonempty subset of the first eight ternary powers. Result 981. The written proof supplies the general/exhaustive justification.

**Source comparison:** Read and compared all 5 archived solution entries. Alternatives are binary-to-ternary conversion and recursive blocks of subset sums.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. The ordered set of sums of distinct powers of3 is encoded by binary indices.

## 1986 AIME Problem 8: Logarithms of proper divisors

Let S be the sum of the base 10 logarithms of all positive proper divisors of 1,000,000. Find the integer nearest to S.

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### Solution

The number $10^6=2^6 5^6$ has 49 positive divisors. The involution $d\mapsto 10^6/d$ pairs logarithms with sum 6; equivalently, summing the identity $\log d+\log(10^6/d)=6$ over all 49 divisors gives twice the full sum as 294. Thus the full sum is 147. Excluding $10^6$ subtracts 6, giving S=141 exactly.

**Answer:** 141.

**Methods:** divisor pairing, logarithms.

**Verification:** Direct exponent summation over $0\le a,b\le 6$ gives 147 for each prime's exponent; deleting(6,6) leaves 141 and product 10^141.

**Source comparison:** All three entries reviewed. The divisor-product or involution proof handles the self-paired divisor 1000 automatically. Index 2 has a sign slip in an intermediate sentence but uses the correct final 144+3-6.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Pairing complementary divisors evaluates their logarithmic sum.

## 1987 AIME Problem 1: Addition without carrying

Count ordered nonnegative integer pairs (m,n) summing to 1492 with no decimal carry.

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### Solution

For a target digit $d$, choose the corresponding digit of $m$ from 0 through $d$; the digit of $n$ is then forced. Leading zeros are allowed and the four positions are independent. Hence the count is $(1+1)(4+1)(9+1)(2+1)=300$.

**Answer:** 300.

**Methods:** digit product rule.

**Verification:** Independent exact check passed: Enumerated all nonnegative summand pairs. Result 300. The written proof supplies the general/exhaustive justification.

**Source comparison:** Read and compared all 1 archived solution entries. The first archived solution uses the same central counting model; this is a newly written exposition after consultation.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Independent digit splits count carry-free ordered decompositions of1492.

### 1987 AIME Problem 3: Numbers equal to their proper-divisor product

For this problem, proper divisors exclude both 1 and the number itself. Call an integer n>1 nice when the product of its distinct proper divisors equals n. Find the sum of the first ten nice integers.

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#### Solution

If $\tau$(n) counts all positive divisors, pairing d with n/d shows their product is n^($\tau$(n)/2). Removing 1 andn leaves n^($\tau$(n)/2-1). Since n>1, equality withn is equivalent to$\tau$(n)=4. Thus n is either $p^3$ for a primep or pq for distinct primesp,q. The ten smallest are 6,8,10,14,15,21,22,26,27,33, whose sum is 182. Checking $p^3 \le 33$ and $pq \le 33$ lists every possibility before the tenth.

**Answer:** 182.

**Methods:** divisor product, divisor count, prime factorization.

**Verification:** The four-divisor forms exhaust all n≤33 and produce exactly the ten listed values, summing 182. Independent exact check passed: Directly multiply all distinct proper divisors through the tenth nice number.

**Source comparison:** Both alternatives identify prime cubes and products of two distinct primes. Index 0 initially says merely two distinct prime divisors, which would allow extra exponents; the intended condition is exponent 1 each. The divisor-product identity gives the full necessity cleanly.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. The product-of-divisors identity characterizes integers with exactly four divisors.

### 1987 AIME Problem 5: A factored equation in two integer squares

Integers x,y satisfy $y^2+3x^2y^2=30x^2+517$. Find $3x^2y^2$.

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#### Solution

Rearrange to $(3x^2+1)(y^2-10)=507=3 \cdot 13^2$. The first factor is a positive divisor congruent to 1 modulo 3, so it is 1,13,or 169. These give $x^2$=0,4,or 56. The last is not a square; the first would force $y^2$=517, also not a square. Therefore $x^2$=4, and the second factor is 39, giving $y^2$=49. The requested product is $3 \cdot 4 \cdot 49$=588.

**Answer:** 588.

**Methods:** factorization, integer squares, divisor cases.

**Verification:** For $x^2$=4,$y^2$=49, the original sides equal 49+588=637 and 120+517=637. Independent exact check passed: Enumerate all integer pairs within the bounds supplied by the positive factorization.

**Source comparison:** The sole alternative gives the correct factorization but omits the factor 1 case corresponding to x=0. The new proof explicitly excludes it.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Completing a product converts an integer equation in two squares to finitely many divisor cases.

## 1987 AIME Problem 7: LCM triples

Count positive ordered triples (a,b,c) with lcm(a,b)=1000 and lcm(a,c)=lcm(b,c)=2000.

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### Solution

Only primes 2 and 5 can occur. For the 2-exponents $(u, v, w)$, the conditions are $\max(u, v) = 3$ and $\max(u, w) = \max(v, w) = 4$. Thus $w = 4$ and $(u, v) \in \{0, 1, 2, 3\}^2$ has maximum 3, giving $4^2 - 3^2 = 7$ choices. For the 5-exponents, every pair has maximum 3, so at least two exponents are 3. There are $1 + 3 \cdot 3 = 10$ choices. Independent prime-exponent choices give $7 \cdot 10 = 70$.

**Answer:** 070.

**Methods:** prime exponent product, hybrid number theory.

**Verification:** The derivation yields the answer in the independently hosted answer field; no separate exhaustive computation claimed.

**Source comparison:** Read and compared all 2 archived solution entries. Alternatives count prime exponents independently or split on the exponent pattern of c.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Prime-exponent choices are counted under three least-common-multiple constraints.

## 1987 AIME Problem 8: Exactly one integer in a moving interval

Find the largest positive integer n for which exactly one integer k satisfies 8/15<n/(n+k)<7/13.

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### Solution

The positive fraction forces n+k>0. Taking reciprocals and subtracting 1 yields 6n/7<k<7n/8, an open interval of lengthn/56. If n>112, its length exceeds 2 and it contains at least two integers. For n=112, it is exactly 96<k<98, containing only 97. Thus 112 is the largest possible n.

**Answer:** 112.

**Methods:** integer intervals, strict inequalities.

**Verification:** At n=112,k=97 the fraction 112/209 lies strictly between the given bounds; the interval-length argument excludes every larger n.

**Source comparison:** All four alternatives yield 112. The first two assert the interval-width bound without explanation; the third supplies an integer-count argument. Open endpoints are essential because an interval of length exactly 2 can contain just one integer.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Uniqueness of an integer in a rational open interval is governed by interval length and endpoints.

### 1987 AIME Problem 11: The longest consecutive-integer representation

Find the largest k such that $3^{11}$ is a sum of k consecutive positive integers.

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#### Solution

If the first integer is a≥1, then $2 \cdot 3^{11}$=k(2a+k-1). Thus k divides $2 \cdot 3^{11}$ and k(k+1)≤$2 \cdot 3^{11}$, in particular k<$3^6$=729. Divisors have the form $3^j$ or $2 \cdot 3^j$, and the largest below 729 is $2 \cdot 3^5$=486. For that value,2a+k-1=$3^6$=729, so a=122. Hence 122+⋯+607 has 486 positive terms and the required sum; the maximum is 486.

**Answer:** 486.

**Methods:** consecutive sums, factor pairs, positivity bounds.

**Verification:** The arithmetic-series sum is 486(122+607)/2=$243 \cdot 729=3^{11}$. Independent exact check passed: Enumerate every feasible length below the positive-sum bound.

**Source comparison:** All three entries read. Indices 0 and 2 use valid factor pairs. Index 1 falsely claims an even number of consecutive terms has even sum and concludes k must be odd, contradicting the actual answer 486.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Consecutive-integer sums correspond to opposite-parity factor pairs of twice the target.

### 1987 AIME Problem 12: An integer just above a cube

Let m be the smallest integer with cube root n+r, where n is a positive integer and 0<r<1/1000. Find n.

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#### Solution

For fixed n, the least possible integer is $n^3$+1. It works exactly when$(n+1/1000)^3$-$n^3$>1. This difference increases with n. At n=18 it is 972054001/$10^9$<1; at n=19 it is 1083057001/$10^9$>1. Therefore the first possible n is 19, realized by m=$19^3$+1. Every larger n gives a larger m, so this also minimizes m globally.

**Answer:** 019.

**Methods:** integer bounds, binomial expansion, monotonicity.

**Verification:** The exact cubic increments at 18 and 19 straddle 1, and their monotonicity settles every smaller n.

**Source comparison:** All four alternatives read. Index 1 uses the exact interval width. Index 0 has an insufficiently justified preliminary bound; index 2 simply discards positive small terms, which cannot prove minimality. Index 3 gives derivative bounds but leaves the endpoint comparison implicit.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. An integer immediately above a cube must lie in a short interval whose width increases with the cube root.

## 1988 AIME Problem 2: A digit-sum iteration

Let T(k) be the square of the sum of the decimal digits of the positive integer k. Define $f_1(k)=T(k)$ and $f_n(k)=T(f_{n-1}(k))$. Find $f_{1988}(11)$.

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### Solution

The successive values are 4,16,49,169,256,169. Since T(169)=256 and T(256)=169, the sequence alternates between 169 and 256 from index 4 onward. Index 1988 is even, so $f_{1988}(11)=169$.

**Answer:** 169.

**Methods:** digit sums, iteration, periodicity.

**Verification:** The cycle follows from $(1+6+9)^2=256$ and $(2+5+6)^2=169$.

**Source comparison:** The single alternative correctly identifies and verifies the eventual two-cycle; the initial indexing places 169 at every even index from 4 onward.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A repeated digit-sum-square operation enters a two-cycle.

## 1988 AIME Problem 5: Divisor probability

A divisor of 10^99 is chosen uniformly. Find numerator plus denominator for its probability of being divisible by 10^88.

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### Solution

Write the divisor as $2^a5^b$ with $0 \le a, b \le 99$. All 10000 exponent pairs correspond to distinct divisors. Divisibility by $10^{88}$ requires both exponents at least 88, giving $12^2 = 144$ pairs. Thus the chance is 9/625, giving 634.

**Answer:** 634.

**Methods:** exponent lattice, probability.

**Verification:** The derivation yields the answer in the independently hosted answer field; no separate exhaustive computation claimed.

**Source comparison:** Read and compared all 1 archived solution entries. The first archived solution uses the same central counting model; this is a newly written exposition after consultation.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Prime-exponent choices count divisors satisfying a prescribed divisibility condition.

### 1988 AIME Problem 8: A functional equation hiding the least common multiple

On positive integer pairs, f satisfies f(x,x)=x, f(x,y)=f(y,x), and (x+y)f(x,y)=y f(x,x+y). Find f(14,52).

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#### Solution

Normalize by setting g(x,y)=f(x,y)/(xy). The third identity becomes g(x,x+y)=g(x,y), and symmetry permits interchange of the arguments. Applying the subtractive Euclidean algorithm therefore reduces g(x,y) to g(d,d), where d=gcd(x,y). At that pair, g(d,d)=d/$d^2$=1/d. Hence f(x,y)=xy/gcd(x,y), uniquely. Since gcd(14,52)=2, f(14,52)=14 · 52/2=364.

**Answer:** 364.

**Methods:** functional equations, Euclidean algorithm, normalization.

**Verification:** The derived formula satisfies all three identities. Subtractive Euclidean reduction of (14, 52) passes through $(14, 38), (14, 24), (10, 14), (4, 10)$, then through $(4, 6), (2, 4), (2, 2)$, reaching $d = 2$.

**Source comparison:** All three alternatives read. The first two perform the recurrence directly. Index 2 verifies that lcm is a solution but appeals to AIME answer uniqueness rather than proving functional uniqueness. The invariant plus descending Euclidean algorithm supplies that proof.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. A two-variable functional equation becomes an invariant under the Euclidean algorithm.

### 1988 AIME Problem 9: The smallest cube ending in eight-eight-eight

Find the least positive integer n whose cube has final three decimal digits 888.

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#### Solution

The congruence modulo 8 forces n even. Modulo 5, $n^3$≡3 gives n≡2, so write n=2+5a. Reducing modulo 25 gives 8+10a≡13, hence a≡3(mod 5) and n=17+25b. Modulo 125 this gives 38+50b≡13, hence b≡2(mod 5) and n≡67(mod 125). Combining with evenness yields n≡192(mod 250). These conditions are also sufficient modulo 8 and 125, hence modulo 1000. The least positive solution is 192.

**Answer:** 192.

**Methods:** modular lifting, Chinese remainder theorem, cubic residues.

**Verification:** $192^3$=7077888, and the full solution class is 192 modulo 250, proving minimality. Independent exact check passed: Exhaust all residues modulo1000 directly.

**Source comparison:** All six alternatives read. Digit-by-digit lifting and CRT approaches agree. Some alternatives test a successful residue without explicitly showing uniqueness; the linear lift congruences here determine each digit uniquely modulo 5.

**Archived entries inspected:** 6 of 6. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Successive lifting modulo5,25,125 and a parity condition determine a cube residue.

## 1989 AIME Problem 3: A repeated decimal digit forced by divisibility

For a positive integer n and a decimal digit d, n/810=0.d25d25d25…. Find n.

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### Solution

The repeating decimal equals (100d+25)/999. Therefore n=750(4d+1)/37. Since 750 is relatively prime to 37, integrality requires 37 to divide 4d+1. For $0 \le d \le 9$, this expression lies between 1 and 37, so it must equal 37. Thus d=9 and n=750.

**Answer:** 750.

**Methods:** repeating decimals, divisibility, digit constraints.

**Verification:** 750/810=25/27=925/999=0.925925…. Independent exact check passed: Exhaust the digit and numerator ranges with exact fractions.

**Source comparison:** All four alternatives read. The denominator-divisibility argument supplies uniqueness. Index 3 incorrectly states that n divides 25 and 30; its conclusion is not supported by that sentence.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. A repeated three-digit decimal is converted to a fraction and constrained by divisibility.

## 1989 AIME Problem 4: One middle term, two perfect powers

Five consecutive positive integers are a<b<c<d<e. The sum b+c+d is a square, and a+b+c+d+e is a cube. Find the smallest possible c.

Published problem and solutions · MAA AMC.

### Solution

The conditions say that 3c is a square and 5c is a cube. If $e_p$ is the exponent of prime p in c, then $e_3$ is odd and divisible by 3, so $e_3 \ge 3$. Similarly $e_5$ is even and congruent to 2 modulo 3, so $e_5 \ge 2$. Every other

$e_p$ is divisible by 6 and may be zero. Consequently $c \geq 3^3 5^2 = 675$. This value works because $3c = 45^2$ and $5c = 15^3$; the five consecutive integers are positive.

**Answer:** 675.

**Methods:** prime valuations, perfect powers, minimality.

**Verification:** $3 \cdot 675 = 2025 = 45^2$ and $5 \cdot 675 = 3375 = 15^3$. Independent exact check passed: Exhaust every possible smaller middle term and test square/cube conditions.

**Source comparison:** All four alternatives read. Index 2 once calls the three-term sum a cube instead of a square. Some alternatives choose the smallest exponents without explaining why additional primes cannot lower the product; the valuation bounds provide that justification.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Simultaneous square and cube conditions prescribe prime valuations of the middle integer.

## 1989 AIME Problem 9: Recovering a fifth root with congruences

A positive integer n satisfies $133^5 + 110^5 + 84^5 + 27^5 = n^5$. Find n.

Published problem and solutions · MAA AMC.

### Solution

Modulo 2 and 3, the equation forces $n \equiv 0$ modulo 6. Modulo 5, fifth powers equal their bases, so $n \equiv 4$ modulo 5 and hence $n \equiv 24$ modulo 30. Modulo 7 the sum is 2. A nonzero residue has fifth power equal to its inverse modulo 7, so $n \equiv 4$ modulo 7. Combining the last two conditions gives $n \equiv 144$ modulo 210. Also $133 < n < 266$, because the sum exceeds $133^5$ and is less than $4 \cdot 133^5 < 266^5$. The unique integer in that interval congruent to 144 modulo 210 is 144.

**Answer:** 144.

**Methods:** power residues, Chinese remainder theorem, bounds.

**Verification:** The four fifth powers sum to $61917364224 = 144^5$. Independent exact check passed: Compute the fifth-power equation as exact integers.

**Source comparison:** All six alternatives read. Modular approaches are effective when paired with explicit bounds. Indices 1–3 invoke rough estimates or an allegedly obvious upper bound; the interval here is elementary and sufficient. Index 5 quotes a large integer root without a derivation.

**Archived entries inspected:** 6 of 6. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Congruences for fifth powers isolate an integer fifth root within a short interval.

## 1989 AIME Problem 14: Real integers written in a complex base

Let $\beta = -3 + i$. Consider four-digit expressions $a_3\beta^3 + a_2\beta^2 + a_1\beta + a_0$ with decimal digits $0 \leq a_j \leq 9$ and $a_3 \neq 0$. Find the sum of all distinct real integers represented.

Published problem and solutions · MAA AMC.

**Solution**

Since $\beta^2=8-6i$ and $\beta^3=-18+26i$, vanishing imaginary part requires $a_1-6a_2+26a_3=0$. As $6a_2-a_1\le 54$, we have $a_3=1$ or 2. For $a_3=1$, the digit bounds on $a_1=6a_2-26$ force $(a_2,a_1)=(5,4)$. For $a_3=2$ they force $(a_2,a_1)=(9,2)$. The real parts are respectively $10+a_0$ and $30+a_0$. Thus the represented integers are exactly 10,…,19 and 30,…,39, with sum 145+345=490.

**Answer:** 490.

**Methods:** complex bases, digit constraints, bounded linear equations.

**Verification:** The complete digit analysis yields precisely twenty integers in two disjoint ten-term intervals. Independent exact check passed: Enumerate every four-digit complex-base numeral using integer real/imaginary components.

**Source comparison:** The single archived alternative correctly reduces the imaginary part to two digit cases. The two resulting integer intervals are disjoint, so summing numerals does not introduce duplicate values.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. The imaginary part of a complex-base numeral imposes a bounded linear equation on its digits.

## 1990 AIME Problem 1: Ranking nonsquares and noncubes

List positive integers that are neither perfect squares nor perfect cubes. Find the five-hundredth.

Published problem and solutions · MAA AMC.

**Solution**

Up to $N$, inclusion-exclusion gives $N-\lfloor\sqrt{N}\rfloor-\lfloor\sqrt[3]{N}\rfloor+\lfloor\sqrt[6]{N}\rfloor$ admissible numbers. At $N=528$ this is $528-22-8+2=500$, while at 527 it is 499. Therefore the five-hundredth term is 528.

**Answer:** 528.

**Methods:** inclusion-exclusion, rank inversion.

**Verification:** Independent exact check passed: Direct filtering of all integers through 599. Result 528. The written proof supplies the general/exhaustive justification.

**Source comparison:** Read and compared all 2 archived solution entries. The first archived solution uses the same central counting model; this is a newly written exposition after consultation.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Inclusion-exclusion counts integers that are neither squares nor cubes.

## 1990 AIME Problem 3: A largest regular-polygon side count

Regular polygons with r and s sides satisfy $r\ge s\ge 3$, and each interior angle of the r-gon is 59/58 times each interior angle of the s-gon. Find the largest possible s.

Published problem and solutions · MAA AMC.

**Solution**

Using the interior-angle formula gives $58(r-2)/r=59(s-2)/s$, hence $r=116s/(118-s)$. Positivity of r and s forces $s<118$, so $s\leq 117$. For $s=117$, the formula gives $r=116\cdot 117$, an integer exceeding s. Thus the bound is attained and the answer is 117.

**Answer:** 117.

**Methods:** regular polygons, rational constraints, attainability.

**Verification:** At $s=117$,$r=13572$, the rearranged angle-ratio equation holds exactly.

**Source comparison:** All three alternatives read. Indices 0 and 2 prove the bound and exhibit an integer solution. Index 1 splits divisibility into cases that are not exhaustive and invokes the contest answer range; those arguments are unnecessary.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory. Regular-polygon angle ratios lead to an integer bound on the number of sides.

## 1990 AIME Problem 5: The smallest multiple with seventy-five divisors

Let n be the smallest positive multiple of 75 having exactly 75 positive divisors. Find n/75.

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**Solution**

For $n=\prod p^e$, the divisor count is $\prod(e+1)=75=3\cdot 5^2$. Since 3 and 5 divide n, there are at least two distinct primes. The only exponent multisets are (24,2), (14,4), and (4,4,2); the one-prime pattern (74) is impossible. In either two-prime case the primes must be 3 and 5, and assigning the larger exponent to 3 minimizes the product. The resulting bounds $3^{24}5^2$ and $3^{14}5^4$ both exceed 32400. In the three-prime case the least possible product assigns exponents 4,4,2 to primes 2,3,5 respectively, giving $n=2^4 3^4 5^2=32400$. Therefore $n/75=432$.

**Answer:** 432.

**Methods:** divisor function, prime-exponent optimization, multiplicative partitions.

**Verification:** The proposed n has $(4+1)(4+1)(2+1)=75$ divisors and is divisible by 75. Independent exact check passed: Test every eligible multiple up to the proposed global minimum directly.

**Source comparison:** The single alternative identifies the correct three-prime pattern but does not compare it with all two-prime patterns. The complete factorization list and comparison establish global minimality.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Factorizations of the divisor count determine the smallest admissible prime-exponent pattern.

## 1990 AIME Problem 10: Products of two root-of-unity groups

Let $A = \{z : z^{18} = 1\}$ and $B = \{w : w^{48} = 1\}$. How many distinct complex numbers occur as $zw$ with $z \in A$ and $w \in B$?

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### Solution

Let $\zeta = e^{2\pi i/144}$. Every product has the form $\zeta^{8a+3b}$, so it is among the 144 powers of $\zeta$. Conversely, fix any integer $k$. Choose $a \in \{0, 1, 2\}$ with $8a \equiv k \pmod 3$, and set $b = (k - 8a)/3$. Then $\zeta^{8a}$ is an eighteenth root and $\zeta^{3b}$ is a forty-eighth root, and their product is $\zeta^k$. Thus all 144 distinct powers occur.

**Answer:** 144.

**Methods:** roots of unity, finite cyclic groups, image counting, least common multiple.

**Verification:** The multiplication map has $18 \cdot 48$ inputs and kernel $\{(u,u^{-1}):u^6=1\}$ of size 6, independently giving 864/6=144 products. Independent exact check passed: Enumerate all864 root-product exponents, independently checking the image size.

**Source comparison:** All three alternatives read. The first uses a Frobenius-number argument that needs explicit reduction of exponent ranges. The second comment incorrectly claims every integer 0 through 143 is literally representable with nonnegative bounded exponents; modular representability is sufficient. The third uses synchronized exponents, which does generate all 144 values but needs an order calculation.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, algebra, number_theory. The number of distinct products of roots of unity is the size of a finite cyclic-group image.

## 1990 AIME Problem 11: A factorial as a shorter consecutive product

Find the largest positive integer n for which n! is the product of n−3 consecutive positive integers.

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### Solution

Any such block must end above n: a block of n−3 positive integers ending at most n has product at most n!/3!<n!. Among blocks ending above n, the smallest is $5 \cdot 6 \cdot \ldots \cdot (n+1)=(n+1)!/24$. Consequently equality with n! requires $(n+1)!/24 \le n!$, or $n \le 23$. At n=23 this smallest block equals 24!/24=23!, so 23 is attained.

**Answer:** 023.

**Methods:** factorials, monotone products, extremal bound.

**Verification:** The twenty consecutive integers 5 through 24 have product 24!/4!=23!. Independent exact check passed: Exhaust consecutive blocks for every n from4 through100; proof controls the infinite tail.

**Source comparison:** Both alternatives read. Index 0 invokes a decreasing root approximation that is not a valid global proof. Index 1 asserts a minimizing endpoint principle without an explicit inequality. Comparing every block to the least block ending above n supplies the missing bound.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Monotonicity of consecutive-integer products bounds when a factorial can equal a shorter block.

### 1990 AIME Problem 13: Leading nines in powers

Among 9^0,...,9^4000, count those beginning with decimal digit 9, given that 9^4000 has 3817 digits and begins with 9.

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#### Solution

Write $d_k$ for the digit length of $9^k$. Multiplication by 9 increases length by either 0 or 1. For $k \geq 1$, $d_k = d_{k-1}$ precisely when $9^k$ begins with 9: the two inequalities become $9 \cdot 10^{d_k - 1} \leq 9^k < 10^{d_k}$. Among 4000 transitions, exactly $3817 - 1 = 3816$ increase the digit length, and the other 184 preserve it. Since $9^0 = 1$ does not begin with 9, the answer is 184.

**Answer:** 184.

**Methods:** telescoping, leading digits.

**Verification:** The derivation yields the answer in the independently hosted answer field; no separate exhaustive computation claimed.

**Source comparison:** Read and compared all 3 archived solution entries. Alternatives group powers by digit length or count length-preserving multiplication steps.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Digit-length changes count powers of nine with a prescribed leading digit.

### 1991 AIME Problem 1: Symmetric sums with an integer restriction

Positive integers x,y satisfy xy+x+y=71 and $x^2y+xy^2$=880. Find $x^2+y^2$.

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#### Solution

Let s=x+y and p=xy. Then s+p=71 and sp=880, so {s,p}={16,55}. Positive integers satisfy xy≥x+y−1, because (x−1)(y−1)≥0. Thus s=55,p=16 is impossible. Therefore s=16,p=55 and $x^2+y^2=s^2-2p$=146. The values x,y=5,11 verify existence.

**Answer:** 146.

**Methods:** symmetric sums, Vieta formulas, integer inequalities.

**Verification:** $5 \cdot 11+5+11=71$ and $5 \cdot 11 \cdot 16=880$. Independent exact check passed: Exhaust positive integer pairs within the first-equation bound.

**Source comparison:** All five alternatives read. Indices 2 and 4 discard the wrong branch only because it would exceed the contest answer range; index 3 leaves that exclusion implicit. The positive-integer inequality eliminates it mathematically.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Symmetric sums reduce the equations to two cases, and positive integrality eliminates one.

## 1991 AIME Problem 5: Coprime factor allocations

How many reduced fractions strictly between 0 and 1 have numerator times denominator equal to 20!?

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### Solution

The eight distinct primes dividing 20! are $2, 3, 5, 7, 11, 13, 17, 19$. Coprimality requires each prime power in 20! to be placed wholly in the numerator or wholly in the denominator, giving $2^8$ ordered allocations. Swapping numerator and denominator pairs them, with no fixed point because the coprime factors cannot be equal. Exactly half have numerator smaller, so the count is 128.

**Answer:** 128.

**Methods:** prime allocation, involution.

**Verification:** The derivation yields the answer in the independently hosted answer field; no separate exhaustive computation claimed.

**Source comparison:** Read and compared all 1 archived solution entries. The first archived solution uses the same central counting model; this is a newly written exposition after consultation.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Coprime numerator and denominator choices distribute entire prime powers between two factors.

## 1991 AIME Problem 6: Recovering a number from a floor sum

A real r satisfies $\Sigma_{k=19}{}^{91}$ floor(r+k/100)=546. Find floor(100r).

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### Solution

The 73 floor values are nondecreasing and differ by at most 1. Since $546=73 \cdot 7+35$, they consist of 38 sevens followed by 35 eights. The last seven occurs at k=56 and the first eight at k=57. Thus $r+56/100<8\leq r+57/100$, giving $7.43\leq r<7.44$. Therefore floor(100r)=743.

**Answer:** 743.

**Methods:** floor functions, threshold counting, interval recovery.

**Verification:** Throughout [7.43,7.44), offsets 19 through 56 give 7 and offsets 57 through 91 give 8, totaling 546.

**Source comparison:** All three entries read. Index 0 is only a warning that a removed solution was incorrect. Index 1 gives the valid jump argument. Index 2 uses Hermite's identity but reverses a displayed monotonic inequality and ambiguously includes the endpoint r+1; the identity requires exactly the hundred offsets 0 through 99.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. A run of equally spaced floor terms has a single jump, fixing the fractional interval.

### 1991 AIME Problem 8: Integer roots via a shifted factorization

For how many real values a does $x^2+ax+6a=0$ have two integer roots, counted with multiplicity?

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#### Solution

If the roots are r,s, then r+s=−a and rs=6a. Eliminating a gives (r+6)(s+6)=36. Unordered positive factor pairs of 36 are (1,36),(2,18),(3,12),(4,9),(6,6), and there are five corresponding negative pairs. Each gives integer r,s and a=12−((r+6)+(s+6)). The positive pair sums 37,20,15,13,12 are distinct, and the negative pair sums are their negatives; hence all ten values of a are distinct. The answer is 10.

**Answer:** 010.

**Methods:** integer roots, Vieta formulas, factor pairs.

**Verification:** The ten parameter values are −25,−8,−3,−1,0,24,25,27,32,49. Independent exact check passed: Enumerate all shifted-factor-bounded integer root pairs and distinct parameters.

**Source comparison:** Both alternatives read. The shifted-root factorization is shorter than the discriminant approach. Counting pairs alone requires checking that different pairs do not yield the same parameter; their explicit sums establish injectivity.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Integer quadratic roots correspond to signed factor pairs of 36 after a Vieta shift.

### 1991 AIME Problem 13: Sock counts from a probability

At most 1991 socks, red or blue, are in a drawer. Two drawn without replacement have the same color with probability 1/2. Maximize the number of red socks.

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#### Solution

Write $N = r + b$. The equation $\binom{r}{2} + \binom{b}{2} = \frac{1}{2}\binom{N}{2}$ simplifies to $(r-b)^2 = N$. Therefore $N = t^2$ and the larger color class has $(t^2+t)/2$ socks. Since $t^2 \le 1991$, $t \le 44$. Choosing $t = 44$, red $r = 44 \cdot 45/2 = 990$ and blue 946 satisfies the equation and the size bound. The maximum is 990.

**Answer:** 990.

**Methods:** sampling without replacement, Diophantine probability.

**Verification:** The derivation yields the answer in the independently hosted answer field; no separate exhaustive computation claimed.

**Source comparison:** Read and compared all 6 archived solution entries. All six forms reduce to the square identity $(r-b)^2=r+b$, with different algebraic parametrizations.

**Archived entries inspected:** 6 of 6. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. A sampling probability imposes a quadratic Diophantine condition on the two sock counts.

## 1991 AIME Problem 15: An attainable vector minimum that is integral

For each positive integer n, let $S_n$ be the minimum of $\Sigma_{k=1}{}^{n}\sqrt{((2k-1)^2+a_k{}^2)}$ over positive real $a_k$ with $\Sigma a_k=17$. Find the unique n for which $S_n$ is an integer.

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### Solution

The triangle inequality for vectors $(2k-1,a_k)$ gives $S_n\geq\sqrt{((\Sigma(2k-1))^2+17^2)}=\sqrt{(n^4+289)}$. Equality is attained by $a_k=17(2k-1)/n^2$, which is positive and sums to 17; hence this bound is the actual minimum. If $S_n=m$ is integral, then $(m-n^2)(m+n^2)=289$. The positive factor pairs are (1,289) and (17,17). The latter gives n=0 and is excluded. The former gives $n^2=144$, so n=12, and indeed $S_{12}=145$.

**Answer:** 012.

**Methods:** triangle inequality, equality cases, perfect squares, factorization.

**Verification:** At n=12, the chosen $a_k$ sum to 17 and all vectors are parallel; $144^2+17^2=145^2$.

**Source comparison:** All four alternatives read. Indices 0 and 1 state a lower bound but do not explicitly construct equality. Index 3 has sign errors in its difference-of-squares factorization. The proportional positive coordinates prove attainment, and factor pairs establish uniqueness.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. The vector triangle inequality gives an attainable minimum, whose integrality becomes a difference of squares.

## 1992 AIME Problem 1: Pairing reduced fractions around five

Find the sum of all positive rational numbers less than 10 whose reduced denominator is 30.

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### Solution

These numbers are $k/30$ with $1 \leq k < 300$ and $\gcd(k, 30) = 1$. In every block of thirty integers the allowed residues are $1, 7, 11, 13, 17, 19, 23, 29$, so there are 80 fractions. The map $k \mapsto 300-k$ pairs them without a fixed point, and each pair of fractions sums to 10. Their total is therefore $40 \cdot 10 = 400$.

**Answer:** 400.

**Methods:** totient function, involution, reduced fractions, residue classes.

**Verification:** The eight allowed residues avoid divisibility by 2,3,5, and complementing a numerator preserves coprimality to 30. Independent exact check passed: Direct exact sum over all reduced numerators.

**Source comparison:** All three alternatives read. The first two correctly count eight reduced residues per unit interval. Index 2 uses $\phi(300)$=80; this happens to apply because 30 and 300 have the same prime divisors, a fact that should be made explicit rather than treated as automatic.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Coprime residue classes count the fractions, and a complementary-numerator involution sums them.

## 1992 AIME Problem 5: Numerators of repeating decimals

For all numbers 0.abcabc… strictly between 0 and 1, count distinct numerators after reducing the fraction.

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### Solution

Every fraction is $u/999$ with $1 \le u \le 998$, and $999 = 3^3 \cdot 37$. A reduced numerator n occurs exactly when some divisor d of 999 satisfies $n < d$ and $\gcd(n, d) = 1$. If n is divisible by neither 3 nor 37, take d=999; there are $\varphi(999) = 648$ such n. If 3 divides n but 37 does not, a coprime denominator must divide 37, so precisely $n = 3, 6, \ldots, 36$ work, adding 12. If 37 divides n, any coprime denominator is at most 27, smaller than n, so none work. The total is 660.

**Answer:** 660.

**Methods:** distinct image, inclusion-exclusion.

**Verification:** Independent exact check passed: Reduced every numerator 1,…,998 over 999 and deduplicated. Result 660. The written proof supplies the general/exhaustive justification.

**Source comparison:** Read and compared all 1 archived solution entries. The consulted source counts reductions of multiples of 81. Characterizing possible reduced denominators makes exhaustiveness explicit.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Reduced numerators of repeating decimals are counted through divisibility and denominator cancellation.

## 1992 AIME Problem 6: Consecutive integers without carries

Count consecutive pairs whose members are both in [1000,2000] and whose decimal addition produces no carry.

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**Solution**

Write the smaller member as 1ABC. If it has no trailing 9, all three digits A,B,C must be 0 through 4, giving $5^3$. If it has exactly one trailing 9, A and B must be 0 through 4, giving $5^2$. With exactly two trailing 9s, A has five choices. The remaining pair 1999,2000 also works. These disjoint cases give $125 + 25 + 5 + 1 = 156$.

**Answer:** 156.

**Methods:** carry states, digits.

**Verification:** Independent exact check passed: Direct decimal-digit carry test on all 1000 consecutive pairs. Result 156. The written proof supplies the general/exhaustive justification.

**Source comparison:** Read and compared all 3 archived solution entries. The reliable partition is by the number of trailing nines of the smaller integer.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Digit carries impose local restrictions on pairs of consecutive integers.

## 1992 AIME Problem 11: The order of two successive reflections

Reflect a line through the origin first in the line making angle $\pi/70$ with the positive x-axis, then in the line making angle $\pi/54$. Starting with y=(19/92)x, find the least positive number of repetitions that restores the original line.

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**Solution**

Reflection in a line of angle $\alpha$ sends a direction $\theta$ to $2\alpha-\theta$ modulo $\pi$. The two reflections therefore send $\theta$ to $\theta+2(\pi/54-\pi/70)=\theta+8\pi/945$. After m repetitions the original unoriented line returns exactly when $8m\pi/945$ is an integer multiple of $\pi$. Since gcd(8,945)=1, this occurs precisely when 945 divides m. The least positive value is 945.

**Answer:** 945.

**Methods:** composition of reflections, rotation order, modular arithmetic.

**Verification:** 945 repetitions rotate by $8\pi$; coprimality excludes every smaller positive repetition count.

**Source comparison:** The single alternative correctly treats lines as directions modulo $\pi$ rather than $2\pi$. The initial slope is irrelevant because every line through the origin has the same return order.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory. Composing reflections gives a rational-angle rotation whose order on unoriented lines is computed modulo pi.

## 1992 AIME Problem 15: Missing factorial trailing-zero counts

A positive integer is attainable if it is the exact number of trailing zeros of some factorial. How many positive integers below 1992 are unattainable?

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**Solution**

Let $z(m) = \sum_{j\geq 1} \lfloor m/5^j \rfloor$. It is constant on each block $5k, \dots, 5k{+}4$, and increases strictly between blocks. At $m = 7975$, $z(m) = 1595 + 319 + 63 + 12 + 2 = 1991$, while $z(7980) = 1596 + 319 + 63 + 12 + 2 = 1992$. Therefore exactly the 1595 positive block values for $k = 1, \dots, 1595$ occur below 1992. Of the 1991 positive target integers, $1991 - 1595 = 396$ are missing.

**Answer:** 396.

**Methods:** monotone image, valuation counting.

**Verification:** Independent exact check passed: Enumerated all distinct trailing-zero counts until the first value 1992. Result 396. The written proof supplies the general/exhaustive justification.

**Source comparison:** Read and compared all 2 archived solution entries. The second method counts skipped values at higher powers of five.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Gaps in the factorial trailing-zero function are counted from its discrete jumps.

## 1993 AIME Problem 4: Equal outer and inner sums

Count integer quadruples 0<a<b<c<d<500 satisfying a+d=b+c and bc−ad=93.

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**Solution**

Set $u = b - a = d - c > 0$ and $v = c - b > 0$. Then $(a, b, c, d) = (a, a + u, a + u + v, a + 2u + v)$ and $bc - ad = u(u + v) = 93$. Thus $(u, u + v)$ is either $(1, 93)$ or $(3, 31)$; the reversed factor pairs violate $v > 0$. The spans d−a are respectively 94 and 34, allowing $499 - 94 = 405$ and $499 - 34 = 465$ positive starting values a. The total is 870.

**Answer:** 870.

**Methods:** gap parametrization, hybrid number theory.

**Verification:** The derivation yields the answer in the independently hosted answer field; no separate exhaustive computation claimed.

**Source comparison:** Read and compared all 6 archived solution entries. Alternatives factor a translation-invariant product using gaps, a shifted origin or centered coordinates.

**Archived entries inspected:** 6 of 6. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Gap variables and a factorization count ordered integer quadruples with fixed symmetric relations.

### 1993 AIME Problem 6: One integer as three consecutive sums

Find the smallest positive integer expressible as the sum of nine consecutive integers, ten consecutive integers, and eleven consecutive integers.

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#### Solution

A sum of nine consecutive integers is divisible by 9, and a sum of eleven is divisible by 11. A sum of ten has form 10b+45, hence is congruent to 5 modulo 10. Therefore the desired number is a positive multiple of 99 ending in 5; the smallest is $99 \cdot 5 = 495$. It works: the nine-term block starts at 51, the ten-term block at 45, and the eleven-term block at 40.

**Answer:** 495.

**Methods:** consecutive sums, congruences, least common multiple.

**Verification:** $9 \cdot 51+36 = 10 \cdot 45+45 = 11 \cdot 40+55 = 495$. Independent exact check passed: Test the arithmetic-series starting-term integrality for every smaller positive target.

**Source comparison:** All five alternatives read. Some assume the individual summands are positive, although the problem only specifies integers; the congruence argument needs no such assumption. Index 3 invokes the contest answer range unnecessarily. Explicit starting terms prove attainability.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Representability as three consecutive sums imposes congruences modulo 9,10,11.

### 1993 AIME Problem 9: Colliding labels in a circular walk

On a circle of 2000 marked points, label one point 1. From label k−1 move k points clockwise to place label k, continuing through 1993. What is the smallest label sharing the point labeled 1993?

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#### Solution

The displacement to label n is $2+3+\cdots+n = n(n+1)/2-1$ modulo 2000. Thus labels n and 1993 coincide exactly when $(1993-n)(1994+n)$ is divisible by $4000 = 32 \cdot 125$. The two factors have odd sum 3977, relatively prime to both 2 and 5. Hence the entire factor 32 lies in one of them, and the entire factor 125 lies in one of them. Assigning these prime powers in the four possible ways and applying CRT gives $n \equiv 1993, 2006, 3881, 118$ modulo 4000. Among positive labels at most 1993, the smallest is 118.

**Answer:** 118.

**Methods:** triangular numbers, Chinese remainder theorem, prime-power allocation.

**Verification:** The triangular-number difference is $(1993 \cdot 1994 - 118 \cdot 119)/2 = 1980000$, divisible by 2000; the four residue classes prove minimality. Independent exact check passed: Enumerate all labels and compare their cumulative step residues.

**Source comparison:** Both alternatives read. Index 0 incorrectly keeps modulus 2000 after clearing the factor 1/2 and uses 16 rather than 32; it reaches the correct answer accidentally. Index 1 correctly works modulo 4000 and lists all four CRT classes.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Cumulative clockwise steps give triangular-number congruences and four CRT classes.

### 1994 AIME Problem 1: Indexing squares one above multiples of three

The increasing sequence 3,15,24,48,… consists of positive multiples of 3 that are one less than a square. Find the remainder of its 1994 th term on division by 1000.

Published problem and solutions · MAA AMC.

#### Solution

The terms are $n^2-1$ with $n\geq 2$ and $n\equiv 1$ or 2 modulo 3. In increasing order the admissible roots occur in pairs $3j-1,3j+1$ for $j=1,2,\ldots$. The 1994 th root is therefore $3\cdot 997+1=2992$. Modulo 1000 its square minus one is $(-8)^2-1=63$.

**Answer:** 063.

**Methods:** periodic residue classes, sequence indexing, modular arithmetic.

**Verification:** Each of the first 997 root pairs contributes two terms, and the second root of pair 997 is 2992. Independent exact check passed: Construct the actual increasing sequence through its1994th term.

**Source comparison:** The sole alternative correctly indexes the two admissible root classes. The exclusion of $n=1$ is necessary because the sequence contains positive terms only.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Square roots congruent to $\pm 1$ modulo 3 index the admissible sequence.

### 1994 AIME Problem 4: A sum of binary logarithm floors

Find the positive integer n satisfying $\Sigma_{k=1}{}^{n}\ \mathrm{floor}(\log_2 k)=1994$.

Published problem and solutions · MAA AMC.

#### Solution

For $2^j\leq k<2^{j+1}$ the summand is j. Thus the sum through 255 is $\Sigma^{j}{}_{=0}{}^{7} j2^j=1538$, and through 511 it is $1538+8\cdot 256=3586$. Hence $256\leq n\leq 511$. Every new term in that range equals 8, so $1538+8(n-255)=1994$, giving $n=312$. The sum strictly increases for $n\geq 2$, proving uniqueness.

**Answer:** 312.

**Methods:** floor sums, dyadic intervals, arithmetico-geometric sums.

**Verification:** $1538+8\cdot 57=1994$, with $n=255+57=312$. Independent exact check passed: Compute each logarithm floor independently by integer bit length.

**Source comparison:** The single alternative correctly groups integers between successive powers of two. Its endpoint counts and resulting linear equation are valid.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Binary magnitude intervals give a piecewise linear formula for a logarithm-floor sum.

## 1994 AIME Problem 5: A weighted sum over digit strings

For each positive integer n, multiply its nonzero decimal digits. Let S be the sum of these products for n=1,…,999. Find the largest prime divisor of S.

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### Solution

Use three-digit strings with leading zeros, assigning weight 1 to a zero and its usual value to every other digit. Independence of the three positions gives total weight $(1+1+2+\cdots+9)^3 = 46^3$ over 000 through 999. Remove the artificial all-zero string, of weight 1. Thus $S = 46^3 - 1 = 45 \cdot 2163 = 3^3 \cdot 5 \cdot 7 \cdot 103$, and the largest prime factor is 103.

**Answer:** 103.

**Methods:** weighted product rule, digits.

**Verification:** The derivation yields the answer in the independently hosted answer field; no separate exhaustive computation claimed.

**Source comparison:** Read and compared all 2 archived solution entries. The first archived solution uses the same central counting model; this is a newly written exposition after consultation.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Independent digit choices factor the total product-of-nonzero-digits sum.

## 1994 AIME Problem 7: Lines with only integral circle intersections

Count real pairs (a,b) for which ax+by=1 and $x^2+y^2=50$ have at least one common point and every common point has integral coordinates.

Published problem and solutions · MAA AMC.

### Solution

The circle's integral points are $(\pm1, \pm7), (\pm7, \pm1), (\pm5, \pm5)$, twelve in all. A qualifying line is either tangent at one of these twelve points or a secant through two of them. There are $\binom{12}{2}$ secants, but the six antipodal pairs give lines through the origin, which cannot have equation $ax + by = 1$. All other secants and all tangents have unique normalized equations of that form. Hence the answer is $66 - 6 + 12 = 72$.

**Answer:** 072.

**Methods:** geometric pairs, exceptional fibers.

**Verification:** The derivation yields the answer in the independently hosted answer field; no separate exhaustive computation claimed.

**Source comparison:** Read and compared all 1 archived solution entries. The finite lattice-point model determines all tangent and secant possibilities; antipodal pairs require a separate exclusion.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, geometry, number_theory. Integer points on a circle determine allowable secant and tangent lines.

## 1994 AIME Problem 10: A prime-cube projection in an integer right triangle

Triangle ABC is right at C and has integer side lengths. The altitude from C meets AB at D with BD=$29^3$. If cosB=m/n in lowest terms, find m+n.

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### Solution

Similarity gives $BC^2$=AB · BD=$29^3$AB. Integer prime valuations imply $29^2$ divides BC, so write BC=$29^2$x and AB=$29x^2$ for a positive integer x. Pythagoras yields $AC^2$=$(29x)^2(x^2-29^2)$. Since an integer square divisible by $(29x)^2$ has its root divisible by 29x, put AC=29 xy with y a positive integer. Then (x−y)(x+y)=$29^2$. Both factors are positive; their only nondegenerate pair is 1,841, since 29,29 would give y=0. Thus x=421, and cosB=BC/AB=29/421. The requested sum is 450.

**Answer:** 450.

**Methods:** integer right triangles, prime valuations, difference of squares, projection theorem.

**Verification:** The resulting sides are $29 \cdot 421 \cdot 420$, $29^2 \cdot 421$, and $29 \cdot 421^2$; the projection $BC^2/AB$ equals $29^3$.

**Source comparison:** All three alternatives read. Index 0 supplies the valid integer-side reduction. Index 1 treats the altitude as an integer without proof and invokes the answer range to discard factor pairs. Index 2 assumes an integral parametrization prematurely. The divisibility argument justifies every integer variable.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory. A right-triangle projection and integer side lengths lead to a prime-square factorization.

## 1994 AIME Problem 11: Distinct heights of a brick tower

Build a tower of 94 bricks, each contributing height 4,10 or 19. Count distinct possible total heights.

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### Solution

Subtract the base height $94 \cdot 4$ and divide by 3. The offset is $t = 2a + 5b$, where $a, b \geq 0$ and $a + b \leq 94$. Whenever $a \geq 5$, replace five height-2 contributions by two height-5 contributions and three zero contributions; this preserves both height and the 94-brick total. Repetition gives a canonical description with $0 \leq a \leq 4$. Such descriptions are unique: t modulo 5 determines a, then t determines b. For each a, the choices are $b = 0, \ldots, 94 - a$, giving $95 - a$ heights. Hence the total is $95 + 94 + 93 + 92 + 91 = 465$.

**Answer:** 465.

**Methods:** sumset, bounded occupancy.

**Verification:** Independent exact check passed: Enumerated all feasible orientation-count pairs and deduplicated heights. Result 465. The written proof supplies the general/exhaustive justification.

**Source comparison:** Read and compared all 2 archived solution entries. Alternative index 1 is the shortest: reduce the count of height-2 offset bricks modulo five, obtaining unique canonical descriptions.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Distinct tower heights are the image of bounded integer combinations of three brick heights.

### 1994 AIME Problem 12: The finest square grid within a fence budget

A fenced 24 by 52 meter rectangle is partitioned completely into congruent axis-parallel square plots using at most 1994 meters of additional internal fence. Find the largest possible number of plots.

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#### Solution

If the square side is s, the integer grid dimensions u=24/s and v=52/s satisfy 13u=6v. Thus u=6t,v=13t for a positive integer t. Internal fence length is 52(u−1)+24(v−1)=624t−76. The budget gives 624t≤2070, so t≤3. The maximum occurs at t=3, producing uv=78$t^2$=702 plots and requiring 1796 meters of fence.

**Answer:** 702.

**Methods:** rectangular grids, integer ratios, resource optimization.

**Verification:** The maximizing grid is 18 by 39, with side 4/3. Its fence length is $17 \cdot 52 + 38 \cdot 24 = 1796$ meters, below the budget of 1994. The next value $t = 4$ requires 2420 meters.

**Source comparison:** Both alternatives read. Index 0 correctly counts internal grid lines. Index 1 uses plus signs where boundary exclusions require minus signs and gives a dimensionally incorrect fence formula; it happens to find the right grid size.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory. Integral grid dimensions and a fence-length budget bound the square subdivision scale.

### 1995 AIME Problem 2: A product of roots without solving them

Find the last three decimal digits of the product of all positive roots of $\sqrt{1995} \cdot x^{(\log_{1995} x)} = x^2$.

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**Solution**

Set u=$\log_{1995}$x, a bijection from positive x to real u. The equation becomes $u^2+1/2=2u$. Its two distinct real roots have sum 2, so the corresponding x-values have product 1995^($u_1+u_2$)=$1995^2$. Modulo 1000 this is $(-5)^2=25$, giving final digits 025.

**Answer:** 025.

**Methods:** logarithmic substitution, Vieta formulas, root products, modular arithmetic.

**Verification:** The transformed discriminant is positive, and the logarithmic substitution preserves exactly all positive solutions.

**Source comparison:** All three alternatives read. Index 1 takes logarithms to base x without first excluding x=1, which indeed is not a solution. Index 2 makes an arithmetic error when simplifying the quadratic roots; their sum remains correct. Vieta avoids unnecessary radicals.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. A logarithmic substitution makes the positive roots exponential images of a quadratic's roots.

## 1995 AIME Problem 6: Divisors below a square root

Let n=2^31 · 3^19. Count divisors of $n^2$ that are smaller than n but do not divide n.

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**Solution**

The $63 \cdot 39$ divisors of $n^2$ pair under $d \mapsto n^2/d$, with unique fixed point n. Hence $(63 \cdot 39 - 1)/2 = 1228$ lie below n. Every proper divisor of n lies among these; there are $32 \cdot 20 - 1 = 639$. Subtraction gives $1228 - 639 = 589$.

**Answer:** 589.

**Methods:** divisor involution, complement.

**Verification:** The derivation yields the answer in the independently hosted answer field; no separate exhaustive computation claimed.

**Source comparison:** Read and compared all 3 archived solution entries. The general two-prime simplification is that the requested count equals the product of the two exponents.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Divisor pairing around the square root counts divisors outside a smaller exponent box.

## 1995 AIME Problem 8: Two simultaneous integer ratios

Count positive integer pairs $y < x \leq 100$ for which $x/y$ and $(x+1)/(y+1)$ are integers.

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**Solution**

The two conditions are equivalent to both $y$ and $y+1$ dividing $x-y$. Since they are coprime, $x-y = ty(y+1)$ for an integer $t \geq 1$. Conversely this expression satisfies both conditions. For a given $y$ the count is $\lfloor (100-y)/(y(y+1)) \rfloor$. Only $y = 1, \ldots, 9$ contribute, with counts $49, 16, 8, 4, 3, 2, 1, 1, 1$. Their sum is 85.

**Answer:** 085.

**Methods:** parametric counting, divisibility.

**Verification:** Independent exact check passed: Checked every bounded integer pair directly. Result 85. The written proof supplies the general/exhaustive justification.

**Source comparison:** Read and compared all 3 archived solution entries. Alternatives derive the same y(y+1) divisibility parametrization or enumerate its small y-cases.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Two divisibility conditions parameterize and count bounded positive integer pairs.

## 1995 AIME Problem 10: A composite term in five consecutive residue steps

Find the largest positive integer that cannot be expressed as a positive multiple of 42 plus a positive composite integer.

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**Solution**

For any integer N>215, consider N−42,N−84,N−126,N−168,N−210. Because 42 is relatively prime to 5, these five integers occupy all residue classes modulo 5. One is divisible by 5, and every one exceeds 5, so that term is composite. Thus every N>215 has the required representation. For N=215, the positive remainders after subtracting positive multiples of 42 are 173,131,89,47,5, all prime. Therefore 215 is impossible and is the largest exception.

**Answer:** 215.

**Methods:** residue covering, composite numbers, extremal construction.

**Verification:** Trial division by primes up to the respective square roots confirms that 173,131,89,47,5 are prime; larger subtracted multiples leave nonpositive remainders. Independent exact check passed: Test every candidate through1000 directly for a positive composite remainder.

**Source comparison:** Both alternatives read. They discuss the noncomposite remainders as primes and overlook the possible remainder 1 in general. The five-term residue argument avoids that issue and supplies a uniform upper bound instead of a case table.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A short arithmetic progression contains a composite multiple of five, bounding the exceptional integers.

## 1995 AIME Problem 11: A self-similar cut prism

An integer-sided rectangular prism has sorted sides $a \le b \le c$ with $b = 1995$. A face-parallel cut produces a smaller similar prism. Count possible triples $(a, b, c)$.

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### Solution

The similar piece has scale factor q<1 and sorted sides qa,qb,qc, two of which are unchanged original side lengths. The unchanged lengths cannot include c, since qc<c; they therefore are a and b, forcing qb=a and qc=b. Consequently $ac = b^2$ and $a < b < c$. Conversely any such factor pair works: cut the c-side to length $qa = a^2/b$, which is positive and less than c; the remaining two dimensions are a,b, as required. Now $1995 = 3 \cdot 5 \cdot 7 \cdot 19$, so $b^2$ has $3^4 = 81$ divisors. Excluding the central pair a=c=b and pairing reciprocal divisors gives $(81 - 1)/2 = 40$ triples.

**Answer:** 040.

**Methods:** factor pairing, hybrid geometry.

**Verification:** The derivation yields the answer in the independently hosted answer field; no separate exhaustive computation claimed.

**Source comparison:** Read and compared all 1 archived solution entries. The first archived solution uses the same central counting model; this is a newly written exposition after consultation.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, geometry, number_theory. Similarity under a planar cut constrains integer prism dimensions and their factor choices.

## 1995 AIME Problem 13: Grouping a sum by rounded fourth roots

Let f(n) be the integer nearest to the positive fourth root of n. Find $\Sigma_{k=1}{}^{1995} 1/f(k)$.

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### Solution

For a positive integer j, f(n)=j exactly when $(j-1/2)^4<n<(j+1/2)^4$. The endpoints are never integers; both have fractional part 1/16, so the number of integer n in a complete interval is their difference, $4j^3+j$. For j=1,…,6 these intervals contain $4(1^3+\cdots+6^3)+(1+\cdots+6)=4 \cdot 441+21=1785$ integers and contribute $\Sigma(4j^2+1)=4 \cdot 91+6=370$. The remaining 210 integers through 1995 have f(n)=7, contributing 210/7=30. The total is 400.

**Answer:** 400.

**Methods:** rounding intervals, floor endpoints, power sums.

**Verification:** $6.5^4=1785+1/16$ and $7.5^4=3164+1/16$, so exactly the integers 1786 through 1995 form the final partial group. Independent exact check passed: Find each nearest fourth root by exact integer midpoint comparisons, then sum individually.

**Source comparison:** Both alternatives read. Index 0 counts integers by flooring an interval length, which is not valid in general; it works here because both endpoints have the same fractional part 1/16. That endpoint fact is made explicit. Index 1 gives only an abbreviated numerical grouping.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Rounding fourth roots groups a reciprocal sum into polynomially sized intervals.

## 1996 AIME Problem 2: Even logarithm floors

Count positive integers $n < 1000$ for which $\lfloor \log_2 n \rfloor$ is positive and even.

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### Solution

The allowed floor values are 2,4,6,8. Value 2j corresponds exactly to $2^{2j} \le n < 2^{2j+1}$, a block of $2^{2j}$ integers. All four blocks lie below 1000, and the next begins at 1024. The count is $4+16+64+256 = 340$.

**Answer:** 340.

**Methods:** interval partition, hybrid algebra.

**Verification:** The derivation yields the answer in the independently hosted answer field; no separate exhaustive computation claimed.

**Source comparison:** Read and compared all 1 archived solution entries. The first archived solution uses the same central counting model; this is a newly written exposition after consultation.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Dyadic intervals count integers whose logarithm floor is a positive even value.

## 1996 AIME Problem 8: Integer pairs of prescribed harmonic mean

Count positive integer pairs x<y whose harmonic mean is 6^20.

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### Solution

Put $H = 6^{20}/2 = 2^{19}3^{20}$. The equation is $xy = H(x+y)$, so $(x-H)(y-H) = H^2$. Both factors are positive, because $1/x + 1/y = 1/H$ forces x,y>H. Thus strict pairs x<y correspond bijectively to factor pairs of $H^2 = 2^{38}3^{40}$ with unequal factors. There are $39 \cdot 41 = 1599$ divisors and one central equal pair, so the answer is $(1599-1)/2 = 799$.

**Answer:** 799.

**Methods:** factor-pair bijection, hybrid number theory.

**Verification:** The derivation yields the answer in the independently hosted answer field; no separate exhaustive computation claimed.

**Source comparison:** Read and compared all 1 archived solution entries. The first archived solution uses the same central counting model; this is a newly written exposition after consultation.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. A shifted factorization counts harmonic-mean solutions through paired divisors.

### 1996 AIME Problem 10: An integer angle from a tangent identity

Find the smallest positive integer x satisfying tan(19x°)=(cos 96°+sin 96°)/(cos 96°−sin 96°).

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#### Solution

Dividing numerator and denominator by cos 96° and using tangent addition transforms the right side into tan(96°+45°)=tan 141°. Therefore 19x≡141 modulo 180. Since $19^2$=361≡1 modulo 180, multiplication by 19 gives x≡2679≡159 modulo 180. The least positive solution is 159.

**Answer:** 159.

**Methods:** tangent addition, modular inverses, integer angles.

**Verification:** $19 \cdot 159$=3021=141+$16 \cdot 180$, and none of these angles is a tangent pole.

**Source comparison:** All three alternatives read. The first two correctly reduce to a congruence. Index 2 changes subtraction to addition in a cosine-addition line; the intended identity is clear but the displayed signs are wrong.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. A tangent addition identity becomes a linear congruence in an integer angle.

### 1996 AIME Problem 14: Cubes crossed by a space diagonal

A box of side lengths 150,324,375 is partitioned into unit cubes. Count cubes whose interiors meet a long diagonal.

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#### Solution

Parametrize the diagonal by $(150t, 324t, 375t)$, 0<t<1. A new cube begins at each distinct interior crossing of a coordinate-integer plane. The three crossing sets have sizes 149,323,374; pairwise overlaps have sizes $\gcd(150, 324) - 1 = 5$, $\gcd(324, 375) - 1 = 2$, $\gcd(150, 375) - 1 = 74$, and the triple overlap has size $\gcd(150, 324, 375) - 1 = 2$. Inclusion-exclusion gives 767 crossing times. They divide the diagonal into 768 open intervals, each lying in exactly one cube interior. The answer is 768.

**Answer:** 768.

**Methods:** inclusion-exclusion, lattice geometry.

**Verification:** Independent exact check passed: Built the exact union of all interior rational crossing times. Result 768. The written proof supplies the general/exhaustive justification.

**Source comparison:** Read and compared all 2 archived solution entries. Both sources use inclusion-exclusion of simultaneous coordinate-plane crossing times.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, geometry, number_theory. Grid-plane crossings along a space diagonal are counted with gcd-based inclusion-exclusion.

## 1997 AIME Problem 1: Differences of squares

Count integers 1 through 1000 that are differences of squares of nonnegative integers.

Published problem and solutions · MAA AMC.

### Solution

Every odd integer $2a+1$ equals $(a+1)^2-a^2$, and every positive multiple $4b$ equals $(b+1)^2-(b-1)^2$. Conversely squares are 0 or 1 modulo 4, so a difference cannot be 2 modulo 4. The qualifying integers are exactly the 500 odds and 250 multiples of 4, totaling 750.

**Answer:** 750.

**Methods:** residue classification, constructive counting.

**Verification:** The derivation yields the answer in the independently hosted answer field; no separate exhaustive computation claimed.

**Source comparison:** Read and compared all 1 archived solution entries. The first archived solution uses the same central counting model; this is a newly written exposition after consultation.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Parity classes characterize and count integers representable as differences of squares.

## 1997 AIME Problem 3: Concatenation mistaken for multiplication

A two-digit integer x written immediately before a three-digit integer y forms a five-digit number equal to nine times xy. Find x+y.

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### Solution

The concatenation condition is $1000x+y=9\,xy$, so $y=1000/(9-1/x)$. Because $10\le x\le 99$, we have $111<1000/9<y\le 10000/89<113$. Thus the integer y must be 112. Substitution gives $1000x+112=1008x$, hence $x=14$ and $x+y=126$.

**Answer:** 126.

**Methods:** digit concatenation, rational bounds, integer uniqueness.

**Verification:** The concatenation 14112 equals $9\cdot 14\cdot 112$. Independent exact check passed: Exhaust all two-digit and three-digit factor pairs.

**Source comparison:** All three alternatives read. Index 0 finds a successful divisor but does not exhaust the possible divisors; index 1 says the ratio is "a little" below nine without explicit bounds. The two-digit range forces y uniquely by a narrow interval.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. A concatenation equation and the two-digit bound force the three-digit factor.

## 1997 AIME Problem 5: Decimal numbers closest to two sevenths

Count four-place decimals 0.abcd for which 2/7 is the closest fraction whose numerator is 1 or 2 and whose denominator is an integer.

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### Solution

The neighboring distinct admissible fractions around 2/7 are $1/4 = 2/8$ and $1/3 = 2/6$. All fractions with numerator 1 also occur with numerator 2 and an even denominator, so there are no omitted intervening candidates. The nearest-fraction cell is between the midpoints 15/56 and 13/42. Multiplying by 10000 gives bounds 2678.57 ... and 3095.23 ..., neither an integer. Thus the allowed integers abcd are 2679 through 3095, numbering 417.

**Answer:** 417.

**Methods:** interval lattice count, nearest neighbor.

**Verification:** The derivation yields the answer in the independently hosted answer field; no separate exhaustive computation claimed.

**Source comparison:** Read and compared all 1 archived solution entries. The first archived solution uses the same central counting model; this is a newly written exposition after consultation.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Nearest allowed rational neighbors determine a decimal interval whose grid points are counted.

## 1997 AIME Problem 6: An equilateral attachment and another regular polygon

An equilateral triangle $A_1A_2B$ is attached externally to a regular n-gon $A_1A_2\ldots A_n$. Find the largest n for which $A_1$,$A_n$,B can be three consecutive vertices of a regular polygon.

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### Solution

To bound the maximum it suffices to consider $n>6$. The equal segments $A_1A_n$ and $A_1B$ then enclose angle $120°+360°/n$, so $A_1$ must be the middle of the three consecutive vertices. If the new polygon has m sides, its interior angle gives $180°-360°/m=120°+360°/n$. Rearranging yields $(m-6)(n-6)=36$, with $m,n>6$. Thus $n-6\le 36$ and $n\le 42$. Equality occurs for $m=7,n=42$: the two equal segments enclose precisely a regular heptagon's interior angle, so they extend to its consecutive sides.

**Answer:** 042.

**Methods:** regular polygon angles, integer factorization, geometric attainability.

**Verification:** For n=42,120°+360°/42=900°/7=180°−360°/7. Independent exact check passed: Enumerate integer polygon angle compatibilities over the proven relevant range.

**Source comparison:** All three alternatives correctly solve the angle equation. The new proof explicitly treats the large-n regime relevant to maximality and explains why the shared vertex is the middle vertex of the second polygon.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory. Regular-polygon angle compatibility reduces to an integer factorization.

## 1998 AIME Problem 1: Counting possible LCM partners

How many positive integers k satisfy lcm(6^6,8^8,k)=12^12?

Published problem and solutions · MAA AMC.

### Solution

The existing least common multiple is $2^{24}3^6$ and the target is $2^{24}3^{12}$. Thus k has no other prime factors, its 3-exponent must be 12, and its 2-exponent can be any integer from 0 to 24. There are 25 choices.

**Answer:** 025.

**Methods:** prime exponent product.

**Verification:** The derivation yields the answer in the independently hosted answer field; no separate exhaustive computation claimed.

**Source comparison:** Read and compared all 2 archived solution entries. The first archived solution uses the same central counting model; this is a newly written exposition after consultation.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Prime-exponent constraints count the integers that complete a prescribed least common multiple.

## 1998 AIME Problem 5: A four-term cycle of signed triangular numbers

For positive k let $A_k$=k(k−1)cos(k(k−1)π/2)/2. Find $|A_{19}+A_{20}+\cdots+A_{98}|$.

Published problem and solutions · MAA AMC.

### Solution

The cosine is (−1)^(k(k−1)/2), so its signs repeat +,+,−,− for k≡0,1,2,3 modulo 4. In each block beginning at k=4j+3, the four signed triangular numbers are $-T_{k-1}$,$+T_k$,$+T_{k+1}$,$-T_{k+2}$, where $T_r$=r(r+1)/2. Their sum is k−(k+2)=−2. The eighty terms from 19 through 98 form twenty such blocks, giving sum −40 and absolute value 40.

**Answer:** 040.

**Methods:** triangular-number parity, periodic signs, telescoping.

**Verification:** The first block is $-171+190+210-231=-2$, and the same difference identity applies to all twenty blocks.

**Source comparison:** The sole alternative correctly pairs consecutive triangular differences and then pairs those results. Its displayed absolute-value expression temporarily omits outer bars, but the final sign treatment is correct.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Triangular-number parity gives a four-term sign cycle and telescoping cancellation.

## 1998 AIME Problem 8: The longest subtractive recurrence

A sequence begins 1000,x and each later term is the term two places earlier minus the preceding term. Stop when the first negative term appears, including it. Which positive integer x makes the sequence longest?

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### Solution

Repeated substitution gives the tenth and eleventh terms as $34x-21000$ and $34000-55x$. Any sequence reaching a twelfth term must have both nonnegative, so $21000/34 \le x \le 34000/55$. The only integer in this interval is 618. For $x=618$ the sequence is 1000,618,382,236,146,90,56,34,22,12,10,2,8,−6, of length 14. Every other positive integer stops by its eleventh term, so 618 is the unique maximizer.

**Answer:** 618.

**Methods:** subtractive recurrences, Fibonacci coefficients, integer intervals, maximal length.

**Verification:** Direct recurrence evaluation gives fourteen terms for 618; the necessary interval for surviving eleven nonnegative terms contains no other integer. Independent exact check passed: Generate every subtractive sequence for$1 \le x \le 1000$; $x>1000$ stops immediately at term3.

**Source comparison:** All three entries read. Entry 0 is an introductory fragment with an incorrect general formula. Entry 1 uses strict inequalities although zero is allowed and misquotes one Fibonacci ratio. Entry 2 relies only on a limiting ratio, which does not prove the finite optimum. Two exact term inequalities plus the full candidate sequence settle maximality.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Consecutive nonnegativity constraints in a Fibonacci-coefficient recurrence isolate the maximizing integer.

## 1998 AIME Problem 14: A sharp bound for a box volume ratio

Positive integers $m \le n \le p$ satisfy $2\,mnp=(m+2)(n+2)(p+2)$. Find the largest possible p.

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**Solution**

The equation implies $2=(1+2/m)(1+2/n)(1+2/p)>1+2/m$, so $m\geq 3$. Put $a=m-2, b=n-2$, positive integers. Solving for p gives $p=2(a+4)(b+4)/(ab-8)$, with $ab-8$ a positive integer. Since $(a-1)(b-1)\geq 0$, we have $a+b\leq ab+1$. Therefore $p=(2\,ab+8(a+b)+32)/(ab-8)\leq 10+120/(ab-8)\leq 130$. Equality is achieved by $a=1, b=9$, giving $(m,n,p)=(3,11,130)$. Thus the maximum is 130.

**Answer:** 130.

**Methods:** integer optimization, volume ratios, product-sum bound, attainment.

**Verification:** $2\cdot 3\cdot 11\cdot 130=8580=(3+2)(11+2)(130+2)$, and the dimensions are in the required order. Independent exact check passed: Enumerate ordered integral dimension pairs within the analytic bound and solve exactly for the third.

**Source comparison:** Both alternatives read. Index 0 eventually supplies the necessary global inequality after an exploratory denominator guess. Index 1 fixes $m=3$ without proving global optimality. The positive integer denominator and sharp product-sum bound establish the maximum uniformly.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. An integer volume ratio is bounded by a factored denominator and a sharp product-sum inequality.

## 1999 AIME Problem 1: Five primes in arithmetic progression

Find the smallest prime that can be the fifth term of a strictly increasing arithmetic progression whose first four terms are also prime.

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**Solution**

Write the progression $p, p+d, \ldots, p+4d$. If d were odd, at least two terms would be even and one would exceed 2, impossible. Thus d is even, and p is odd. If 3 did not divide d, one of the first three terms would be divisible by 3. If it were either of the first two, the term three places later would be a larger multiple of 3; if it were the third, it would exceed 3. Hence 3 divides d, so $d\geq 6$. Also $p\neq 3$, since then every term would be divisible by 3; thus $p\geq 5$. The fifth term is at least $5+4\cdot 6=29$, attained by 5,11,17,23,29.

**Answer:** 029.

**Methods:** prime progressions, small-prime obstructions, minimality.

**Verification:** All five terms of 5,11,17,23,29 are prime, and the common difference is 6. Independent exact check passed: Exhaust all progressions that could improve the proposed fifth prime.

**Source comparison:** Both alternatives read. Their claims about parity and divisibility are mostly asserted rather than proved; index 1's digit commentary is especially imprecise. The residue argument establishes the lower bound and verifies an attaining progression despite the primary answer field being blank.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Small-prime obstructions bound the common difference of a five-term prime progression.

### 1999 AIME Problem 3: A quadratic square condition with four solutions

Find the sum of all positive integers n for which $n^2-19n+99$ is a perfect square.

Published problem and solutions · MAA AMC.

#### Solution

Write $n^2-19n+99=k^2$ with $k\geq 0$. Completing the square gives $(2k-(2n-19))(2k+(2n-19))=35$. Since $4k^2=(2n-19)^2+35$, both factors are positive. The ordered factor pairs are (1,35),(5,7),(7,5),(35,1), giving n=18,10,9,1 respectively. All four are positive and satisfy the equation, so their sum is 38.

**Answer:** 038.

**Methods:** perfect-square conditions, difference of squares, factor pairs.

**Verification:** The corresponding squares are 81,9,9,81 for n=1,9,10,18. Independent exact check passed: Direct square test through100; factorization proves no later solutions.

**Source comparison:** All three alternatives are valid. The factorization method exhausts the possibilities without checking an interval of individual integers.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Completing the square converts a quadratic perfect-square condition into factor pairs of 35.

### 1999 AIME Problem 5: Possible digit-sum changes

Let S(x) be the decimal digit sum and $T(x)=|S(x+2)-S(x)|$ for positive integers x. Count distinct T-values at most 1999.

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#### Solution

In adding 2, each carry reduces the digit-sum change by 9. Thus $S(x+2)-S(x) = 2-9k$ for some $k \geq 0$. Every $k$ occurs: $k = 0$ at $x = 1$, and $k \geq 1$ at $x = 10^k - 2$. Therefore possible absolute changes are 2 and $9k-2$ for $k \geq 1$, with no repetitions. The latter are at most 1999 exactly when $k \leq \lfloor 2001/9 \rfloor = 222$. Total: 223.

**Answer:** 223.

**Methods:** carry invariant, distinct image.

**Verification:** The derivation yields the answer in the independently hosted answer field; no separate exhaustive computation claimed.

**Source comparison:** Read and compared all 1 archived solution entries. The first archived solution uses the same central counting model; this is a newly written exposition after consultation.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Decimal carry chains characterize and count the possible digit-sum differences.

### 1999 AIME Problem 7: Four-position divisor switches

Switches labeled $2^x3^y5^z$ for $0 \le x, y, z \le 9$ cycle $A \to B \to C \to D \to A$. Processing every label once advances all switches whose labels divide it. Starting at A, how many finish at A?

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#### Solution

A switch with exponents x,y,z advances once for each multiple label, namely $(10-x)(10-y)(10-z)$ times. It returns to A iff this product is divisible by 4. Each factor ranges independently over 1,…,10. The complementary products have either no factor of 2, giving $5^3 = 125$ triples, or exactly one factor of 2, giving $3 \cdot 3 \cdot 5^2 = 225$ (choose its location and one of 2,6,10). Hence $1000 - 125 - 225 = 650$ switches return to A.

**Answer:** 650.

**Methods:** valuation complement, divisor incidence.

**Verification:** Independent exact check passed: Checked the advance count for all 1000 switch labels. Result 650. The written proof supplies the general/exhaustive justification.

**Source comparison:** Read and compared all 1 archived solution entries. The first archived solution uses the same central counting model; this is a newly written exposition after consultation.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Divisor multiplicities in an exponent box determine switch states modulo four.

### 1999 AIME Problem 13: Distinct tournament win totals

Forty teams play each pair once with equally likely independent winners. If the probability that all win totals differ is $m/n$ in lowest terms, find $\log_2 n$.

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#### Solution

The 40 distinct win totals must be exactly 0,…,39. A team with 39 wins beats all others; removing it leaves distinct totals 0,…,38. Induction shows that a ranking uniquely determines every result, so favorable outcomes number 40!. There are $2^{\binom{40}{2}} = 2^{780}$ equally likely tournaments. The power of 2 in 40! is $20 + 10 + 5 + 2 + 1 = 38$. After cancellation the denominator is $2^{780-38} = 2^{742}$, giving 742.

**Answer:** 742.

**Methods:** transitive tournament, valuation.

**Verification:** The derivation yields the answer in the independently hosted answer field; no separate exhaustive computation claimed.

**Source comparison:** Read and compared all 1 archived solution entries. The first archived solution uses the same central counting model; this is a newly written exposition after consultation.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Distinct tournament scores force a total order, and factorial valuations reduce the probability.

## 2000 AIME I Problem 1: Zero-free factors of powers of ten

Find the smallest positive integer $n$ such that, whenever positive integers $a, b$ satisfy $ab = 10^n$, at least one of $a, b$ has a zero in its decimal expansion.

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### Solution

If a factor contains both 2 and 5, it ends in zero. Otherwise the factors must be $2^n$ and $5^n$. For $1 \le n \le 7$, both powers are zero-free: the powers of 2 are $2, 4, 8, 16, 32, 64, 128$, and those of 5 are $5, 25, 125, 625, 3125, 15625, 78125$. For $n = 8$, however, $5^8 = 390625$ has a zero. Thus the minimum is 8.

**Answer:** 008.

**Methods:** prime factorization, decimal representation.

**Verification:** Exhaustively checked every divisor factorization for powers 10^1 through 10^8.

**Source comparison:** The sole source uses the same prime-allocation reduction; its finite digit check is valid.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Allocating prime powers to two factors reduces the decimal-digit condition to powers of 2 and 5.

## 2000 AIME I Problem 2: A reflected pentagon

Let $A = (u, v)$, where $u, v$ are integers with $0 < v < u$. Obtain $B, C, D, E$ successively by reflecting $A$ across $y = x$, $B$ across the $y$-axis, $C$ across the $x$-axis, and $D$ across the $y$-axis. If pentagon $ABCDE$ has area 451, find $u + v$.

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### Solution

The points are $B = (v, u), C = (-v, u), D = (-v, -u), E = (v, -u)$. Rectangle $BCDE$ has area $4uv$; triangle $ABE$ has area $u(u - v)$. Hence $u(u + 3v) = 451 = 11 \cdot 41$. Since $u < u + 3v$, the possible factor pairs are $(1, 451)$ and $(11, 41)$. They give $v = 150$ and $v = 10$, respectively. Only $(u, v) = (11, 10)$ satisfies $v < u$, so the answer is 21.

**Answer:** 021.

**Methods:** coordinate area, reflections, factorization.

**Verification:** Exhaustive bounded integer search of the reflected-polygon area equation.

**Source comparison:** Both downloaded solutions were checked: area decomposition and the shoelace formula yield the same factor equation.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory. Coordinate reflections give an area polynomial whose integral solutions are controlled by factorization.

## 2000 AIME I Problem 5: Two urns with a fixed total

Two urns each contain both black and white balls, with 25 balls in total. Independent uniform draws, one per urn, are both black with probability 27/50. Write the probability of two whites as reduced $m/n$ and find $m+n$.

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### Solution

Let the urn sizes be $a \geq b$. Since $50 \mid ab$ and $a+b=25$, both are multiples of 5, so $(a,b)=(20,5)$ or $(15,10)$. The black counts must have products 54 or 81, respectively. The bounds imposed by a white ball in each urn force black counts $(18,3)$ or $(9,9)$. In either case the white probability is $1/25$, so the answer is 26.

**Answer:** 026.

**Methods:** probability, divisibility.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer. Audit findings: Source 1: Arithmetic slip: the largest feasible denominator is 156, not 154. The candidate multiple 150 and final answer remain valid. Source 2: The exhibited urn sizes establish existence but need an exclusion argument for uniqueness.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Independent draws from two urns give a probability equation whose integer population sizes are constrained by factorization.

## 2000 AIME I Problem 6: A gap between means

How many integer pairs $(x,y)$ satisfy $0<x<y<10^6$ and $\frac{x+y}{2}-\sqrt{xy}=2$?

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### Solution

The equation is $(\sqrt{y}-\sqrt{x})^2=4$, hence $\sqrt{y}=\sqrt{x}+2$. Squaring gives $4\sqrt{x}=y-x-4$, so $\sqrt{x}$ is rational. A rational square root of an integer is an integer: in a reduced fraction its denominator must divide its numerator after squaring. Write $x=a^2, y=(a+2)^2$. The bounds are precisely $1 \leq a \leq 997$, giving 997 pairs.

**Answer:** 997.

**Methods:** radicals, integer squares.

**Verification:** Exhaustive check over the full x range of the derived square condition; exhaustion relies on the analytic reduction.

**Source comparison:** All five entries were reviewed. Several omit the rational-square-root justification. Entry 3 contains a mistaken derived square, and entry 4 has inconsistent factors in its squaring steps. The integer-square parametrization above supplies the missing implication.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. The mean equation forces two integer square roots differing by 2.

## 2000 AIME I Problem 11: Independent prime exponents

Sum $a/b$ over ordered pairs of coprime positive divisors of 1000. If the sum is $S$, find $\lfloor S/10 \rfloor$.

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### Solution

For each prime $p \in \{2, 5\}$, coprimality allows its exponent on only one side. The exponent of $p$ in the ratio therefore ranges uniquely over $-3, -2, \ldots, 3$. Independence gives $S = (\sum_{j=-3}^{3} 2^j)(\sum_{j=-3}^{3} 5^j) = \frac{127}{8}\frac{19531}{125} = 2480.437$. Thus the answer is 248.

**Answer:** 248.

**Methods:** weighted counting, product principle, prime exponents.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer. Audit findings: Source 3: The numerical approximation is unnecessary; the exact finite sum used here avoids rounding.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Coprime divisor pairs are counted by independently assigning each prime-power contribution.

## 2000 AIME II Problem 2: Lattice points on a difference of squares

Count all ordered integer pairs $(x, y)$ satisfying $x^2 - y^2 = 2000^2$.

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### Solution

The factors $x - y, x + y$ have equal parity and product $2000^2$, so both are even. Put $u = (x - y)/2$ and $v = (x + y)/2$. Then $uv = 10^6 = 2^6 5^6$, and conversely every integer factor pair gives exactly one point $(x, y) = (u + v, v - u)$. There are $7 \cdot 7 = 49$ choices of positive $u$ and another 49 negative choices, with $v$ determined. Hence there are 98 points.

**Answer:** 098.

**Methods:** difference of squares, divisor counting, bijections.

**Verification:** Brute-force integer-square test for every y in the analytically bounded range, counting both signs of x.

**Source comparison:** The sole source was checked. Its factor allocation is sound; the explicit inverse here clarifies why the factor of 2 accounts for signs without an additional factor for order.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, combinatorics. A parity-preserving factorization gives a bijection between lattice points and signed divisors.

### 2000 AIME II Problem 4: Prescribed odd and even divisor counts

Find the least positive integer having exactly six positive odd divisors and twelve positive even divisors.

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#### Solution

Write $N = 2^a m$ with $m$ odd. The odd and even divisor counts are $d(m)$ and $a\,d(m)$, so $a = 2$ and $d(m) = 6$. The possible prime-exponent patterns of $m$ are $(5)$ or $(2, 1)$. Their least values are $3^5 = 243$ and $3^2 \cdot 5 = 45$: the larger exponent belongs on the smaller prime to minimize the product. Thus $N = 4 \cdot 45 = 180$.

**Answer:** 180.

**Methods:** divisor function, minimal prime factorization.

**Verification:** Direct divisor enumeration for every positive integer through the claimed minimum.

**Source comparison:** All three source entries were reviewed. Entry 1 mistakenly states exponents 3 and 2 before giving the correct product with exponents 2 and 1. Entry 2 asserts the minimum without proving the two exponent possibilities.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. The odd part determines six divisors, while the 2-adic exponent determines the even-to-odd ratio.

### 2000 AIME II Problem 11: Integral equal legs restrict base directions

An isosceles trapezoid $ABCD$ has integer-coordinate vertices, $A = (20, 100)$, $D = (21, 107)$, no horizontal or vertical side, and only $AB \parallel CD$. The sum of the absolute values of all possible slopes of $AB$ is $m/n$ in lowest terms. Find $m + n$.

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#### Solution

Translate $A$ to the origin, and put $d = \overrightarrow{AD} = (1, 7)$, $e = \overrightarrow{BC}$. Since the legs have equal lengths, $e$ is an integer vector of squared length 50, hence is one of $(\pm 1, \pm 7), (\pm 7, \pm 1), (\pm 5, \pm 5)$. Parallel bases force

$\overrightarrow{AB}$ to be parallel to $d-e$. The four choices $(\pm 1, \pm 7)$ yield a parallelogram, degeneracy, or a horizontal or vertical base. The remaining eight give slopes $-1, -4/3, 3/4, 1, -1/2, -3, 1/3, 2$. Each is attainable: set $B = 2(d-e)$ and $C = B + e$; the resulting quadrilateral is convex, has unequal parallel bases and equal nonparallel legs. The absolute slopes sum to $119/12$, so the answer is 131.

**Answer:** 131.

**Methods:** integer coordinates, sum of two squares, vector geometry.

**Verification:** Enumerated all integer leg vectors of squared norm 50 and summed distinct admissible slopes.

**Source comparison:** Both downloaded entries were checked. The reflection formulation and direct leg-vector enumeration agree. The explicit construction here supplies sufficiency for every listed direction.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory. Equal integer-coordinate leg lengths restrict the second leg to representations of 50 as a sum of two squares.

## 2000 AIME II Problem 14: Borrowing in factorial base

For $N = 16! - 32! + 48! - \cdots - 1984! + 2000!$, let $N = \sum_{i=1}^{j} f_i i!$ be its factorial-base expansion, where $0 \le f_i \le i$ and $f_j > 0$. Find $\sum_{i=1}^{j} (-1)^{i+1} f_i$.

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### Solution

The identity $k\,k! = (k+1)! - k!$ implies $(b+1)! - a! = \sum_{k=a}^{b} k\,k!$. Group the original expression as $16! + \sum_{m=1}^{62} ((32m+16)! - (32m)!)$. Its factorial digits are therefore $f_{16} = 1$, $f_k = k$ on each interval $32m \le k \le 32m + 15$, and zero elsewhere. These intervals are disjoint and all digits are allowed, so no carrying occurs. Each interval contributes eight pairs $-2q + (2q+1) = 1$, while $f_{16}$ contributes $-1$. The result is $62 \cdot 8 - 1 = 495$.

**Answer:** 495.

**Methods:** mixed radix, telescoping, factorial digits.

**Verification:** Direct mixed-radix division of the full exact factorial integer, with reconstruction check.

**Source comparison:** All three entries were reviewed. The first two give correct telescoping and borrowing arguments. Entry 2 repeatedly misprints factorial indices and later writes a 32-term block where the nonzero block has 16 terms; its final digit pattern is correct.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. The telescoping identity k k!=(k+1)!-k! explicitly describes factorial-base borrowing.

## 2001 AIME I Problem 1: Two digits dividing their number

Find the sum of all positive two-digit integers divisible by both of their digits.

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**Solution**

Write the number as $10a + b$, with $1 \le a, b \le 9$; zero cannot be a divisor. Divisibility by $a$ gives $b = ka$, while divisibility by $b$ then gives $k \mid 10$. Since $b \le 9$, the possibilities are $k = 1, 2, 5$. These produce $11a$ for $1 \le a \le 9$, $12a$ for $1 \le a \le 4$, and 15 for $a = 1$. Their sum is $11 \cdot 45 + 12 \cdot 10 + 15 = 630$.

**Answer:** 630.

**Methods:** digits, divisibility.

**Verification:** Exhaustive check of all 90 two-digit integers against both divisibility conditions.

**Source comparison:** All four entries were reviewed. The quotient-divides-10 reduction proves completeness more directly than the three enumeration variants.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Divisibility by the tens digit forces the units digit to be its multiple, and the remaining quotient divides 10.

## 2001 AIME I Problem 8: Largest number with doubled reinterpreted digits

Call $N > 0$ admissible if reading its base-7 digits as a decimal numeral gives $2N$. Find the largest admissible integer.

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**Solution**

If its digits are $a_i \in \{0, \dots, 6\}$, the condition is $a_0 + 4a_1 = 2a_2 + \sum_{i \ge 3}(10^i - 2 \cdot 7^i)a_i$. The left side is at most 30, whereas every coefficient on the right for $i \ge 3$ is at least 314. Thus there are at most three digits and $4a_1 + a_0 = 2a_2$. To maximize $N$, take the greatest feasible leading digit, $a_2 = 6$; any $a_2 \le 5$ gives $N \le 293$. With $a_2 = 6$, maximizing the next digit gives $a_1 = 3, a_0 = 0$. Hence $N = (630)_7 = 315$, which indeed doubles to decimal 630.

**Answer:** 315.

**Methods:** base representations, digit bounds, optimization.

**Verification:** Exhaustive check of all N below 7^3; the analytic place-value argument supplies the global bound.

**Source comparison:** All three entries were reviewed. Entry 0 supplies the necessary digit-length bound. Entries 1 and 2 appeal to the AIME answer range instead, which is not a mathematical proof of the global maximum.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Comparing base-7 and base-10 place values bounds the digit length and reduces maximization to one linear digit equation.

## 2001 AIME I Problem 11: Two numberings of a rectangular array

In a $5 \times N$ array choose one point per row. Row-major labels $x_i$ and column-major labels $y_i$ obey $x_1 = y_2, x_2 = y_1, x_3 = y_4, x_4 = y_5, x_5 = y_3$. Find the smallest $N$.

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**Solution**

If the chosen columns are $c_i$, then $x_i = (i-1)N + c_i$ and $y_i = 5c_i - 5 + i$. The first two relations give $N = 24c_2 - 19$, $c_1 = 5c_2 - 3$. Eliminating $c_4, c_5$ from the last three gives $124c_3 = 117N + 51$. Reduction modulo 31 forces $c_2 \equiv 7 \pmod{31}$, so $N \geq 149$. At $N = 149$, the columns $(32, 7, 141, 88, 107)$ satisfy all five relations and lie in $[1, N]$. Thus 149 is attained.

**Answer:** 149.

**Methods:** permutations, congruences.

**Verification:** Analytic derivation checked against the archived answer; not independently machine-checked.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Two grid labelings impose a permutation of row indices and divisibility constraints on the number of columns.

## 2001 AIME II Problem 1: Digit adjacency as a directed graph

Find the leading three digits of the largest positive integer whose every adjacent two-digit block is a perfect square.

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**Solution**

Allowed edges between successive digits are $1 \to 6$, $2 \to 5$, $3 \to 6$, $4 \to 9$, $6 \to 4$, and $8 \to 1$. This directed graph has no cycles. Its longest directed path is $8 \to 1 \to 6 \to 4 \to 9$, producing the only five-digit candidate, 81649. Every other path gives fewer digits. The answer is 816.

**Answer:** 816.

**Methods:** directed graphs, extremal paths.

**Verification:** Analytic derivation checked against the archived answer; not independently machine-checked.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Allowed adjacent two-digit squares form a directed digit graph, whose longest valid word gives the largest integer.

## 2001 AIME II Problem 5: Triangle-free ordered sets

Every ten-element subset of $\{4, 5, \ldots, n\}$ must contain three side lengths of a nondegenerate triangle. Find the largest such $n$.

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### Solution

For increasing $b_1 < \cdots < b_{10}$ to avoid triangles, necessarily and sufficiently $b_{i+2} \geq b_i + b_{i+1}$. Sufficiency holds because the two largest entries below $b_{i+2}$ maximize their sum. Starting with the smallest possible pair $4, 5$ gives the coordinatewise least forbidden set: $4, 5, 9, 14, 23, 37, 60, 97, 157, 254$. Thus no forbidden ten-set fits below 254, while this one fits at 254. The answer is 253.

**Answer:** 253.

**Methods:** extremal principle, greedy bounds, recurrences.

**Verification:** Analytic derivation checked against the archived answer; not independently machine-checked.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Triangle-free increasing integer sequences obey a Fibonacci-type growth bound, yielding an extremal subset threshold.

## 2001 AIME II Problem 10: Counting exponent gaps

How many different numbers $10^j - 10^i$, with $0 \leq i < j \leq 99$, are divisible by 1001?

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### Solution

Since 1001 is coprime to 10, divisibility depends only on $d = j - i$. Modulo 1001, $10^3 \equiv -1$, while direct checking of $10, 10^2, 10^3$ shows the order is 6. Thus $d = 6, 12, \ldots, 96$. For each $d$ there are $100 - d$ pairs, totaling $\sum_{k=1}^{16}(100 - 6k) = 784$. Different pairs give different decimal strings (a block of nines followed by zeros), so no further quotient is needed.

**Answer:** 784.

**Methods:** residue classes, summation.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** The first archived proof has a divisibility typo in its necessity step; the order-of-10 argument here supplies the missing justification. Audit findings: Source 2: The period-six argument is sound, but one congruence and the index names are mistyped.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. The multiplicative order of 10 modulo 1001 constrains exponent gaps, which are then counted.

### 2001 AIME II Problem 14: Unit roots of a sparse polynomial

List the unit-modulus roots of $z^{28} - z^8 - 1 = 0$ by arguments $0 \le \theta_1 < \cdots < \theta_{2n} < 360^\circ$. Find $\theta_2 + \theta_4 + \cdots + \theta_{2n}$.

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#### Solution

Put $w = z^8$. Since $|w| = |w + 1| = 1$, subtracting squared moduli gives $\Re w = -1/2$. Thus either $(z^8, z^{28}) = (e^{120^\circ i}, e^{60^\circ i})$ or its conjugate. In the first case $\theta = 15^\circ + 45^\circ k$ for $0 \le k \le 7$; the second power condition becomes $60^\circ + 180^\circ k \equiv 60^\circ \pmod{360^\circ}$, so $k$ is even. These arguments are $15, 105, 195, 285$ degrees; conjugation adds $75, 165, 255, 345$ degrees. In increasing order the even-indexed arguments are the latter four, summing to 840.

**Answer:** 840.

**Methods:** unit complex roots, angle congruences.

**Verification:** Checked every 15-degree candidate in the original complex equation; the analytic congruence proof establishes exhaustiveness.

**Source comparison:** The sole source was reviewed. Its unit-circle geometry is correct; the explicit congruence calculation here proves completeness of the listed roots.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Equal unit moduli constrain two powers to equilateral-triangle vertices, followed by simultaneous angle congruences.

### 2002 AIME I Problem 3: Reversed two-digit ages

One person is currently 25, and another is older. In a positive integer number $n$ of years their ages are two-digit reversals. Count the possible ordered pairs consisting of the older person's present age and $n$.

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#### Solution

Write the future younger age as $10a + b$. The older age is $10b + a$, so $1 \le a < b \le 9$, and $10a + b > 25$. Conversely each such digit pair uniquely determines both requested integers. There are $\binom{9}{2} = 36$ pairs; exclude eight with $a = 1$ and three with $a = 2, b \le 5$. The result is 25.

**Answer:** 025.

**Methods:** bijections, digit counting.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Reversed two-digit ages impose a digit inequality, and admissible ages and waiting times are counted.

## 2002 AIME I Problem 4: A telescope followed by a prime-square factorization

For $a_k = 1/(k^2+k)$, positive integers $m < n$ satisfy $\sum_{k=m}^{n-1} a_k = 1/29$. Find $m+n$.

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### Solution

Since $1/(k(k+1)) = 1/k - 1/(k+1)$, the condition is $1/m - 1/n = 1/29$. In particular $m < 29$. Rearranging gives $(29-m)(29+n) = 29^2$. The positive factor $29-m$ lies between 1 and 28, so the only possibility is $29-m=1$. Thus $m=28, n=812$, and the sum is 840.

**Answer:** 840.

**Methods:** telescoping, Diophantine factorization.

**Verification:** Summed all 784 original rational terms exactly for the claimed endpoints.

**Source comparison:** Both entries were reviewed. The first needs an explicit coprimality step when inferring a divisor of 29; the shifted-product proof makes the uniqueness immediate.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. The reciprocal quadratic terms telescope; a shifted product then uses the primality of 29.

## 2002 AIME I Problem 7: Three fractional digits of an enormous power

Find the first three digits after the decimal point of $(10^{2002}+1)^{10/7}$.

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### Solution

Write the number as $10^{2860}(1+10^{-2002})^{10/7}$. Taylor expansion with second derivative bounded above by $30/49$ on positive arguments gives $10^{2860} + 10^{859}/7 + R$, where $0 < R < (15/49)10^{-1144}$. Since $10^6 \equiv 1 \pmod 7$ and $859 \equiv 1 \pmod 6$, the fractional part of the middle term is $3/7$. The positive remainder is smaller than $429/1000 - 3/7 = 3/7000$, so the number has fractional part strictly between $428/1000$ and $429/1000$. The requested digits are 428.

**Answer:** 428.

**Methods:** binomial approximation, remainder bounds, modular arithmetic.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** The sole source was reviewed, including its modular alternative. It correctly identifies the decisive term but only informally bounds the tail. The explicit remainder bound here proves that no carry changes the third digit.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. A rigorously bounded binomial tail reduces the fractional digits to a modular power divided by 7.

## 2002 AIME I Problem 8: The first repeated Fibonacci-type endpoint

Consider nondecreasing positive integer sequences satisfying $a_n = a_{n-1} + a_{n-2}$ for $n > 2$. Find the smallest value of $a_9$ attained by at least two such sequences.

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### Solution

Writing $a_1 = u, a_2 = v$ with $1 \le u \le v$, recurrence expansion gives $a_9 = 13u + 21v$. If two initial pairs give the same value and the first has smaller u, their difference must be $(21t, -13t)$ for an integer $t \ge 1$. Nondecreasing initial terms for the second pair require $v - 13t \ge u + 21t$, hence $v \ge u + 34t \ge 35$. Thus every collision has value at least $13 + 21 \cdot 35 = 748$. Equality is attained by initial pairs $(1, 35)$ and $(22, 22)$, so the answer is 748.

**Answer:** 748.

**Methods:** linear recurrences, Diophantine representations, minimal collision.

**Verification:** Enumerated all positive ordered initial pairs with ninth term at most 748; found the unique first collision.

**Source comparison:** The sole source was reviewed. Its primitive step and initial-term inequality are correct; including a general positive multiple t makes the lower bound cover all possible collisions.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. The endpoint is a linear form in the initial terms, and two representations differ by a primitive lattice vector.

## 2002 AIME I Problem 9: Three arithmetic progressions partition the integers

Three progressions start at $1, 2, 3$ and have positive integer steps $h, t, u$. They partition the positive integers exactly once. Sum $100h + 10t + u$ over all valid triples.

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### Solution

No step is 1, and $h \neq 2$. To cover 4, either $h = 3$ or $t = 2$. If $h = 3$, avoiding its residue class forces $3 \mid t$, $u$; covering $5, 6$ then forces $t = u = 3$. Otherwise $t = 2$ covers every even integer, so $h, u$ are even. We cannot have $u = 2$, since its odd progression would eventually meet the first. Covering 5 forces $h = 4$, and covering 7 forces $u = 4$. Both $(3, 3, 3)$ and $(4, 2, 4)$ work. Their encoded sum is 757.

**Answer:** 757.

**Methods:** covering systems, extremal first gap.

**Verification:** Analytic derivation checked against the archived answer; not independently machine-checked.

**Source comparison:** The first archived solution rejects larger h using the AIME answer range, which is not a proof. The first-uncovered-integer argument here closes that gap. Audit findings: Source 0: Excluding larger step sizes merely from the three-digit answer format is not a proof. The original solution here proves exhaustion by the first uncovered integer. Source 2: The assertion that the smallest step must divide the other steps needs additional justification beyond the stated congruence argument.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Disjoint arithmetic progressions covering all positive integers force reciprocal-density and congruence restrictions.

## 2002 AIME I Problem 11: Unfolding a reflected ray in a cube

A ray starts at a vertex of a cube of side 12. Its first reflection is on the opposite face at offsets 7 and 5 from the two adjacent edges through the corresponding vertex. It reflects specularly until reaching another cube vertex. If the path length is $m\sqrt{n}$ with square-free $n$, find $m+n$.

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### Solution

Unfold successive reflected cubes across their faces. The ray becomes the straight line $t(12,7,5)$, and cube vertices become points whose three coordinates are multiples of 12. The first coordinate forces integer t; then $7t$ being divisible by 12 forces $12 \mid t$. The smallest positive possibility is $t = 12$, and it also makes $5t$ divisible by 12. Thus the path length is $12\sqrt{12^2+7^2+5^2} = 12\sqrt{218}$. Since $218 = 2 \cdot 109$ is square-free, the answer is 230.

**Answer:** 230.

**Methods:** reflection unfolding, lattice intersections, least common multiples.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** Both entries were reviewed. Entry 1 gives a rigorous unfolded ray. Entry 0 describes the required 12 scale units as reflections, which can obscure additional side-face reflections; the unfolded parameter avoids this counting issue.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory. Unfolding makes the reflected path straight, and a vertex hit becomes simultaneous coordinate divisibility.

## 2002 AIME I Problem 14: Integer means after deletion

A set of distinct positive integers contains 1, has maximum 2002, and has integral mean after deleting any single member. Find its largest possible size.

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### Solution

Write the size as $r+1$ and total as $T$. Integrality says every member is congruent to $T \pmod r$, hence to $1 \pmod r$. Therefore $r \mid 2001$ and the $(r+1)$st possible positive member is $1+r^2 \le 2002$. Thus

$r \le 44$ and $r \mid 3 \cdot 23 \cdot 29$, giving $r \le 29$. The set $\{1, 30, 59, \ldots, 813, 2002\}$ has 30 members, all 1 (mod 29), and satisfies the deletion condition. The maximum is 30.

**Answer:** 030.

**Methods:** extremal sets, congruence invariants.

**Verification:** Analytic derivation checked against the archived answer; not independently machine-checked.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Integral means after every deletion force a common residue class, whose size is bounded inside a fixed interval.

## 2002 AIME II Problem 1: Differences of reversed three-digit integers

Let $x, y$ be three-digit positive integers whose decimal digits are reversals of one another. How many distinct values of $|x - y|$ occur?

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### Solution

Write $x = 100a + 10b + c$, $y = 100c + 10b + a$, where $1 \le a, c \le 9$ and $0 \le b \le 9$. Then $|x - y| = 99|a - c|$. The absolute digit difference takes every value from 0 to 8, giving exactly 9 distinct results.

**Answer:** 009.

**Methods:** decimal digits, digit reversal.

**Verification:** Exhaustively reversed every eligible three-digit integer and counted distinct differences.

**Source comparison:** The sole source was reviewed; the digit cancellation and nine attainable differences are correct.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Digit reversal cancels the middle digit and leaves 99 times the outer-digit difference.

## 2002 AIME II Problem 3: An integral geometric progression with a square gap

Positive integers $a < b < c$ form a geometric progression and satisfy $\log_6 a + \log_6 b + \log_6 c = 6$. If $b - a$ is a perfect square, find $a + b + c$.

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### Solution

The logarithms give $abc = 6^6$. A geometric progression has $ac = b^2$, hence $b^3 = 6^6$ and $b = 36$. Since $0 < a < 36$ and $36 - a$ is a positive square, $a$ is one of $35, 32, 27, 20, 11$. We also need $c = 1296/a$ integral. Only 27 divides 1296, giving $c = 48$. Thus the sum is $27 + 36 + 48 = 111$.

**Answer:** 111.

**Methods:** geometric progressions, integer divisibility.

**Verification:** Exhaustively checked every smaller integer first term against the square-gap and integral geometric-mean constraints.

**Source comparison:** Both entries were reviewed. Their square-gap enumeration and middle-term reduction are correct.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. The product fixes the middle term, leaving a short list of square gaps compatible with divisibility.

## 2002 AIME II Problem 5: When a power fails to divide another

Find the sum of all integers $a = 2^n 3^m$, with $n, m \geq 0$, for which $a^6$ does not divide $6^a$.

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### Solution

Failure means $6n > 2^n 3^m$ or $6m > 2^n 3^m$. For $n \geq 5$, $2^n \geq 6n$ by induction from $32 \geq 30$; for $m \geq 3$, $3^m \geq 6m$ similarly. In the first inequality only $n = 1, 2, 3, 4$ remain, and each forces $m = 0$, yielding $a = 2, 4, 8, 16$. In the second only $m = 1, 2$ remain, each forcing $n = 0$, yielding $a = 3, 9$. These six distinct values sum to 42.

**Answer:** 042.

**Methods:** prime valuations, exponential bounds.

**Verification:** Exhaustively tested exponent pairs through 20; the analytic monotonic bounds certify all larger pairs are excluded.

**Source comparison:** Both entries were reviewed. Entry 0 gives the correct strict inequalities but invokes growth informally; induction supplies the finite cutoff. Entry 1 uses non-strict inequalities and confuses the logical alternatives, so its reasoning is invalid despite the correct list.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Prime valuations reduce divisibility to two exponential-versus-linear inequalities with explicit monotonic bounds.

## 2002 AIME II Problem 7: A square sum divisible by 200

Find the smallest positive integer $k$ for which $1^2 + 2^2 + \cdots + k^2$ is divisible by 200.

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### Solution

The condition is $1200 \mid k(k+1)(2k+1)$. A factor of 3 is automatic. Since exactly one of k and k+1 is even, divisibility by 16 requires $k \equiv 0$ or $15 \pmod{16}$. No two of the three factors share a factor 5, so

divisibility by 25 requires $k \equiv 0, 24$, or 12 (mod 25). Combining the two lists gives $k \equiv 0, 112, 175, 224, 287, 399$ (mod 400). Thus the smallest positive choice is 112.

**Answer:** 112.

**Methods:** valuations, Chinese remainder theorem, power sums.

**Verification:** Directly accumulated square sums until the first multiple of 200, without congruence reduction.

**Source comparison:** Both entries were reviewed. The first correctly derives the six congruence classes. Entry 1 misprints one residue in its cycle of multiples of 16; this does not affect the stated class 399.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. The three factors in the square-sum formula have disjoint odd-prime divisibility, leading to small congruence classes.

## 2002 AIME II Problem 8: The first missing integer quotient

Find the least positive integer $k$ never equal to $\lfloor 2002/n \rfloor$ for an integer $n \neq 0$.

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### Solution

A positive quotient requires $n > 0$, and $\lfloor 2002/n \rfloor = k$ is equivalent to $2002/(k+1) < n \leq 2002/k$. For $1 \leq k \leq 44$ this interval has length $2002/(k(k+1)) \geq 2002/1980 > 1$, so it contains an integer. Values $k = 45, 46, 47, 48$ are attained at $n = 44, 43, 42, 41$, respectively. For $k = 49$, the interval is $(2002/50, 2002/49] = (40.04, 40 + 42/49]$, containing no integer. The answer is 49.

**Answer:** 049.

**Methods:** floor quotients, interval lengths.

**Verification:** Enumerated every possible positive nonzero floor quotient and found the first missing value.

**Source comparison:** All four entries were reviewed. Entry 0 gives a sound no-skipping argument. Entry 1 checks only the preceding value after experimentation and does not prove minimality. Entry 2 mishandles the strict lower endpoint using a ceiling; entry 3 provides only an estimate before referring to entry 0. The interval-length proof covers every smaller k.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. A floor quotient equals k exactly when a reciprocal interval contains an integer, giving a guaranteed initial range and a short boundary check.

## 2003 AIME I Problem 1: The largest factorial factor

Write $((3!)!)!/3! = k\,n!$ with positive integers k,n and n as large as possible. Find $k + n$.

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**Solution**

The number is $720!/6 = 120 \cdot 719!$. Thus $n = 719$ is feasible with $k = 120$. Since the number is smaller than 720!, no $n \geq 720$ is feasible. The answer is $120 + 719 = 839$.

**Answer:** 839.

**Methods:** factorials, size bounds.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** The sole source was reviewed; its divisibility and size argument is complete.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A factorial quotient lies strictly below the next factorial but contains the preceding one as a divisor.

## 2003 AIME I Problem 7: Integral distances in an isosceles triangle

Points $A, B, C$ are collinear in that order with $AB = 9, BC = 21$. A noncollinear point D satisfies $AD = CD$, and both AD and BD are integers. Find the sum of all possible perimeters of triangle ACD.

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**Solution**

The perpendicular bisector of AC contains D. The midpoint of AC is 15 from A and 6 from B. If $x = AD = CD$ and $y = BD$, subtracting the squared-distance equations gives $x^2 - y^2 = 15^2 - 6^2 = 189$. The positive factor pairs for $(x - y, x + y)$ are $(1, 189), (3, 63), (7, 27), (9, 21)$, yielding $x = 95, 33, 17, 15$. The last gives zero height and is excluded; each other pair gives positive height and a valid D. The perimeter sum is $3 \cdot 30 + 2(95 + 33 + 17) = 380$.

**Answer:** 380.

**Methods:** integer lengths, difference of squares.

**Verification:** Enumerated every possible equal-side length within the factor-pair bound and tested the second distance by integer square roots.

**Source comparison:** All three entries were reviewed. The perpendicular-bisector and cosine-law methods agree. Entry 1 reverses the sign after its correct Stewart equation; the required identity is AD squared minus BD squared equals 189.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory. Subtracting two right-triangle equations leaves a small difference-of-squares factorization.

## 2003 AIME I Problem 8: An arithmetic triple followed by a geometric triple

Four increasing positive integers have their first three in arithmetic progression and their last three in geometric progression. The last exceeds the first by 30. Find their sum.

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**Solution**

Write the first three terms as $a-d, a, a+d$, with integers $a > d > 0$. The fourth is $(a+d)^2/a$, so the endpoint condition is $30 = 3d + d^2/a$. Because $0 < d^2/a < d$, this gives $3d < 30 < 4d$, hence $d = 8$ or 9. Solving $a = d^2/(30-3d)$ gives the noninteger $32/3$ for $d = 8$ and $a = 27$ for $d = 9$. The sequence is 18, $27, 36, 48$, whose sum is 129.

**Answer:** 129.

**Methods:** mixed progressions, integer bounds.

**Verification:** Exhaustively checked steps 1 through 9 and middle terms through 10000; the analytic inequality narrows the unbounded problem to the two final steps.

**Source comparison:** All three entries were reviewed. Entries 0 and 2 provide valid bounds. Entry 1 explicitly relies on presumed AIME uniqueness instead of checking all factor cases, so its standalone argument is incomplete.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. The endpoint difference bounds the common arithmetic step between two consecutive integers, leaving one integral solution.

## 2003 AIME I Problem 14: The smallest denominator containing a decimal block

For reduced positive fractions $m/n < 1$, find the smallest denominator n whose decimal expansion contains the consecutive digits 251 in that order.

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**Solution**

If the block starts after k preceding digits, the fractional part of $10^k m/n$ begins with 251 and has reduced denominator at most n. It therefore suffices to minimize denominators of fractions $u/v$ in $[251/1000, 252/1000)$. Setting $h = 4u - v$, these inequalities become $v \le 250h < 2v$. Thus $h \ge 1$ and $v > 125$. If $v \le 127$, necessarily $h = 1$, so $v + 1 = 4u$. Neither $v = 126$ nor any smaller v works, while $v = 127$, $u = 32$ does. Since $32/127$ lies in the required interval and is reduced, the answer is 127.

**Answer:** 127.

**Methods:** decimal blocks, rational approximation, denominator bounds.

**Verification:** Exhaustive integer comparison for all fractions of denominator at most 127 in the target decimal interval.

**Source comparison:** Both entries were reviewed. Entry 0 correctly justifies moving the block to the start. Entry 1 omits that reduction and asserts a residue restriction without a full lower-bound argument; the quarter-distance integer h supplies it.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Shifting a decimal block cannot increase its reduced denominator, and approximation to one quarter gives a sharp lattice bound.

## 2003 AIME II Problem 1: A product equal to six times a sum

Three positive integers have product N equal to six times their sum. One integer is the sum of the other two. Find the sum of all possible N.

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### Solution

Write the integers as $a, b, a+b$. Then $ab(a+b) = 12(a+b)$, so $ab = 12$. Up to order the pairs are (1, 12), $(2, 6), (3, 4)$, with sums 13,8,7. Their products N are $12 \cdot 13, 12 \cdot 8, 12 \cdot 7$, all distinct. The total is $12(13+8+7) = 336$.

**Answer:** 336.

**Methods:** integer factorization, cancellation.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** The sole source was reviewed; its three factor pairs exhaust the possibilities.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. The additive relation among three integers cancels a common positive factor and fixes the product of two.

## 2003 AIME II Problem 2: Largest distinct-digit multiple of eight

Let N be the largest positive multiple of 8 with no repeated decimal digit. Find $N$ mod 1000.

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### Solution

At most ten digits can occur. The greatest possible first seven digits of a ten-digit candidate are 9876543. A completion using the remaining digits 0,1,2 is divisible by 8 exactly when its last three-digit block is. Among the six permutations, only 120 is divisible by 8. Thus the largest candidate with this maximal prefix is 9876543120; no shorter number or smaller prefix can exceed it. Its remainder is 120.

**Answer:** 120.

**Methods:** divisibility by eight, lexicographic optimization.

**Verification:** Checked all six suffix permutations, distinct digits, and divisibility of the full maximal-prefix candidate.

**Source comparison:** The sole source was reviewed. Its construction is correct; the explicit feasible maximal prefix justifies global optimality.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Lexicographic maximization fixes a prefix while divisibility depends only on the last three digits.

### 2003 AIME II Problem 10: A nonsquare radical sum with fixed difference

Two positive integers differ by 60, and the sum of their square roots is the square root of a nonsquare integer. Find their largest possible sum.

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#### Solution

Write the integers as a and $a+60$. Squaring the given radical sum shows $\sqrt{a(a+60)}$ is rational, hence an integer t. Thus $(a+30-t)(a+30+t)=900$. The two positive factors have equal parity; since their product is even, both are even. Let the smaller be v. Their sum, which equals $2a+60$, is $v+900/v$, decreasing for $0<v\leq 30$. The smallest possible v is 2, giving $a=196, a+60=256$; this is excluded because the radical sum is 30, the square root of a square. The next admissible v is 6 (v=4 gives an odd other factor), yielding $a=48, a+60=108$. Their radical sum is $10\sqrt{3}=\sqrt{300}$, so the maximal allowed integer sum is 156.

**Answer:** 156.

**Methods:** radical integrality, factor pairs, optimization.

**Verification:** Brute-force exact radical-integrality tests for every possible smaller integer within the analytic factor-pair bound.

**Source comparison:** The sole source was reviewed. Its factor-pair method is valid; the monotonicity argument here explains why the next admissible smaller factor supplies the global maximum.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Squaring forces the integer product to be a square; a difference-of-squares factorization orders the possible sums.

### 2003 AIME II Problem 12: Minimum occupancy with percentage constraints

Twenty-seven candidates receive integer vote counts totaling $s$. For each candidate, its vote count exceeds its exact percentage of all votes by at least one. Find the smallest $s$.

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#### Solution

A count $v$ must satisfy $v(1-100/s)\geq 1$. If $s\leq 133$, then any $v\leq 4$ fails, so every candidate needs at least five votes, forcing $s\geq 135$, a contradiction. At $s=134$, the counts $4,5,\ldots,5$ satisfy the inequality, since even $4(1-100/134)>1$. Thus the minimum is 134.

**Answer:** 134.

**Methods:** pigeonhole principle, extremal occupancy.

**Verification:** Analytic derivation checked against the archived answer; not independently machine-checked.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer. Audit findings: Source 1: An intermediate denominator changes sign incorrectly; retain the positive denominator when deriving the inequality.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Vote-count lower bounds arise from exact percentages and integer rounding, followed by a sharp total-size constraint.

### 2004 AIME I Problem 1: Four descending consecutive digits

A four-digit positive integer has consecutive digits in strictly decreasing order. Find the sum of all possible remainders modulo 37.

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#### Solution

Writing the final digit as d, the number is $3210 + 1111d$ with $0 \le d \le 6$. Since $3210 \equiv 28$ and $1111 \equiv 1 \pmod{37}$, the remainders are $28, 29, \ldots, 34$. Their sum is $7(28 + 34)/2 = 217$.

**Answer:** 217.

**Methods:** decimal digits, modular arithmetic.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** Both entries were reviewed. Their arithmetic-progression reductions are equivalent and correct.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. The seven digit strings form an arithmetic progression with step congruent to 1 modulo 37.

### 2004 AIME I Problem 5: Strict daily rates with a larger aggregate score

A contestant attempts 500 points over two days and scores a positive integer each day. Its first-day success rate is below 160/300, its second-day rate below 140/200, and it does not attempt 300 points on day one. Find m+n for the largest possible aggregate rate m/n in lowest terms.

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#### Solution

Let q be the first-day attempted points and let a,b be the two scores. Then $a < (8/15)q$ and $b < (7/10)(500-q)$, so $a+b < 350-q/6 < 350$. Since the scores are integers, their sum is at most 349. This is attainable by attempting 2 points and scoring 1 on day one, then attempting 498 and scoring 348 on day two: both strict rate inequalities hold. The maximal aggregate rate is 349/500, yielding 849.

**Answer:** 849.

**Methods:** weighted averages, integer optimization, strict inequalities.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** Both entries were reviewed. Entry 0 gives the correct global bound but incorrectly says every q from 2 through 5 attains 349; q=3 or 5 does not. Entry 1 constructs a valid optimum but does not explicitly prove the universal bound. The proof here combines the bound with one verified witness.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. A weighted rate bound caps the integer total, and one allocation attains that cap.

### 2004 AIME I Problem 8: Regular star polygons

Count similarity classes of connected regular 1000-vertex star polygons with equal edge lengths and equal counterclockwise turning angles less than $180^\circ$, requiring a genuine edge crossing.

Published problem and solutions · MAA AMC.

#### Solution

Equal turns and lengths make successive edges a geometric progression of unit directions. Closure and distinct vertices therefore give the regular star $\{1000/k\}$, with $\gcd(k, 1000) = 1$. The counterclockwise turn condition chooses $1 \le k < 500$; $k = 1$ is the convex polygon and must be excluded. Coprime residues pair as $k, 1000 - k$, so the number is $\varphi(1000)/2 - 1 = 400/2 - 1 = 199$.

**Answer:** 199.

**Methods:** cyclic permutations, totient, geometry hybrid.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, geometry, number_theory. Star polygon connectivity is a coprimality condition on a cyclic step, followed by symmetry and crossing exclusions.

### 2004 AIME I Problem 15: Reverse dynamics and forbidden operation runs

Iterate $f(1) = 1$, $f(x) = x/10$ when $10 \mid x$, and $f(x) = x + 1$ otherwise. Let $d(x)$ be the first index with value 1, starting at index one. If $m$ integers have $d(x) = 20$, find the sum of distinct prime factors of $m$.

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#### Solution

Reverse the trajectory: the first reverse step is $1 \to 10$, followed by 18 operations $A : y \mapsto y - 1$ or $B : y \mapsto 10y$. An initial run of nine $A$'s would return to 1, and after a $B$ ten consecutive $A$'s are impossible because subtraction would use a multiple-of-ten predecessor. These are the only restrictions. Among $2^{18}$ words, the first bad type contributes $2^9$; the second contributes $8 \cdot 2^7$, since the block $BA^{10}$ has eight positions and cannot occur twice or coexist with the first bad type. Distinct legal inverse words have

distinct starting integers because the forward map is deterministic. Thus $m = 2^{18} - 2^9 - 8 \cdot 2^7 = 2^9 \cdot 509$, and 509 is prime. The answer is 511.

**Answer:** 511.

**Methods:** inverse processes, forbidden patterns, injectivity.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. An integer map is inverted by counting admissible digit-operation words of prescribed stopping time.

## 2004 AIME II Problem 3: Hidden cubes

A rectangular block made of unit cubes has exactly 231 cubes invisible when three mutually adjacent faces are viewed. Find the least possible total number of cubes.

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### Solution

For side lengths $a+1, b+1, c+1$, the unseen block has dimensions $a, b, c$, so $abc = 231$. Assume $a \le b \le c$. Then $a \le \sqrt[3]{231} < 7$, hence $a = 1$ or 3. If $a = 1$, the possibilities $(b, c) = (1, 231), (3, 77), (7, 33), (11, 21)$ have minimum total $2 \cdot 12 \cdot 22 = 528$. If $a = 3$, the only ordered possibility with $b \ge 3$ is $(b, c) = (7, 11)$, giving $4 \cdot 8 \cdot 12 = 384$.

**Answer:** 384.

**Methods:** finite case analysis, geometry hybrid.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** The first archived source asserts that balancing factors minimizes the product without proving the discrete case. Explicit factor enumeration here justifies the minimum.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, geometry, number_theory. Invisible cubes form an interior rectangular block, whose factor triples are optimized for the total cube count.

## 2004 AIME II Problem 6: Integer shares in a three-stage distribution

Three monkeys successively take portions of a banana pile. They keep respectively $3/4, 1/4, 1/12$ of their portions and split each remainder equally between the other two. Every transfer is integral, the last monkey empties the pile, and final holdings have ratio $3 : 2 : 1$. Find the smallest initial pile.

Published problem and solutions · MAA AMC.

**Solution**

The three taken portions must be $8x, 8y, 24z$ for positive integers x,y,z. The final holdings are $6x + 3y + 11z$, $x + 2y + 11z$, and $x + 3y + 2z$. Equating the first two to three and two times the last gives $3x - 6y + 5z = 0$ and $-x - 4y + 7z = 0$. Elimination yields $9x = 11z$ and $9y = 13z$. Thus $z = 9t$, $x = 11t, y = 13t$ with positive integer t. The total is $(8 \cdot 11 + 8 \cdot 13 + 24 \cdot 9)t = 408t$. Taking t=1 gives holdings $204, 136, 68$, so the minimum is 408.

**Answer:** 408.

**Methods:** integer linear systems, primitive ratios, divisibility.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** All four entries were reviewed. The units-based and normalized-fraction systems are valid. Entry 0 lists 72 as a basic divisibility requirement for the third portion, whereas the split itself requires 24; entry 2 incorrectly says 11 divides that portion. The explicit primitive parameterization removes these bookkeeping errors.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Choosing units that make each split integral turns the final-ratio conditions into a primitive integer null vector.

### 2004 AIME II Problem 8: Divisor counts as exponent allocations

How many divisors of $2004^{2004}$ have exactly 2004 positive divisors?

Published problem and solutions · MAA AMC.

**Solution**

Write a divisor as $2^a 3^b 167^c$. The divisor-count condition is $(a+1)(b+1)(c+1) = 2004 = 2^2 \cdot 3 \cdot 167$. Allocate the two factors of 2 among three labeled slots in $\binom{4}{2} = 6$ ways and allocate each other prime in three ways. All resulting exponents are at most 2003, so the original upper bounds are automatic. The answer is $6 \cdot 3 \cdot 3 = 54$.

**Answer:** 054.

**Methods:** stars and bars, multiplicative counting.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Divisor-count factorizations distribute the prime factors of 2004 among exponent increments subject to bounds.

## 2004 AIME II Problem 10: Two binary ones modulo nine

Choose uniformly an integer below $2^{40}$ with exactly two binary ones. Find numerator plus denominator of the probability that it is divisible by 9.

Published problem and solutions · MAA AMC.

### Solution

Write the integer uniquely as $2^a + 2^b$, $0 \le a < b < 40$. Since the order of 2 modulo 9 is 6 and $2^3 \equiv -1$, divisibility is equivalent to $b - a \equiv 3 \pmod 6$. The favorable count is $37+31+25+19+13+7+1 = 133$, out of $\binom{40}{2} = 780$. The fraction is reduced, so the answer is 913.

**Answer:** 913.

**Methods:** binary digits, residue classes, probability.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer. Audit findings: Source 2: A residue of minus one modulo nine is mistyped as plus one in the exponent classification.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Pairs of binary positions are counted according to powers of 2 modulo 9.

## 2004 AIME II Problem 14: Distinct digit lengths of a sum

A string of $n$ sevens is split into decimal integers by inserting plus signs. How many values of $n$ can give a total of 7000?

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### Solution

Only blocks of lengths one, two, three can occur. If their counts are $a, b, c$, then $a + 11b + 111c = 1000$ and $n = a + 2b + 3c = 1000 - 9k$, where $k = b + 12c$. For fixed $k$, cost $11b + 111c = 11k - 21c$ is minimized at $c = \lfloor k/12 \rfloor$, $b = k \bmod 12$. Thus feasibility is exactly $111\lfloor k/12 \rfloor + 11(k \bmod 12) \le 1000$. For $0 \le k \le 111$, all work except $107, 109, 110, 111$; values $k > 111$ give $n \le 0$. Hence there are $112 - 4 = 108$ lengths.

**Answer:** 108.

**Methods:** projection of integer solutions, greedy optimization.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer. Audit findings: Source 1: One displayed three-digit value is miscomputed. The transformation interpretation also requires careful treatment of overlapping occurrences.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Splits of repeated sevens impose a short integer equation for part lengths and attainable digit totals.

## 2005 AIME I Problem 2: Arithmetic progressions reaching 2005

For each positive integer k, consider $1, 1+k, 1+2k, \ldots$. How many values of k make 2005 a term?

Published problem and solutions · MAA AMC.

### Solution

The condition is $1 + jk = 2005$ for some positive integer j, equivalently $jk = 2004$. Thus k may be any positive divisor of $2004 = 2^2 \cdot 3 \cdot 167$. There are $(2+1)(1+1)(1+1) = 12$ choices.

**Answer:** 012.

**Methods:** arithmetic progressions, divisor function.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** Both entries were reviewed. The divisor enumeration is correct. Entry 1 initially phrases the condition as remainder 1, which fails for k=1; its note repairs that exception. Divisibility of 2004 includes it automatically.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. The required common differences are exactly the divisors of the target minus the first term.

## 2005 AIME I Problem 3: Exactly three small proper divisors

How many positive integers have exactly three positive proper divisors, all less than 50?

Published problem and solutions · MAA AMC.

### Solution

Exactly three proper divisors means four divisors in total, so the integer is either $p^3$ or pq for distinct primes p,q. In the cube case the largest proper divisor is $p^2 < 50$, allowing $p = 2, 3, 5, 7$, hence four integers. In the pq case both primes must be below 50. The fifteen such primes are $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47$, giving $\binom{15}{2} = 105$ products. The cases are disjoint, so the total is 109.

**Answer:** 109.

**Methods:** divisor patterns, prime counting.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** The sole source was reviewed. Its prime-power classification, prime list, and unordered pair count are complete.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, combinatorics. The four-divisor exponent patterns reduce the count to prime pairs and prime cubes.

### 2005 AIME I Problem 4: A rectangular formation five above a square

A band has five more members than a square formation can hold, and exactly fills a rectangle with seven more rows than columns. Find its greatest possible size.

Published problem and solutions · MAA AMC.

#### Solution

Let the rectangle have n columns and let the square side count be s. Then $n(n+7) = s^2 + 5$, so $(2n + 7 - 2s)(2n + 7 + 2s) = 69$. Both factors are positive odd integers; their pairs are $(1, 69)$ and $(3, 23)$. Their sum equals $4n + 14$, which is largest for $(1, 69)$, giving $n = 14, s = 17$. The resulting band size is $14 \cdot 21 = 294$.

**Answer:** 294.

**Methods:** completing the square, Diophantine factorization.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** All seven entries were reviewed. The bounding-squares and factor-pair proofs are sound. Entry 4 contains incorrect signs, a repeated variable, and a factor 68 instead of 69. Entry 6 uses the AIME answer range as an unjustified global bound; the complete factorization avoids that reliance.

**Archived entries inspected:** 7 of 7. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Completing the square converts the two formation requirements to factor pairs of 69.

### 2005 AIME I Problem 12: Parity changes only at squares

Let $S(n) = \sum_{k=1}^{n} \tau(k)$, where $\tau$ counts positive divisors. Among $1 \le n \le 2005$, let $a$ and $b$ count odd and even values of $S(n)$. Find $|a - b|$.

Published problem and solutions · MAA AMC.

#### Solution

Pair divisors $d$ and $k/d$: $\tau(k)$ is odd exactly for squares. Therefore $S(n) \equiv \lfloor \sqrt{n} \rfloor \pmod 2$. Its odd blocks are $[(2j-1)^2, (2j)^2 - 1]$ for $1 \le j \le 22$, of lengths $4j - 1$. Their total is $a = \sum_{j=1}^{22} (4j - 1) = 990$, so $b = 1015$ and $|a - b| = 25$.

**Answer:** 025.

**Methods:** parity, involutions, block counting.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Divisor-count parity changes only at squares, so intervals between squares determine the parity balance of cumulative sums.

### 2005 AIME II Problem 4: Union of divisor sets

Count positive integers dividing at least one of $10^{10}, 15^7, 18^{11}$.

Published problem and solutions · MAA AMC.

#### Solution

The three divisor sets have sizes $121, 64, 276$. Intersections are divisor sets of gcds, so their pairwise sizes are $8, 11, 8$, and their triple intersection is $\{1\}$. Inclusion-exclusion yields $121+64+276-8-11-8+1 = 435$.

**Answer:** 435.

**Methods:** inclusion exclusion, divisor sets.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Inclusion-exclusion on divisor sets uses gcds to count their intersections.

### 2005 AIME II Problem 5: Reciprocal logarithms force powers

Count ordered integer pairs $(a, b)$ with $2 \le a, b \le 2005$ satisfying $\log_a b + 6 \log_b a = 5$.

Published problem and solutions · MAA AMC.

#### Solution

Let $t = \log_a b > 0$. Then $t + 6/t = 5$, so $(t-2)(t-3) = 0$. Thus $b = a^2$ or $b = a^3$. The square family has $a = 2, \dots, 44$, giving 43 pairs; the cube family has $a = 2, \dots, 12$, giving 11. No ordered pair belongs to both because $a^2 = a^3$ would require $a = 1$. The total is 54.

**Answer:** 054.

**Methods:** logarithmic ratios, integer powers.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** All three entries were reviewed. Their quadratic substitutions are valid. The explanatory note in entry 0 correctly distinguishes overlap of b values from overlap of ordered pairs.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. A quadratic in the logarithmic ratio reduces the integer pairs to square and cube families.

## 2005 AIME II Problem 6: Fixed points of a shuffle

A stack labeled $1, \ldots, 2n$ top to bottom is cut into two equal halves. Restack by alternately placing the next card from the lower half and then the upper half on top. At least one card from each half stays fixed, including card 131. Find $2n$.

Published problem and solutions · MAA AMC.

### Solution

A card $j \le n$ from the upper half ends at position $2(n-j)+1$ from the top; a card $n+j$ from the lower half ends at position $2(n-j)+2$. Since 131 is odd it must be from the upper half, and $131 = 2(n-131)+1$ gives $n = 196$. The lower-half fixed-point equation $n + j = 2(n-j) + 2$ then gives $j = 66$, valid. Thus the required stack has 392 cards.

**Answer:** 392.

**Methods:** permutations, fixed points.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. The shuffle is a piecewise linear permutation, and fixed-point equations constrain the deck size.

## 2005 AIME II Problem 9: A swapped sine-cosine power identity

For how many positive integers $n \le 1000$ does $(\sin t + i\cos t)^n = \sin(nt) + i\cos(nt)$ hold for every real t?

Published problem and solutions · MAA AMC.

### Solution

Since $\sin t + i\cos t = ie^{-it}$, the identity is $i^n e^{-int} = ie^{-int}$. It holds for every t exactly when $i^n = i$, equivalently $n \equiv 1 \pmod 4$. The values are $1, 5, \ldots, 997$, totaling 250.

**Answer:** 250.

**Methods:** complex exponentials, roots of unity, congruences.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** All eight entries were reviewed. The exponential and argument calculations are valid. Entry 4 tests only t=0, proving necessity but not sufficiency; entry 5 drops an exponent in an intermediate factorization. The displayed equivalence supplies both directions.

**Archived entries inspected:** 8 of 8. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. The expression is i times a unit exponential, so the universal identity becomes a fourth-root-of-unity congruence.

### 2005 AIME II Problem 13: Integer polynomial values constrain integer roots

An integer-coefficient polynomial satisfies $P(17) = 10$, $P(24) = 17$, and $P(n) = n + 3$ for two distinct integers $n_1, n_2$. Find $n_1 n_2$.

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#### Solution

Neither 17 nor 24 is such a solution. For any integer solution n, the integer-polynomial divisibility rule gives $n - 17 \mid P(n) - P(17) = n - 7$ and $n - 24 \mid P(n) - P(24) = n - 14$. Subtracting the respective divisors shows that both $n - 17$ and $n - 24$ divide 10. They differ by 7, so among the signed divisors $\pm 1, \pm 2, \pm 5, \pm 10$, the only ordered pairs are $(2, -5)$ and $(5, -2)$. Thus the only possible solutions are 19 and 22. Since two distinct solutions are given, their product is 418.

**Answer:** 418.

**Methods:** integer polynomial congruences, signed divisors.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** All three entries were reviewed. Entry 2 gives a valid degree-free argument. Entry 0 writes a quotient A without clarifying it is an integer polynomial rather than a constant; entry 1 assumes P is quadratic without justification. The divisibility proof works for every degree.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Polynomial congruence preservation makes two integer differences divisors of the same small constant.

### 2006 AIME I Problem 3: Deleting the leading digit

Find the smallest positive integer whose value becomes one twenty-ninth as large when its leading decimal digit is removed.

Published problem and solutions · MAA AMC.

#### Solution

Write the number as $d10^k + t$, where $1 \le d \le 9$, $k \ge 1$, and $0 \le t < 10^k$. The condition gives $d10^k = 28t$, so $7 \mid d$ and hence $d = 7$. Consequently $t = 10^k/4$, which is integral first at $k = 2$. The corresponding

number is $700 + 25 = 725$; every larger $k$ gives a larger number.

**Answer:** 725.

**Methods:** decimal representation, divisibility.

**Verification:** Exhaustive decimal-deletion test of every integer below the claimed minimum and the candidate itself.

**Source comparison:** All three entries read. The unrestricted positional argument proves minimality. The alternatives rely on the contest answer range; the last also misprints 29 times 25 as 729.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A positional numeral equation forces a leading digit by divisibility.

## 2006 AIME I Problem 4: Valuation of a product of factorials

Let $z$ be the number of trailing decimal zeros of $\prod_{n=1}^{100} n!$. Find $z$ modulo 1000.

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### Solution

Each factorial contains at least as many factors of two as of five, so $z$ is the total five-adic valuation. A multiple $5j \le 100$ appears in $101 - 5j$ factorials; a multiple $25j \le 100$ contributes a second factor in $101 - 25j$ factorials. Thus $z = \sum_{j=1}^{20}(101 - 5j) + \sum_{j=1}^{4}(101 - 25j) = 970 + 154 = 1124$. The required remainder is 124.

**Answer:** 124.

**Methods:** prime valuation, factorials.

**Verification:** Constructed the full exact product of 100 factorials and repeatedly divided by ten, independently of the valuation sum.

**Source comparison:** The source correctly reverses the order of counting prime factors; no powers of 125 occur.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Trailing zeros are found by summing prime valuations over factorials.

## 2006 AIME I Problem 9: Integral geometric progressions and logarithms

A geometric progression begins with positive integer $a$ and has positive integer ratio $r$. If $\sum_{k=0}^{11} \log_8(ar^k) = 2006$, how many pairs $(a, r)$ exist?

Published problem and solutions · MAA AMC.

**Solution**

Combining logarithms gives $a^{12}r^{66} = 2^{6018}$, or $a^2r^{11} = 2^{1003}$. Unique factorization forces $a = 2^u$, $r = 2^v$ with $u, v \geq 0$. Thus $2u + 11v = 1003$. Exactly the odd values $v = 1, 3, \ldots, 91$ are admissible, and each determines one nonnegative integer $u$. There are 46 pairs.

**Answer:** 046.

**Methods:** logarithms, prime exponents, linear Diophantine equations.

**Verification:** Exhaustive exponent-pair enumeration within bounds supplied by the exact prime-factor reduction.

**Source comparison:** All three entries reviewed. The exponent argument is correct; the third mistakenly writes the final exponent pair as (541,1) rather than (496,1), and sometimes calls exponents a and r.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. A logarithmic product becomes a two-variable prime-exponent equation.

## 2006 AIME I Problem 13: Valuation layers

For even $x$, let $g(x)$ be the largest power of two dividing $x$. Set $S_n = \sum_{k=1}^{2^{n-1}} g(2k)$. Find the greatest $n < 1000$ for which $S_n$ is a square.

Published problem and solutions · MAA AMC.

**Solution**

For $1 \leq j < n$, exactly $2^{n-j-1}$ terms have $g = 2^j$; the final term has $g = 2^n$. Hence $S_n = (n-1)2^{n-1} + 2^n = 2^{n-1}(n+1)$. If $n$ is even, its exponent of two is odd, impossible. If $n$ is odd, $2^{n-1}$ is a square and we need $n+1$ to be a square. The largest even square at most 1000 is $30^2 = 900$, so $n = 899$.

**Answer:** 899.

**Methods:** contribution counting, valuation layers.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer. Audit findings: Source 1: The recurrence is valid with n at least two and S_1=2; the written artificial S_0=1 is not compatible with the actual definition. Source 2: The recurrence is explicitly guessed without proof, and smallest/largest language is reversed. A valuation count proves the formula directly. Source 3: The closing least/greatest wording and a summation index are mistyped; the valuation argument otherwise works.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Counting valuation classes reduces a sum of powers of two to a square criterion.

## 2006 AIME II Problem 3: Three-adic valuation of an odd factorial

Find the greatest exponent $k$ for which $3^k$ divides $1 \cdot 3 \cdot 5 \cdots 199$.

Published problem and solutions · MAA AMC.

### Solution

The product is $200!/(2^{100}100!)$. Legendre counting therefore gives $k = (66+22+7+2)-(33+11+3+1) = 49$, where the four terms in each sum count multiples of $3, 9, 27, 81$.

**Answer:** 049.

**Methods:** prime valuation, double factorial.

**Verification:** Constructed the exact product of the first 100 odd integers and counted factors of three directly.

**Source comparison:** All four entries reviewed. The factorial quotient and direct odd-multiple counts agree. The last entry contains incorrect residue wording and says 27 gives 3 before correctly using 4 in its final sum.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. The product of odd integers is a quotient of factorials.

## 2006 AIME II Problem 7: Complementary decimal numbers

Count ordered positive integer pairs $(a, b)$ with $a + b = 1000$ and no zero digit in either number.

Published problem and solutions · MAA AMC.

### Solution

There are 999 choices for $a$. Exclude the 99 multiples of 10, which also make $b$ end in zero. Among the rest, a zero can occur only in a genuine tens position: $a = 101, \ldots, 109; 201, \ldots, 209; \ldots; 901, \ldots, 909$, giving 81 cases, and likewise 81 for $b$. These two classes are disjoint because their complements have tens digit nine. Thus $999 - 99 - 81 - 81 = 738$.

**Answer:** 738.

**Methods:** complement, digit counting.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Column carries count zero-free digit pairs with prescribed sum.

## 2006 AIME II Problem 13: Five representations by consecutive odd integers

How many integers $N < 1000$ have exactly five representations as a sum of one or more consecutive positive odd integers, counting distinct sequence lengths?

Published problem and solutions · MAA AMC.

### Solution

A length-$j$ sum beginning at $2t+1$ is $N = j(2t+j)$ with $t \geq 0$. Thus representations correspond exactly to factor pairs $u \leq v$ of $N$ with equal parity. For odd $N$, their number is $\lceil \tau(N)/2 \rceil$; for $4 \mid N$, it is $\lceil \tau(N/4)/2 \rceil$; for $N \equiv 2 \pmod 4$, it is zero. Five representations require a divisor count of 9 or 10. The possible exponent patterns are $p^8, p^2q^2, p^9, p^4q$ with distinct primes $p, q$. The pure prime powers exceed the relevant bounds. Odd $N < 1000$ therefore give $225, 441, 405, 567, 891$. For $N/4 < 250$, the possibilities are $36, 100, 196, 225, 48, 80, 112, 176, 208, 162$. The lists are exhaustive by the prime-power bounds in the two remaining patterns. Thus there are $5 + 10 = 15$ values.

**Answer:** 015.

**Methods:** divisor pairs, consecutive sums, prime exponent patterns.

**Verification:** Enumerated every consecutive positive odd sequence with sum below 1000, independently of divisor-count classification.

**Source comparison:** Both entries reviewed. Their lists are correct, but they omit mentioning the pure prime-power exponent patterns; the size bounds exclude those, as made explicit here.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, combinatorics. Representations are in bijection with same-parity factor pairs, followed by divisor-count classification.

## 2006 AIME II Problem 14: Reciprocal digit frequency

Let $S_n$ be the sum of reciprocals of all nonzero decimal digits appearing in $1, \ldots, 10^n$. Find the least positive $n$ for which $S_n$ is integral.

Published problem and solutions · MAA AMC.

### Solution

Pad $0, \ldots, 10^n - 1$ to $n$ digits. Each nonzero digit appears $n10^{n-1}$ times, and $10^n$ contributes one extra 1. Thus $S_n = 1 + n10^{n-1}H_9$. Since $H_9 = 7129/2520$ in lowest terms and $2520 = 2^3 \cdot 3^2 \cdot 5 \cdot 7$, integrality requires $63 \mid n$. The candidate $n = 63$ supplies all the factors of two and five as well, so it is minimal.

**Answer:** 063.

**Methods:** contribution counting, digit frequency.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Uniform digit-occurrence counts determine integrality of a reciprocal-digit sum.

## 2007 AIME I Problem 1: Square multiples of twenty-four

How many positive perfect squares below $10^6$ are divisible by 24?

Published problem and solutions · MAA AMC.

### Solution

If $m^2$ is divisible by $2^3 \cdot 3$, then $m$ is divisible by $2^2 \cdot 3 = 12$, and this condition is sufficient. Since $1 \le m < 1000$, the possibilities are $m = 12, 24, \ldots, 996$, numbering 83.

**Answer:** 083.

**Methods:** prime exponents, perfect squares.

**Verification:** Exhaustively tested divisibility for every eligible positive square root.

**Source comparison:** Both entries reviewed. The second loosely calls the squares multiples of 12, but its actual list correctly takes squares of multiples of 12.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Prime valuations characterize the square roots of admissible multiples.

## 2007 AIME I Problem 4: The next planetary alignment

Three coplanar planets travel uniformly in circular orbits in the same direction, with periods $60, 84, 140$ years. They and the central star are initially collinear. After how many years will this first recur?

Published problem and solutions · MAA AMC.

### Solution

Collinearity permits angular differences that are integral multiples of $\pi$, regardless of which side of the star a planet occupies. Consecutive relative angular speeds are $2\pi(1/60 - 1/84) = \pi/105$ and $2\pi(1/84 - 1/140) = \pi/105$. Thus both required differences are integral half-turns exactly when $t/105$ is an integer. The first positive time is 105.

**Answer:** 105.

**Methods:** relative angular speed, periodicity.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** The sole source correctly works modulo a half-turn, rather than requiring every planet to return to its initial location.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Relative angular speeds lead to a common half-turn period.

### 2007 AIME I Problem 5: A periodic rounding map

Convert an integer Fahrenheit temperature in $[32, 1000]$ to Celsius using $C = 5(F-32)/9$, round to the nearest integer, convert back, and round again. Count temperatures recovered exactly.

Published problem and solutions · MAA AMC.

#### Solution

Write $F - 32 = 9q + r$, $0 \le r < 9$. The round trip depends only on $r$ and sends it to $\operatorname{round}((9/5)\operatorname{round}(5r/9))$. Checking the nine residues yields the fixed set $\{0, 2, 4, 5, 7\}$. The range $0 \le F - 32 \le 968$ contains 107 complete nine-blocks and the residues $0, \dots, 5$ of one last block. Hence the count is $107 \cdot 5 + 4 = 539$. No half-integer rounding ambiguity occurs.

**Answer:** 539.

**Methods:** residue classes, finite states.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Residue classes count temperatures fixed by two rounds of nearest-integer conversion.

### 2007 AIME I Problem 7: A floor-ceiling logarithm indicator

Find $\sum_{k=1}^{1000} k(\lceil \log_{\sqrt{2}} k \rceil - \lfloor \log_{\sqrt{2}} k \rfloor)$ modulo 1000.

Published problem and solutions · MAA AMC.

#### Solution

The parenthesis is zero when the logarithm is integral and one otherwise. If $k = (\sqrt{2})^m$ is an integer, then $m$ is even, so these exceptions are exactly $1, 2, 4, \dots, 512$. The sum is therefore $1000 \cdot 1001/2 - (2^{10} - 1) = 499477$, with remainder 477.

**Answer:** 477.

**Methods:** floor and ceiling, indicator functions.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** The sole source identifies the same exceptions; the parity of the exponent explicitly justifies why odd powers of the square root cannot occur.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. The floor-ceiling difference deletes precisely the powers of two from an arithmetic sum.

### 2007 AIME I Problem 11: Rounding square roots in blocks

For each positive integer $p$, let $b(p)$ be its square root rounded to the nearest integer. Find $\sum_{p=1}^{2007} b(p) \pmod{1000}$.

Published problem and solutions · MAA AMC.

#### Solution

We have $b(p) = k$ precisely for $k^2 - k + 1 \le p \le k^2 + k$, a block of $2k$ integers. Blocks $k = 1, \dots, 44$ are complete; $p = 1981, \dots, 2007$ contributes 27 copies of 45. Thus the sum is $2\sum_{k=1}^{44} k^2 + 27 \cdot 45 = 59955$, whose residue is 955.

**Answer:** 955.

**Methods:** block counting, weighted sums.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Counting the integers in each nearest-square-root interval evaluates a weighted sum.

### 2007 AIME II Problem 2: Divisibility becomes a composition

Count ordered positive triples $(a, b, c)$ with $a \mid b$, $a \mid c$, and $a + b + c = 100$.

Published problem and solutions · MAA AMC.

#### Solution

Write $b = ax, c = ay$ with $x, y \ge 1$. For each divisor $d = 100/a \ge 3$, the condition is $x + y = d - 1$, with $d - 2$ solutions. Therefore the count is $\sum_{d \mid 100, d \ge 3}(d - 2) = 2 + 3 + 8 + 18 + 23 + 48 + 98 = 200$.

**Answer:** 200.

**Methods:** compositions, divisor decomposition.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Divisibility reduces the ordered-triple count to compositions indexed by divisors.

## 2007 AIME II Problem 5: Grid cells below a rational line

Count unit grid squares whose interiors lie in the first quadrant and below $9x + 223y = 2007$.

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### Solution

The intercept rectangle has $223 \cdot 9 = 2007$ cells. Its diagonal crosses $223 + 9 - \gcd(223, 9) = 231$ cells: one initial cell and one new cell for each interior grid-line crossing. Since the intercepts are coprime, no interior horizontal and vertical crossings coincide. Half of the remaining cells lie on each side by central symmetry. Thus the answer is $(2007 - 231)/2 = 888$.

**Answer:** 888.

**Methods:** lattice geometry, symmetry.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, geometry, number_theory. A lattice-square count is a sum of floor-function row lengths below a line.

## 2007 AIME II Problem 7: Multiples inside a cube-root interval

Exactly 70 positive integers $n$ satisfy both $\lfloor \sqrt[3]{n} \rfloor = k$ and $k \mid n$ for some integer $k$. Find the largest possible value of $n/k$ among them.

Published problem and solutions · MAA AMC.

### Solution

Necessarily $k \geq 1$. The multiples lie from $k^3$ through $(k+1)^3 - 1 = k(k^2 + 3k + 3)$, inclusive, with step $k$. Their number is $k^2 + 3k + 3 - k^2 + 1 = 3k + 4$. Hence $k = 22$, and the largest quotient is $k^2 + 3k + 3 = 553$.

**Answer:** 553.

**Methods:** floor functions, arithmetic progressions.

**Verification:** Directly enumerated the cube-root intervals through k=39; the analytic count proves no further k works.

**Source comparison:** Both entries reviewed. The direct endpoint count and two-sided inequality yield the same unique value of k.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Exact endpoints count a short arithmetic progression.

## 2007 AIME II Problem 12: The largest term controls a geometric sum

An increasing geometric sequence consists of integral powers of 3. It satisfies $\sum_{j=0}^{7} \log_3 x_j = 308$ and $56 \le \log_3(\sum_{j=0}^{7} x_j) \le 57$. Find $\log_3 x_{14}$.

Published problem and solutions · MAA AMC.

### Solution

Write $x_j = 3^{a+jd}$ with integers $a$ and $d \ge 1$. The first condition gives $2a + 7d = 77$. If $M = a + 7d$, the sum satisfies $3^M < S < 3^M \sum_{j=0}^{\infty} 3^{-j} = \frac{3}{2}\, 3^M$. Since $3^{56} \le S \le 3^{57}$, an integer $M \le 55$ is too small and $M \ge 57$ is too large. Thus $M = 56$. Solving yields $a = 21$, $d = 5$, so $a + 14d = 91$.

**Answer:** 091.

**Methods:** geometric sums, logarithms, dominant term bounds.

**Verification:** Exact large-integer substitution of the proposed progression into both original conditions.

**Source comparison:** All four entries reviewed. The base-three and geometric-tail ideas are sound. The first uses an approximation where a bound is needed; the last repeatedly discards small terms and does not establish uniqueness. The strict tail bound repairs both issues and also permits negative initial exponents.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. A geometric tail bound fixes the leading exponent exactly.

## 2007 AIME II Problem 13: Pascal propagation modulo three

Place eleven binary digits in the bottom row of a triangular array. Each upper entry is the sum of its two children. Count assignments making the top entry divisible by 3.

Published problem and solutions · MAA AMC.

### Solution

The top is $\sum_{j=0}^{10} \binom{10}{j} x_j$, by Pascal's identity or path counting. Modulo three, $(1+t)^{10} \equiv (1+t^9)(1+t)$, so only coefficients $j = 0, 1, 9, 10$ survive, each equal to one. Their sum must be zero or three, giving $\binom{4}{0} + \binom{4}{3} = 5$ choices. The other seven bits are free, so the answer is $5 \cdot 2^7 = 640$.

**Answer:** 640.

**Methods:** binomial propagation, modular generating functions.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Pascal coefficients modulo three control a binary-assignment count.

### 2008 AIME I Problem 3: Integer speeds from two distance totals

Biking, jogging, and swimming speeds $b, j, s$ are positive integers. A two-hour bike ride, three-hour jog, and four-hour swim cover 74 km; four hours biking, two jogging, and three swimming cover 91 km. Find $b^2 + j^2 + s^2$.

Published problem and solutions · MAA AMC.

#### Solution

The equations are $2b + 3j + 4s = 74$ and $4b + 2j + 3s = 91$. Elimination gives $s = 125 - 8b$ and $j = 10b - 142$. Positivity forces $14.2 < b < 15.625$, so the integer $b$ must be 15. Then $j = 8$, $s = 5$, and the required sum is $225 + 64 + 25 = 314$.

**Answer:** 314.

**Methods:** integer linear systems, positivity bounds.

**Verification:** Exhaustively checked all positive integral rates within bounds from the first distance equation.

**Source comparison:** All four entries reviewed. The first two rigorously isolate the solution. The third contains parity errors and unjustified congruence steps; the fourth appeals to plausible athletic speeds rather than proving uniqueness.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Positivity sharply bounds the single free parameter in an integer linear system.

### 2008 AIME I Problem 4: A shifted difference of squares

Positive integers $x, y$ satisfy $x^2 + 84x + 2008 = y^2$. Find $x + y$.

Published problem and solutions · MAA AMC.

#### Solution

Completing the square gives $(y - x - 42)(y + x + 42) = 244$. Both factors are positive and have the same parity. As $244 = 4 \cdot 61$, the only same-parity positive factor pair is $(2, 122)$. Hence $y = 62$, $x = 18$, and $x + y = 80$.

**Answer:** 080.

**Methods:** difference of squares, factor pairs.

**Verification:** Direct square test through the factorization-derived global bound, finding the unique positive solution.

**Source comparison:** All six entries read. The factorization proof is complete. Several alternatives merely test candidates or invoke the stated uniqueness. The fifth has a sign error in its expansion; the last writes x=12 even though its own relation gives x=18.

**Archived entries inspected:** 6 of 6. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Completing the square reduces the equation to same-parity factor pairs.

### 2008 AIME I Problem 6: Summing an arithmetic row repeatedly

The top row is $1, 3, \ldots, 99$. Each row below sums adjacent entries until one remains. Count all entries divisible by 67.

Published problem and solutions · MAA AMC.

#### Solution

At row $r$ and position $k$, with $1 \le r \le 50$ and $1 \le k \le 51 - r$, induction gives $a_{r,k} = 2^{r-1}(r + 2k - 2)$. The parenthetical factor lies between $r$ and $100 - r$, so divisibility by 67 means it equals 67. This has exactly one integral $k$ for each odd $r \le 33$ and none otherwise. There are 17 such rows.

**Answer:** 017.

**Methods:** recurrences, binomial propagation.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Binomial coefficients express triangular sums and permit counting their divisibility by 67.

### 2008 AIME I Problem 7: Occupied hundred-blocks

For $0 \le i \le 999$, let $S_i = \{100i, \ldots, 100i + 99\}$. How many of these sets contain no perfect square?

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#### Solution

The first 26 blocks $S_0, \ldots, S_{25}$ all contain squares: up to $50^2$, consecutive square gaps are at most 99, and $50^2$ lies in $S_{25}$. Above $50^2$, successive squares differ by more than 100, so $51^2, \ldots, 316^2$ occupy 266 new distinct blocks. These are all squares below 100000. Hence $1000 - 26 - 266 = 708$ blocks are empty.

**Answer:** 708.

**Methods:** complement, gap bounds.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Intervals containing consecutive squares are counted with duplicate-bin correction.

## 2008 AIME II Problem 4: Balanced ternary support

Write 2008 as a sum of distinct nonnegative powers of 3, each with coefficient 1 or $-1$. Find the sum of the exponents that occur.

Published problem and solutions · MAA AMC.

### Solution

In balanced ternary, the lowest digit is the unique element of $\{-1, 0, 1\}$ congruent to the number modulo three; subtract it and divide by three to continue. This also proves uniqueness. Applying the rule gives $2008 = 3^7 - 3^5 + 3^4 - 3^3 + 3^2 + 3^0$. Hence the exponent sum is $7 + 5 + 4 + 3 + 2 + 0 = 21$.

**Answer:** 021.

**Methods:** balanced ternary, numeral systems.

**Verification:** Exact reconstruction from the signed powers; the proof supplies uniqueness.

**Source comparison:** The sole source converts ordinary ternary digits by carrying; the residue rule here makes uniqueness explicit.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Signed base-three digits are fixed successively by residues modulo three.

## 2008 AIME II Problem 8: A trigonometric telescope and divisibility

Put $a = \pi/2008$. Find the smallest positive integer $n$ for which $2\sum_{k=1}^{n} \cos(k^2 a)\sin(ka)$ is an integer.

Published problem and solutions · MAA AMC.

### Solution

Product-to-sum gives $2\cos(k^2 a)\sin(ka) = \sin(k(k+1)a) - \sin(k(k-1)a)$. The sum therefore equals $\sin(n(n+1)\pi/2008)$. An integer sine must be $-1, 0$, or $1$, so its argument must be a multiple of $\pi/2$. Thus $1004 = 4 \cdot 251$ must divide $n(n+1)$. The prime 251 forces the first candidate to be $n = 250$, but $250 \cdot 251$ is not divisible by four. The next candidate $n = 251$ works because 252 is divisible by four. Hence the answer is 251.

**Answer:** 251.

**Methods:** telescoping, trigonometric identities, divisibility.

**Verification:** Directly summed the original trigonometric series at every candidate index through 251.

**Source comparison:** Both entries reviewed. The complex-exponential derivation is equivalent to product-to-sum and has a cubic exponent typo in one expanded telescoping line.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Quadratic-angle product-to-sum terms telescope to one sine.

## 2008 AIME II Problem 15: Consecutive cubes and a second square

Find the largest integer $n$ such that $n^2$ is the difference of consecutive integer cubes and $2n + 79$ is a perfect square.

Published problem and solutions · MAA AMC.

### Solution

For positive $n$, write $n^2 = (t+1)^3 - t^3 = 3t^2 + 3t + 1$. Then $(2n-1)(2n+1) = 3(2t+1)^2$. The two positive odd factors are coprime, so one is a square and the other three times a square. The possibility $2n-1 = 3u^2$ would make the square $2n+1$ congruent to two modulo three, impossible. Hence $2n-1 = b^2$. Writing $2n + 79 = a^2$ gives $(a-b)(a+b) = 80$. Both factors are positive even integers, so $a - b \geq 2$ and $a + b \leq 40$, giving $b \leq 19$ and $n \leq 181$. Equality works: $2(181) + 79 = 21^2$ and $181^2 = 105^3 - 104^3$. Negative n cannot exceed this positive example, so the maximum is 181.

**Answer:** 181.

**Methods:** coprime factors, difference of squares, Pell-type equations.

**Verification:** Directly checked both original square/cube conditions through the analytically established maximum, without using the factor-pair candidates.

**Source comparison:** All three entries reviewed. The first gives a genuine global bound. The Pell and candidate-testing alternatives assume the answer is below 1000 from contest format and thus do not prove maximality; the final candidate list is also incomplete.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Coprime factors of a Pell-type expression force a bounded difference of squares.

## 2009 AIME I Problem 1: Extremal three-digit geometric progressions

A three-digit integer is called geometric when its three distinct digits, in order, form a geometric progression. Find the largest such integer minus the smallest.

Published problem and solutions · MAA AMC.

### Solution

Write the digits as $a, b, c$. None can be zero, since a zero ratio would repeat zeros, and $b^2 = ac$. The smallest possible first digit is one; the smallest distinct middle digit then is two, producing 124. For the largest numeral take $a = 9$, which is feasible. Then $b^2 = 9c$ forces $3 \mid b$; distinctness excludes $b = 9$, so the largest possible $b$ is six, giving $c = 4$ and numeral 964. The difference is $964 - 124 = 840$.

**Answer:** 840.

**Methods:** digit constraints, geometric progressions.

**Verification:** Exhaustively tested all 900 three-digit numbers for distinct geometric digits.

**Source comparison:** All three entries reviewed. The geometric-mean condition gives a complete extremal argument; the shortest entry leaves its candidate exclusions implicit.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Digit bounds and a geometric-mean identity find the extreme numerals.

### 2009 AIME I Problem 6: Integer values of a stepped power function

How many positive integers $N < 1000$ occur as $x^{\lfloor x \rfloor}$ for some real $x$ where the expression is defined?

Published problem and solutions · MAA AMC.

#### Solution

For $x < 0$, either $-1 < x < 0$ gives a negative reciprocal or $x \leq -1$ gives magnitude at most one, with no positive integer value at the boundary; thus there are no new values. For $0 < x < 1$, the value is one. If $k \leq x < k+1$ with integer $k \geq 1$, the image is $[k^k, (k+1)^k)$. For $k = 1, 2, 3, 4$ the integer counts are $1, 5, 37, 369$; these ranges are disjoint. For $k \geq 5$ all values exceed 1000. Hence the total is $1 + 5 + 37 + 369 = 412$.

**Answer:** 412.

**Methods:** floor functions, piecewise monotonicity.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** Both entries reviewed. The positive-domain counts are correct; neither source discusses negative real x, which contributes no additional positive integer values as checked here.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. The floor partitions the domain into disjoint monotone-image intervals.

### 2009 AIME I Problem 7: An exponential recurrence that telescopes

Let $a_1 = 1$ and $5^{a_{n+1} - a_n} - 1 = 1/(n + 2/3)$ for $n \geq 1$. Find the least integer $k > 1$ such that $a_k$ is integral.

Published problem and solutions · MAA AMC.

#### Solution

The recurrence is $5^{a_{n+1} - a_n} = (3n+5)/(3n+2)$. Multiplying from $n = 1$ through $k - 1$ gives $5^{a_k - 1} = (3k+2)/5$, hence $5^{a_k} = 3k + 2$. For $k > 1$, the first candidate powers are 25 and 125. The former is one modulo three and cannot equal $3k + 2$; the latter gives $k = (125 - 2)/3 = 41$. Thus the answer is 41.

**Answer:** 041.

**Methods:** telescoping products, logarithms, modular arithmetic.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** Both entries reviewed. Their telescoping and induction approaches are equivalent and complete.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Multiplication of adjacent exponential ratios produces an arithmetic progression inside a logarithm.

## 2009 AIME I Problem 11: Integral triangle areas on a line

Choose distinct nonnegative lattice points $P, Q$ on $41x + y = 2009$. Count triangles $OPQ$ of positive integer area, where $O = (0, 0)$.

Published problem and solutions · MAA AMC.

### Solution

Write the two x-coordinates as $u, v \in \{0, \ldots, 49\}$. Their determinant is $2009(u-v)$, so the area is integral exactly when $u, v$ have the same parity. Each parity class contains 25 values. Every distinct pair yields positive area, hence the count is $2\binom{25}{2} = 600$.

**Answer:** 600.

**Methods:** parity, geometry hybrid.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer. Audit findings: Source 2: The inference that the base is an integral multiple of twice the primitive lattice length requires the lattice-spacing condition; integrality of area alone is insufficient.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, geometry, number_theory. Lattice-point pairs are filtered by the parity condition for integral triangle area.

## 2009 AIME I Problem 13: Infinite integrality forces a two-cycle

Every term of a sequence is a positive integer, and $a_{n+2} = (a_n + 2009)/(1 + a_{n+1})$ for $n \geq 1$. Find the smallest possible $a_1 + a_2$.

Published problem and solutions · MAA AMC.

**Solution**

Subtract consecutive rearranged recurrence equations to obtain $(a_{n+3} - a_{n+1})(a_{n+2} + 1) = a_{n+2} - a_n$. Thus the integer differences $d_n = a_{n+2} - a_n$ satisfy $|d_{n+1}| \le |d_n|/2$. If $d_1$ were nonzero, all subsequent differences would remain nonzero by the same identity, contradicting this bound. Hence $d_1 = 0$, so $a_3 = a_1$ and the original equation gives $a_1 a_2 = 2009 = 7^2 \cdot 41$. Conversely any such factor pair repeats periodically and works. The factor pairs are $(1, 2009), (7, 287), (41, 49)$; the minimum sum is 90.

**Answer:** 090.

**Methods:** integer descent, nonlinear recurrence, periodic sequences.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** All three entries reviewed. The first integer-difference descent is rigorous. The interval argument in the second incorrectly claims all intervening integers occur; the third explicitly offers only a periodicity guess. The quantitative contraction makes the necessity precise.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. A contracting integer difference must vanish identically.

## 2009 AIME I Problem 14: Minimizing a second moment with fixed first and fourth moments

A sequence of 350 entries, each in $\{1, 2, 3, 4\}$, satisfies $\sum a_i = 513$ and $\sum a_i^4 = 4745$. Find the smallest possible $\sum a_i^2$.

Published problem and solutions · MAA AMC.

**Solution**

Let $w, x, y, z$ count the entries equal to $1, 2, 3, 4$. Subtracting the count equation from the two moment equations gives $x + 2y + 3z = 163$ and $15x + 80y + 255z = 4395$. Eliminating $x$ gives $5y + 21z = 195$. Thus $z \equiv 0 \pmod 5$ and $0 \le z \le 9$, so $(y, z)$ is $(39, 0)$ or $(18, 5)$. These give $(w, x) = (226, 85)$ or $(215, 112)$ respectively. Their second moments are 917 and 905, so the minimum is 905, attained by the latter frequencies.

**Answer:** 905.

**Methods:** frequency variables, integer optimization, moments.

**Verification:** Enumerated every feasible frequency vector using the count and moment constraints, and minimized the second moment.

**Source comparison:** The sole source correctly reduces the constraints to two feasible frequency vectors. The explicit second moments here certify which is minimal.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory, combinatorics. Frequency variables reduce a finite-valued sequence optimization to two feasible integer points.

## 2009 AIME II Problem 4: The smallest winning score

For some number of children, the child ranked $k$ eats exactly $n+2-2k$ grapes, a nonnegative amount. Their total is 2009. Find the smallest possible winning count $n$.

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### Solution

If $n \le 88$, the largest possible total occurs at $n = 88$ with every positive term included: $88+86+\cdots+2 = 44 \cdot 45 = 1980 < 2009$. Hence $n \ge 89$. For $n = 89$, the sum of all positive odd terms through 89 is $45^2 = 2025$. Deleting the tail $7+5+3+1 = 16$ leaves $89+87+\cdots+9 = 2009$. This is a valid contest with 41 children, so the minimum is 89.

**Answer:** 089.

**Methods:** arithmetic progressions, extremal bounds.

**Verification:** Enumerated all nonnegative arithmetic-score contests with winner count at most 89.

**Source comparison:** All four entries reviewed, including the statement-clarification entry. The maximum-sum construction is rigorous. The final factor-pair discussion notes an invalid reversed participant count, though the valid 41-child solution is correct.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. A maximal arithmetic-progression sum gives a sharp lower bound, attained by deleting a tail.

## 2009 AIME II Problem 7: A sum with strictly increasing dyadic denominators

Write $\sum_{i=1}^{2009} (2i-1)!!/(2i)!!$ in lowest terms with denominator $2^a b$, where $b$ is odd. Find $ab/10$.

Published problem and solutions · MAA AMC.

### Solution

The ith summand equals $\binom{2i}{i}/2^{2i}$, so its reduced denominator is a power of two. Since $(2i)! = 2^i i!(2i-1)!!$, its numerator binomial has valuation $i-\nu_2(i!)$. Thus the reduced denominator exponent is $e_i = i+\nu_2(i!)$, strictly increasing with i. On the common denominator $2^{e_{2009}}$, the last numerator is odd and every earlier numerator is even. No further factor of two cancels, so $b = 1$ and $a = 2009 + \nu_2(2009!)$. Legendre gives $\nu_2(2009!) = 1004+502+251+125+62+31+15+7+3+1 = 2001$. Hence $a = 4010$ and $ab/10 = 401$.

**Answer:** 401.

**Methods:** prime valuation, unique minimum valuation, central binomial coefficients.

**Verification:** Summed all 2009 rational summands exactly and checked the fully reduced denominator.

**Source comparison:** All four entries reviewed, including comments in the first. The strict valuation argument is sound. Several alternatives invoke contest answer bounds to assert b=1, or infer a sum valuation from a non-strict lower bound; those do not suffice. The unique odd numerator here explicitly prevents cancellation.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. A unique largest reduced power-of-two denominator cannot cancel in the sum.

### 2009 AIME II Problem 9: An injection leaves two boundary slices

Let $m, n$ count positive integer solutions to $4x + 3y + 2z = 2009$ and $4x + 3y + 2z = 2000$, respectively. Find $(m - n) \bmod 1000$.

Published problem and solutions · MAA AMC.

#### Solution

Adding three to $y$ injects the second solution set into the first. In the first equation $y$ is odd, and the image consists exactly of solutions with $y \geq 5$. The missing slices are $y = 1$ and $y = 3$. They give $2x + z = 1003$ and $2x + z = 1000$, with 501 and 499 positive solutions. Thus $m - n = 1000$, giving 0.

**Answer:** 000.

**Methods:** injections, boundary counting.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. A residue-sensitive lattice-point count compares two restricted compositions.

### 2009 AIME II Problem 11: Exactly fifty logarithmic neighbors

Positive integers $m \geq n$ have exactly fifty positive integers $k$ satisfying $|\log m - \log k| < \log n$. Find the sum of all possible products $mn$.

Published problem and solutions · MAA AMC.

#### Solution

The inequality is $m/n < k < mn$. Exactly fifty integers occur precisely when $mn - 51 \leq m/n < mn - 50$, or $50n/(n^2-1) < m \leq 51n/(n^2-1)$; $n = 1$ gives none. Since $m \geq n$, the upper bound forces $n^2 - 1 \leq 51$, so $2 \leq n \leq 7$. For $n = 2$ the interval is $(100/3, 34]$, giving $m = 34$; for $n = 3$ it is $(150/8, 153/8]$, giving $m = 19$. For $n = 4, 5, 6, 7$ the intervals $(200/15, 204/15]$, $(250/24, 255/24]$, $(300/35, 306/35]$, and $(350/48, 357/48]$ contain no integer. The sum is $68 + 57 = 125$.

**Answer:** 125.

**Methods:** logarithmic inequalities, integer intervals.

**Verification:** Exhaustively counted integer points in every analytically bounded open interval.

**Source comparison:** The sole source gives the same exact endpoint treatment and finite bound on n.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Strict logarithmic bounds translate into a narrow interval for an integer parameter.

### 2010 AIME I Problem 1: One square divisor in a pair

Choose two distinct positive divisors of $2010^2$ uniformly. Find numerator plus denominator of the probability that exactly one is a square.

Published problem and solutions · MAA AMC.

#### Solution

Since $2010 = 2 \cdot 3 \cdot 5 \cdot 67$, there are $3^4 = 81$ divisors, of which $2^4 = 16$ have all exponents even. The favorable unordered pairs number $16 \cdot 65$, so $P = 1040/\binom{81}{2} = 26/81$. The answer is 107.

**Answer:** 107.

**Methods:** prime exponents, probability.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Divisor exponent choices count squares and nonsquares before sampling a pair.

### 2010 AIME I Problem 2: A product of repeated nines

Find $\prod_{k=1}^{999}(10^k - 1)$ modulo 1000.

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#### Solution

For $k \geq 3$, $10^k - 1 \equiv -1 \pmod{1000}$. There are 997 such factors, so the product is congruent to $9 \cdot 99 \cdot (-1)^{997} = -891 \equiv 109 \pmod{1000}$.

**Answer:** 109.

**Methods:** modular arithmetic.

**Verification:** Direct modular multiplication of all 999 factors.

**Source comparison:** Both entries reviewed. The first explicitly counts the 997 factors, supplying the sign that the abbreviated second explanation leaves implicit.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. All but the first two factors are minus one modulo one thousand.

### 2010 AIME I Problem 5: Equality in a difference-of-squares bound

Positive integers $a > b > c > d$ satisfy $a + b + c + d = 2010$ and $a^2 - b^2 + c^2 - d^2 = 2010$. How many distinct values can $a$ take?

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#### Solution

Since $a-b, c-d \geq 1$, we have $(a+b)(a-b)+(c+d)(c-d) \geq a+b+c+d$. Equality forces $a-b = c-d = 1$. Hence $a + c = 1006$, $b = a - 1$, $d = c - 1$. The remaining conditions are $c \geq 2$ and $a - 1 > c$, equivalent to $504 \leq a \leq 1004$. Every integer in this interval works, so there are 501.

**Answer:** 501.

**Methods:** equality cases, integer inequalities.

**Verification:** Complete analytic proof checked against the archived answer; no independent numerical check claimed.

**Source comparison:** Both entries reviewed. The first explicitly forces both unit gaps. The second bounds the first gap but leaves sufficiency and the second gap implicit.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Integer gaps are at least one, so equality forces both gaps to equal one.

### 2010 AIME I Problem 8: The smallest disk enclosing floor-defined squares

Let $R = \{(x, y) : \lfloor x \rfloor^2 + \lfloor y \rfloor^2 = 25\}$. The smallest closed disk containing $R$ has radius $\sqrt{m}/n$ in reduced squarefree form. Find $m + n$.

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#### Solution

The relevant integer pairs are $(\pm 5, 0), (0, \pm 5), (\pm 3, \pm 4), (\pm 4, \pm 3)$, each indexing a half-open unit square. Any closed containing disk also contains their closures. Centering a disk at $(1/2, 1/2)$, its squared distance to a vertex is at most $25 + |u| + |v| + 1/2$, maximized at $|u| + |v| = 7$. Thus radius $\sqrt{130}/2$ covers every cell. Conversely closure vertices $(5, 4)$ and $(-4, -3)$ are distance $\sqrt{130}$ apart, so every containing disk has radius at least half that. Hence $m + n = 130 + 2 = 132$.

**Answer:** 132.

**Methods:** lattice points, minimum enclosing disk, central symmetry.

**Verification:** Enumerated all closure-cell vertices, their distances from the proposed center, and all pairwise distances.

**Source comparison:** The sole source gives the correct radius but calls the half-open cells symmetric and does not explicitly prove that another center cannot improve it. Passing to the closure and using a diametral pair supplies both details.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory. Lattice solutions identify square cells, and antipodal closure points certify the enclosing radius.

## 2010 AIME I Problem 10: Digits allowed to carry

Count representations $2010 = 1000a_3 + 100a_2 + 10a_1 + a_0$ with every integer $a_i$ between 0 and 99.

Published problem and solutions · MAA AMC.

### Solution

Choose $a_3, a_1$ first with $1000a_3 + 10a_1 \le 2010$. The remainder has a unique quotient and remainder on division by 100, which supply $a_2, a_0$ within the permitted bounds. For $a_3 = 0, 1$ all 100 choices of $a_1$ work. For $a_3 = 2$ only $a_1 = 0, 1$ work. Thus the count is 202.

**Answer:** 202.

**Methods:** mixed radix, quotient remainder bijection.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer. Audit findings: Source 2: One congruence omits a division by ten, and the two exceptional ordered pairs are transposed/mistyped. Source 5: The multiple-of-ten condition belongs to a_0, not a_1. The regrouped place-value equation also omits a factor ten.

**Archived entries inspected:** 6 of 6. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Carry choices count redundant decimal representations with bounded coefficients.

## 2010 AIME I Problem 12: A multiplicative two-color threshold

Find the least $m \ge 3$ such that every two-coloring of $\{3, \dots, m\}$ has a monochromatic solution $ab = c$, allowing $a = b$.

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### Solution

At $m = 243$, color 3 red. Avoiding monochromatic products forces 9 blue, 81 red, and 27 blue (since $3 \cdot 27 = 81$). Then $243 = 3 \cdot 81 = 9 \cdot 27$ cannot have either color. For $m = 242$, color $[3, 8] \cup [81, 242]$ red and $[9, 80]$ blue. Products of two small red entries lie in $[9, 64]$; any product using a large red entry exceeds 242; products of two blue entries are at least 81 and cannot be blue. Thus 243 is sharp.

**Answer:** 243.

**Methods:** Ramsey theory, extremal coloring.

**Verification:** Analytic derivation checked against the archived answer; not independently machine-checked.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. A multiplicative two-coloring problem requires both a forcing argument and an avoiding coloring.

### 2010 AIME I Problem 14: A monotone sum of decimal digit counts

For positive integers $n$, define $f(n) = \sum_{k=1}^{100} \lfloor \log_{10}(kn) \rfloor$. Find the largest $n$ with $f(n) \leq 300$.

Published problem and solutions · MAA AMC.

#### Solution

Each summand is nondecreasing in n, so f is nondecreasing. At $n = 109$, products for $k = 1, \ldots, 9$ have three digits, those for $k = 10, \ldots, 91$ have four, and those for $k = 92, \ldots, 100$ have five. Thus $f(109) = 9 \cdot 2 + 82 \cdot 3 + 9 \cdot 4 = 300$. At $n = 110$, the middle block ends at $k = 90$, giving $f(110) = 9 \cdot 2 + 81 \cdot 3 + 10 \cdot 4 = 301$. Monotonicity excludes every larger n, so the answer is 109.

**Answer:** 109.

**Methods:** digit counts, monotonicity, floor logarithms.

**Verification:** Evaluated the 100 original floor-log terms exactly using decimal lengths at the boundary integers.

**Source comparison:** All four entries reviewed. The first incorrectly says strictly increasing, although nondecreasing is enough. Approximate averaging and broad claims about unit increments are unnecessary; two exact evaluations establish maximality.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Monotonicity makes two adjacent exact evaluations a complete maximality proof.

### 2010 AIME II Problem 1: The largest multiple with distinct even digits

Let $N$ be the largest positive multiple of 36 whose decimal digits are distinct and even. Find $N$ modulo 1000.

Published problem and solutions · MAA AMC.

#### Solution

There are only five available digits, $0, 2, 4, 6, 8$. A five-digit candidate has digit sum 20, so cannot be divisible by nine. For a four-digit candidate, deleting one digit from that set must leave a multiple of nine; only deleting two gives sum 18. The largest arrangement is 8640, whose final two digits are divisible by four, so it is divisible by 36. It exceeds all shorter candidates. Thus the remainder is 640.

**Answer:** 640.

**Methods:** digit restrictions, divisibility tests.

**Verification:** Enumerated every decimal string of distinct even digits, excluding leading zero, and directly tested divisibility by 36.

**Source comparison:** The sole source identifies the correct candidate. The explicit digit-sum argument proves that no larger four-digit arrangement or five-digit number works.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Divisibility by nine limits the digit set before lexicographic maximization.

## 2010 AIME II Problem 3: Valuation of all pair differences

Let $K = \prod_{1 \le a < b \le 20} (b - a)$. Find the largest exponent $n$ for which $2^n \mid K$.

Published problem and solutions · MAA AMC.

### Solution

A difference $d$ occurs $20-d$ times. Count each factor of two by its level: $v_2(K) = \sum_{j=1}^{4} \sum_{1 \le d \le 19,\, 2^j \mid d} (20-d)$. The four sums are $90, 40, 16, 4$, totaling 150. This avoids multiplying the large product.

**Answer:** 150.

**Methods:** valuation layers, contribution counting.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Multiplicities of pairwise differences determine a factorial-product valuation.

## 2010 AIME II Problem 6: The first reducible quartic

Find the smallest positive integer $n$ for which $x^4 - nx + 63$ factors into two nonconstant integer-coefficient polynomials.

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### Solution

Because the polynomial is monic, a factorization can be taken monic. A linear factor gives an integer root $r \mid 63$ and $n = r^3 + 63/r$. For positive n, r must be positive. The choices $r = 1, 3$ give 64 and 48, and $r \ge 7$ gives n above 343. Otherwise two monic quadratics have form $(x^2 + ax + b)(x^2 - ax + d)$. Coefficients require $bd = 63$, $b + d = a^2$, and $n = a(b - d)$. The factors b,d must be positive; their unordered pairs are $(1, 63), (3, 21), (7, 9)$. Only sums 64 and 16 are squares, yielding positive n values 496 and 8. Indeed $x^4 - 8x + 63 = (x^2 + 4x + 9)(x^2 - 4x + 7)$. Thus the minimum is 8.

**Answer:** 008.

**Methods:** integer polynomial factorization, rational root theorem.

**Verification:** Exhausted integral constant-factor pairs and integer roots in all factorization types; the coefficient bound |a|<=12 covers b+d<=126.

**Source comparison:** Both entries reviewed. The first covers both possible degree patterns but abbreviates the linear case. The second omits linear-cubic factorizations entirely. Both are addressed explicitly here.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Both linear-cubic and quadratic-quadratic factorization patterns must be checked.

### 2010 AIME II Problem 10: Quadratics modulo interchange of roots

Count quadratic polynomials with integer coefficients and integer roots whose constant coefficient is 2010.

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#### Solution

Write $f(x) = a(x-r)(x-s)$, so $ars = 2010$. Since 2010 has four distinct prime factors, assigning each prime to $|a|, |r|, |s|$ gives $3^4$ ordered magnitude triples. There are four sign assignments with positive product, hence 324 ordered signed triples. Swapping $r, s$ leaves the polynomial unchanged and fixes exactly two triples, with $r = s = 1$ or $r = s = -1$ and $a = 2010$. Burnside gives $(324 + 2)/2 = 163$.

**Answer:** 163.

**Methods:** Burnside lemma, prime allocations.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory, algebra. Integer-root quadratics are counted through signed divisor pairs and leading-coefficient choices.

### 2010 AIME II Problem 12: Two isosceles triangles with equal area and perimeter

Two noncongruent integer-sided isosceles triangles have equal areas and perimeters, and their base lengths are in ratio $8 : 7$. Find their least possible common perimeter.

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#### Solution

Write the bases as $8t, 7t$ with positive integer t. Equal perimeters make their difference t twice the difference of equal-side lengths, so t is even; put $t = 2u$. With common semiperimeter s, the side triples are $(s - 8u, s - 8u, 16u)$ and $(s - 7u, s - 7u, 14u)$, and s is integral. Heron gives equal squared areas $64u^2 s(s - 16u) = 49u^2 s(s - 14u)$. Thus $15s = 338u$. Since 15 and 338 are coprime, the least positive

scale is $u = 15$, $s = 338$. The resulting triples $(218, 218, 240)$ and $(233, 233, 210)$ satisfy the triangle inequalities, so the minimum perimeter is 676.

**Answer:** 676.

**Methods:** Heron formula, parity, primitive integer scaling.

**Verification:** Exhausted all integer-sided isosceles pairs with base ratio 8:7 and perimeter through 676, comparing exact squared areas.

**Source comparison:** Both entries reviewed, including the parity note. The main scaling calculation is correct, but integrality of the parameter in bases 16u and 14u requires the parity argument supplied here.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory, algebra. Parity and Heron reduce the primitive integral solution to a coprime scaling condition.

## 2011 AIME I Problem 5: Sliding sums force periodic residues

Label a regular nonagon by $1, \ldots, 9$, requiring every three consecutive labels to have sum divisible by 3. Identify rotations but not reflections. Count arrangements.

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### Solution

Subtract consecutive triple-sum congruences to obtain $a_{i+3} \equiv a_i \pmod 3$. Since each residue class occurs exactly three times, the three position classes modulo three must receive the three residues in a permutation. There are 3! residue patterns on labeled vertices and $(3!)^3$ assignments of actual labels. Every rotation orbit has size nine because all labels are distinct. Thus the number is $3!(3!)^3/9 = 144$.

**Answer:** 144.

**Methods:** sliding windows, periodicity, rotations.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Residue-class patterns around a nonagon and rotation classes determine admissible labelings.

## 2011 AIME I Problem 7: Splitting a power into equal smaller powers

For how many positive integers $m$ can one power $m^{x_0}$ equal a sum of exactly 2011 powers $m^{x_i}$ with all exponents nonnegative?

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**Solution**

The case $m = 1$ is impossible. For $m > 1$, reduction modulo $m - 1$ gives $1 \equiv 2011$, so $m - 1 \mid 2010$. Conversely, let $k = 2010/(m-1)$. Start with $m^k$ and repeatedly replace one positive-exponent term $m^j$ by $m$ copies of $m^{j-1}$, following a descending chain for $k$ splits. Each split adds $m - 1$ terms, ending at exactly 2011 nonnegative-exponent powers. Thus valid $m$ correspond to divisors of $2010 = 2 \cdot 3 \cdot 5 \cdot 67$, of which there are 16.

**Answer:** 016.

**Methods:** refinement, invariants, m-ary trees.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer. Audit findings: Source 3: Pattern examples suggest the splitting construction but do not establish necessity; reduction modulo m-1 is required. Source 4: Splitting proves sufficiency. Necessity still needs a merging argument or the residue invariant. Source 5: The constructed family proves sufficiency but its form cannot simply be assumed for every representation.

**Archived entries inspected:** 6 of 6. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Splitting powers and binary digit sums determine the possible lengths of power representations.

## 2011 AIME I Problem 10: Obtuse triangles in a regular polygon

Three vertices are chosen uniformly from a regular $n$-gon. Their triangle is obtuse with probability 93/125. Find the sum of all possible $n$.

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**Solution**

An obtuse triangle has a unique circular gap greater than a semicircle. The vertex after that gap is a unique anchor, and its other two vertices must be among the next $h = \lfloor (n-1)/2 \rfloor$ vertices. Thus $P = n\binom{h}{2}/\binom{n}{3}$. If $n = 2t$, this simplifies to $3(t-2)/(2(2t-1))$, giving $t = 188$ and $n = 376$. If $n = 2t-1$, it is $3(t-2)/(2(2t-3))$, giving $t = 64$ and $n = 127$. The sum is 503.

**Answer:** 503.

**Methods:** cyclic gaps, geometric probability.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer. Audit findings: Source 1: The obtuse-triangle formula has an off-by-one error and does not produce the two values subsequently claimed. Trial values also do not prove exhaustiveness.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, geometry, number_theory. Count obtuse vertex triangles of a regular polygon and solve the resulting rational equation.

## 2011 AIME I Problem 11: Summing a power-residue orbit without its order

Let $R$ be the set of distinct residues of $2^n$ modulo 1000 for integers $n \geq 0$. Find the remainder modulo 1000 of the sum of the elements of $R$.

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### Solution

Let $r_j = r_i$ be the first repetition in the sequence $r_n \equiv 2^n \pmod{1000}$, with $i < j$. Since $1000 \mid 2^i(2^{j-i} - 1)$ and the second factor is odd, $i \geq 3$. Cancellation modulo 125 gives $2^{j-i} \equiv 1 \pmod{125}$, hence $2^{j-i+3} \equiv 8 \pmod{1000}$. If $i > 3$, this repeats $r_3$ before index $j$, a contradiction. Thus $i = 3$ and $r_j = 8$. Every distinct residue occurs exactly once among $r_0, \ldots, r_{j-1}$, since the subsequent sequence repeats its cycle. Their sum is congruent to $1 + 2 + \cdots + 2^{j-1} = 2^j - 1 \equiv 7 \pmod{1000}$.

**Answer:** 007.

**Methods:** modular dynamics, geometric series, eventual periodicity.

**Verification:** Generated the full modular orbit up to its first repetition and summed every distinct residue.

**Source comparison:** All five entries were reviewed. Order computations are unnecessary. Some entries only guess that 8 is the first repeated residue or incompletely exclude smaller orders; the first-repetition argument supplies the missing justification without determining the cycle length.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. The transient and periodic parts of powers modulo 1000 control their distinct-residue sum.

## 2011 AIME I Problem 15: Bounding integer roots of a depressed cubic

The polynomial $x^3 - 2011x + m$ has three integer roots. Find the sum of their absolute values.

Published problem and solutions · MAA AMC.

### Solution

None of the roots is zero, since otherwise the other two would be $\pm\sqrt{2011}$. Their sum is zero, so after reversing all signs if necessary write them as $u, v, -s$, with $u, v > 0$ and $s = u + v$. Vieta gives $uv = s^2 - 2011$. Thus $s^2 > 2011$, while $uv \leq s^2/4$ gives $3s^2 \leq 8044$. Hence $45 \leq s \leq 51$. The discriminant of $t^2 - st + (s^2 - 2011)$ must be a square. For these seven values of $s$, it is respectively $1969, 1696, 1417, 1132, 841, 544, 241$. Only $841 = 29^2$ is square. Thus $s = 49$ and $\{u, v\} = \{10, 39\}$; these roots do satisfy the required coefficients. Their absolute values sum to $2s = 98$.

**Answer:** 098.

**Methods:** Vieta formulas, integer roots, discriminant, finite bounding.

**Verification:** Enumerated every sorted integer-root triple in the analytically proved magnitude bound 51.

**Source comparison:** All six entries were reviewed. The sources find the root triple, but several stop at its discovery without excluding all other cases; one proposed valuation discussion is unrelated and contains a year/constant typo. The explicit bound and all seven discriminants prove the required uniqueness.

**Archived entries inspected:** 6 of 6. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Vieta reduces integral roots to seven discriminant checks, proving uniqueness.

## 2011 AIME II Problem 3: Integer angles in arithmetic progression

The interior angle measures of a convex 18-gon, in increasing order, form a strictly increasing arithmetic progression of integers. Find its smallest angle in degrees.

Published problem and solutions · MAA AMC.

### Solution

Write the angles as $a, a+d, \dots, a+17d$, with positive integer $d$. Their sum is $16 \cdot 180$, so $2a+17d = 320$. Thus $d$ is even. The largest angle equals $160+17d/2 < 180$, yielding $d < 40/17$. The only positive even possibility is $d = 2$, and $a = 160 - 17 = 143$.

**Answer:** 143.

**Methods:** polygon angle sum, arithmetic progression, parity.

**Verification:** Exhaustive integer search over all convex arithmetic-progressive angle lists.

**Source comparison:** All four entries were reviewed. The angle-sum equation supplies the parity condition sometimes left implicit in median arguments. The final entry incorrectly describes every angle as 160 before listing a progression; the intended average is 160.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory, algebra. The polygon angle sum and convexity bound an even integral common difference.

## 2011 AIME II Problem 14: Permutations respecting residue partitions

Count permutations $a_1, \dots, a_{30}$ of $1, \dots, 30$ satisfying $a_{i+m} \equiv a_i \pmod{m}$ for $m = 2, 3, 5$ wherever defined. Give the residue modulo 1000.

Published problem and solutions · MAA AMC.

### Solution

For each modulus, values in any one position residue class share a value residue. Equal class sizes and bijectivity force a permutation of the residue classes. Choose these residue permutations independently in $2!, 3!, 5!$ ways. The Chinese remainder theorem then specifies a unique value modulo 30 at each position and guarantees all thirty values are distinct. Therefore $N = 2!3!5! = 1440$, giving 440.

**Answer:** 440.

**Methods:** Chinese remainder theorem, permutation products.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer. Audit findings: Source 2: The relevant CRT hypothesis is pairwise coprimality; a collective gcd of one is not sufficient in general. Source 3: The optional extension to sixty symbols is incorrect: after choosing residue permutations, each of thirty residue classes has an independent two-element permutation.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Residue-class permutations modulo 2,3,5 combine through the Chinese remainder theorem.

## 2011 AIME II Problem 15: Floor equality on three intervals

Let $P(x) = x^2 - 3x - 9$ and choose $x$ uniformly on $[5, 15]$. The probability that $\lfloor \sqrt{P(x)} \rfloor = \sqrt{P(\lfloor x \rfloor)}$ has form $(\sqrt{a} + \sqrt{b} + \sqrt{c} - d)/e$. Find $a + b + c + d + e$.

Published problem and solutions · MAA AMC.

### Solution

The right side must be integral. For integer inputs 5 through 14, $P(k)$ is square only at $k = 5, 6, 13$, with square roots $1, 3, 11$. Since $P$ is increasing, the successful intervals run from each $k$ until $P(x)$ reaches the next square $4, 16, 144$. Their upper endpoints are $(3 + \sqrt{61})/2, (3 + \sqrt{109})/2, (3 + \sqrt{621})/2$. Each upper endpoint is less than $k + 1$, so the floor of $x$ stays at $k$ throughout. Sum their lengths and divide by ten to obtain $(\sqrt{61} + \sqrt{109} + \sqrt{621} - 39)/20$. The requested sum is 850.

**Answer:** 850.

**Methods:** continuous probability, floor stratification.

**Verification:** Analytic derivation checked against the archived answer; not independently machine-checked.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer. Audit findings: Source 1: The substitution mishandles the floor: floor(2x-3) is not 2 floor(x)-3. The subsequent intervals actually use the latter expression, so the written transformed equation is invalid. Source 3: The answer format is used to assert interval bounds without proof. The bounds should be checked directly.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, algebra, number_theory. Continuous probability reduces to integer square values of a polynomial on floor cells.

## 2012 AIME I Problem 1: Divisibility in both digit directions

Count three-digit integers $abc$ with $a, c \neq 0$ such that both $abc$ and $cba$ are divisible by four.

Published problem and solutions · MAA AMC.

**Solution**

The conditions are $2b + c \equiv 0 \pmod 4$ and $2b + a \equiv 0 \pmod 4$. For even $b$, each of $a, c$ is 4 or 8; for odd $b$, each is 2 or 6. Every one of the ten choices for $b$ therefore permits four pairs $(a, c)$, giving 40.

**Answer:** 040.

**Methods:** digit counting, residues.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer. Audit findings: Source 2: The initial even-middle-digit factor is mistyped as four rather than five.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Digit choices are counted under two divisibility conditions.

## 2012 AIME I Problem 4: A shared horse and the first integer meeting point

Two travelers start together along a road with mileposts at every positive integer distance. Their walking speeds are 4 and 5/2 mph; their shared horse carries either at 6 mph. The slower walker rides first to post one and leaves the horse. Thereafter each rider passes the other traveler, leaves the horse at the next post, and walks on. At their first meeting at a milepost, the distance is $n$ miles and elapsed time is $t$ minutes. Find $n + t$.

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**Solution**

Call the faster walker $B$ and the slower walker $S$. If both meet at post $n$ and $B$ has ridden a total of $a$ miles, the horse has traveled $n$ miles and $S$ has ridden $n - a$. All riding segments end at integer posts, including the current segment at the meeting, so $a$ is integral. Their times in hours are

$$T_B = \frac{a}{6} + \frac{n-a}{4}, \qquad T_S = \frac{n-a}{6} + \frac{2a}{5}.$$

Equality gives $5n = 19a$, so any positive milepost meeting is at least nineteen miles away.

It remains to prove that the specified handoff rule actually reaches this candidate. At a handoff post $n$, let $a$ be $B$'s accumulated riding distance. If $B$ next rides, its catch-up position is $u = 19(n-a)/14$; if $S$ next rides, it is $u = 19a/5$. These follow by using the same travel-time formulas at a variable position $u$: in the first case $B$'s riding distance becomes $a + u - n$, and in the second it stays $a$. For a nonintegral catch position, the horse is left at $\lfloor u \rfloor + 1$.

Starting after the first ride, the successive handoff states $(n, a)$ are

$$(1,0), (2,1), (4,1), (5,2), (8,2), (9,3), (12,3), (13,4), (16,4), (17,5).$$

The corresponding next catch positions are

$$\frac{19}{14}, \frac{19}{5}, \frac{57}{14}, \frac{38}{5}, \frac{57}{7}, \frac{57}{5}, \frac{171}{14}, \frac{76}{5}, \frac{114}{7}, 19.$$

Each noninteger value produces precisely the next listed post, so this is a valid execution of the rule. The final catch is at post nineteen. Its elapsed time is $5/6 + 14/4 = 13/3$ hours, or 260 minutes. Thus $n + t = 279$.

**Answer:** 279.

**Methods:** travel-time invariants, integer meeting points, discrete recurrence.

**Verification:** Exact rational simulation of the handoff recurrence reaches post 19 with five miles ridden by the faster walker.

**Source comparison:** The sole stored solution is explicitly marked nonrigorous because it does not establish the candidate handoff pattern. The new proof supplies the exact catch-position recurrence and every state through the first integer meeting.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Travel-time invariants give a divisibility obstruction, and a discrete handoff recurrence proves attainability.

## 2012 AIME I Problem 5: Consecutive binary numbers with equal weight

Let $B$ contain all thirteen-bit strings with eight ones and five zeros, allowing leading zeros. Count ordered pairs $(u, v) \in B^2$ with numeric difference $u - v = 1$.

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### Solution

If $v$ ends in $t$ ones, adding one changes its one-count by $1 - t$. Equal one-counts therefore force $t = 1$, so $v$ ends in 01 and $u$ in 10. Their shared eleven-bit prefix must contain seven ones. This condition is sufficient and unique, yielding $\binom{11}{7} = 330$.

**Answer:** 330.

**Methods:** binary carries, bijections.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Binary carrying determines when consecutive integers preserve their digit weight, followed by subset counting.

## 2012 AIME I Problem 6: A positive imaginary part selects a root-of-unity denominator

Complex numbers satisfy $z^{13} = w$ and $w^{11} = z$. Suppose $\operatorname{Im} z = \sin(m\pi/n)$ for coprime positive integers $m < n$. Determine $n$.

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**Solution**

The specified sine is positive, so $z \neq 0$. Substitution gives $z^{143} = z$, hence $z^{142} = 1$. Its positive-imaginary roots are $z = e^{ik\pi/71}$ for $1 \leq k \leq 70$. Two angles in $(0, \pi)$ have the same sine exactly when they are equal or supplementary. Therefore $m/n = k/71$ or $(71 - k)/71$. Since 71 is prime and both numerators lie between one and seventy, either fraction has reduced denominator 71. Thus $n = 71$.

**Answer:** 071.

**Methods:** roots of unity, sine symmetry, reduced fractions.

**Verification:** The denominators of k/71 and (71−k)/71 are seventy-one for all 1≤k≤70.

**Source comparison:** The sole stored proof inspected. The new exposition adds the equal-or-supplementary sine alternative, so the given rational angle is not assumed to equal the root argument uniquely.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. A power cycle gives roots of unity, and sine symmetry fixes the reduced rational angle denominator.

## 2012 AIME I Problem 10: Order the squares with a prescribed three-digit suffix

For every perfect square whose last three decimal digits are 256, delete those digits, allowing the result zero from the square 256. Arrange the distinct resulting nonnegative integers in increasing order. Find the tenth one modulo 1000.

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**Solution**

Use the positive square root $u$. The suffix condition is $u^2 \equiv 256 \pmod{1000}$. Modulo eight it says $u \equiv 0 \pmod 4$. Modulo 125, the factors $(u-16)(u+16)$ have gcd dividing 32, so only one can contain factors of five; consequently $u \equiv \pm 16 \pmod{125}$. Combining with divisibility by four gives precisely

$$u \equiv 16 \text{ or } -16 \pmod{500}.$$

The square and then the truncated result increase strictly with the positive root. The root sequence begins

$$16, 484, 516, 984, 1016, 1484, 1516, 1984, 2016, 2484.$$

Thus the tenth result is $(2484^2 - 256)/1000 = 6170$, whose remainder is 170.

**Answer:** 170.

**Methods:** prime-power congruences, Chinese remainder theorem, ordered residue classes.

**Verification:** Enumeration of all roots 1≤u≤2484 confirms the ten listed roots and the truncated value 6170.

**Source comparison:** All five stored entries inspected. They use factorization, digit arithmetic, CRT, or a reparameterization. The final entry incorrectly states that negative parameter values give negative outputs; the root-based ordering avoids this duplicate-sign issue. The first square 256 and its truncated value zero are included explicitly.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Prime-power congruences characterize every square root, then ordered residue classes give the tenth value.

## 2012 AIME I Problem 11: Reachability in a lattice semigroup

From $(0,0)$ allow jumps $(7,2),(2,7),(-5,-10),(-10,-5)$. Let $M$ count reachable lattice points with $|x|+|y|\le 100$. Find $M$ mod 1000.

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### Solution

Necessarily $x+y=3u$, $x-y=5v$ for integers $u,v$ of the same parity. Conversely the jumps in $(u,v)$ are $(3,\pm1),(-5,\pm1)$. Choose large nonnegative $C\equiv u \pmod 3$ and put $A=(u+5C)/3$. Then $3A-5C=u$, while $L=A+C\ge|v|$ and $L\equiv u\equiv v \pmod 2$. Among these $L$ jumps choose $(L+v)/2$ plus signs, obtaining the target with nonnegative counts. Hence the invariants are sufficient. Now $|x|+|y|=\max(|3u|,|5v|)$, so $-33\le u\le 33$, $-20\le v\le 20$. Equal parities give $33\cdot 21+34\cdot 20=1373$ points, hence 373.

**Answer:** 373.

**Methods:** lattice invariants, reachability, parity.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** The first archived sufficiency argument does not clearly enforce nonnegative jump counts. Taking a sufficiently large solution and assigning signs here proves actual reachability.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory, geometry. A lattice reachability semigroup is characterized by congruences, and a coordinate transformation turns the diamond region into a rectangle for counting.

## 2012 AIME I Problem 15: Multiplication changes every circular separation

Reseat $n$ people on a numbered circle by $k\mapsto ak \pmod n$. Require this to be a permutation and require every pair's circular separation to change even up to reversal. Count possible $1<n<1000$.

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### Solution

Permutation requires $\gcd(a,n)=1$. A nonzero difference $d$ must never satisfy $ad\equiv d$ or $ad\equiv -d \pmod n$, equivalent to $\gcd(a-1,n)=\gcd(a+1,n)=1$. Thus three consecutive integers must all be coprime to $n$, impossible if $2\mid n$ or $3\mid n$. Conversely if $\gcd(n,6)=1$, choose $a=2$, and all three gcds are one. Inclusion-exclusion on $1,\dots,999$ gives $999-499-333+166=333$ values, of which $n=1$ is excluded. The result is 332.

**Answer:** 332.

**Methods:** cyclic permutations, modular invariants.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Circular permutations are characterized by three coprimality conditions and counted by inclusion-exclusion.

## 2012 AIME II Problem 1: Count lattice points on a positive Diophantine segment

Count ordered positive integer pairs $(m, n)$ satisfying $20m + 12n = 2012$.

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### Solution

Divide by four: $5m + 3n = 503$. Modulo three this forces $m \equiv 1 \pmod 3$, and positivity of $n$ forces $m \leq 100$. Conversely every $m = 1 + 3j$ for $0 \leq j \leq 33$ gives $n = (503 - 5m)/3$, a positive integer. This is a bijection with thirty-four integers $j$, so there are 34 ordered pairs.

**Answer:** 034.

**Methods:** linear Diophantine equations, lattice-point counting, congruences.

**Verification:** Direct enumeration of positive pairs gives thirty-four.

**Source comparison:** All four stored alternatives inspected. They use either congruences or a lattice slope. The last initially states a false divisibility-by-twenty criterion before adding the missing tens-digit condition; the congruence proof here is exact from the outset.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Congruence parameterization and endpoint counting give the number of ordered positive lattice points.

## 2012 AIME II Problem 12: Independent forbidden residues

A positive integer is $p$-safe if its distance from every multiple of $p$ exceeds two. Count integers at most 10000 that are simultaneously 7-, 11-, and 13-safe.

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### Solution

The allowed residues modulo $p$ are $3, \ldots, p-3$, a total of $p-5$. By the Chinese remainder theorem there are $2 \cdot 6 \cdot 8 = 96$ safe residues per block of 1001. Thus $1, \ldots, 10010$ contains 960 safe numbers. Among the last ten integers only $10006, 10007$ qualify (their residues are respectively $-4, -3$ modulo all three moduli). Subtract these to obtain 958.

**Answer:** 958.

**Methods:** Chinese remainder theorem, residue counting.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Independent allowed residues are combined by CRT and counted with a boundary correction.

### 2013 AIME I Problem 2: Uniform digit sums modulo five

Count five-digit integers divisible by five whose first and last digits agree and whose digit sum is divisible by five.

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#### Solution

The first and last digits must both be 5, so write the integer $5abc5$. Choose $a, b$ arbitrarily in 100 ways. Exactly two choices of $c \in \{0, \dots, 9\}$ give $a + b + c \equiv 0 \pmod 5$. The count is 200.

**Answer:** 200.

**Methods:** residue uniformity, digit counting.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Digit counting uses uniform residue classes modulo five.

### 2013 AIME I Problem 8: A function domain imposes an integer divisor

For positive integers $m > 1, n$, the real domain of $\arcsin(\log_m(nx))$ has length $1/2013$. Find the least possible $m + n$, reduced modulo 1000.

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#### Solution

The logarithm must lie in $[-1, 1]$, so the domain is $[1/(mn), m/n]$. Its length condition gives

$$n = 2013m - \frac{2013}{m}.$$

Thus $m$ must divide $2013 = 3 \cdot 11 \cdot 61$. Conversely every divisor $m > 1$ gives a positive integer $n$ and the required domain. The target $m + n = 2014m - 2013/m$ strictly increases for positive $m$, because its

first term increases and its subtracted term decreases. The smallest allowed divisor is $m = 3$, yielding $n = 5368$ and $m + n = 5371$. The remainder is 371.

**Answer:** 371.

**Methods:** function domains, divisors, monotonicity.

**Verification:** Enumerating divisors of 2013 confirms the least admissible pair is (3,5368).

**Source comparison:** All three stored entries inspected. The divisor route is exact; the first entry uses an informal near-linearity argument, replaced here by strict monotonicity. The last begins with a mistaken bound on f rather than its arcsine input and uses a longer factor allocation.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Inverse-trigonometric domain constraints reduce the minimum to a divisor and monotonicity argument.

## 2013 AIME I Problem 10: Conjugate integer roots and distinct cubics

Consider all cubics $x^3 - ax^2 + bx - 65$ with nonzero integers $a, b$ that have a root $r + si$ for nonzero integers $r, s$. For each distinct coefficient pair $(a, b)$, take the sum of its three roots. Find the sum of these values over all admissible pairs.

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### Solution

The conjugate $r - si$ is also a root. The third root is $q = a - 2r$, an integer, and coefficient comparison gives

$$q(r^2 + s^2) = 65, \qquad a = q + 2r, \qquad b = r^2 + s^2 + 2qr.$$

Since $r, s \neq 0$, $q$ is a positive divisor of 65 other than 65. For $q = 1, 5, 13$, the possible $r$ are respectively

$$\{\pm 1, \pm 4, \pm 7, \pm 8\}, \quad \{\pm 2, \pm 3\}, \quad \{\pm 1, \pm 2\}.$$

These lists follow by subtracting positive squares from $65, 13, 5$; only square complements are retained. For each listed $r$, the value of $s^2$ is fixed, and its two signs produce the same polynomial. Every listed $a$ is odd and nonzero; the formulas give nonzero $b$ as well. Distinct choices cannot duplicate a polynomial: its unique real root fixes $q$, and then $a$ fixes $r$.

By Vieta each root sum is $a = q + 2r$. Pair $r$ with $-r$ in each list. The signed terms cancel, leaving $8 \cdot 1 + 4 \cdot 5 + 4 \cdot 13 = 80$.

**Answer:** 080.

**Methods:** complex conjugates, Vieta formulas, sum of two squares, finite classification, involution.

**Verification:** Exact integer enumeration finds sixteen distinct (a,b) pairs and sums a to eighty.

**Source comparison:** Inspected the sole stored argument. The new exposition adds why the third root is integral, excludes q=65, verifies the nonzero coefficient restrictions, and proves that the coefficient-pair count has no duplicates.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory, combinatorics. Distinct coefficient pairs require a finite classification modulo conjugation and a signed-pair cancellation.

## 2013 AIME I Problem 11: Three remainder conditions after an lcm reduction

A positive integer $N$ is divisible by $14, 15, 16$. There exist three distinct positive integers below 14 for which dividing $N$ leaves remainder 3. Find the sum of the distinct prime divisors of the least such $N$.

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### Solution

Write $N = 1680k$, since $\operatorname{lcm}(14, 15, 16) = 1680$. Every positive integer below fourteen except $9, 11, 13$ divides 1680, so none of those other integers can leave remainder three. Thus the three divisors must be $9, 11, 13$. The conditions reduce to

$$k \equiv 2 \pmod 3, \qquad k \equiv 10 \pmod{11}, \qquad k \equiv 1 \pmod{13}.$$

The first follows by dividing $6k \equiv 3 \pmod 9$ by three and reducing the modulus to three. Combine the first two to get $k = 32 + 33t$. Modulo thirteen this gives $6 + 7t \equiv 1$, hence $t \equiv 3 \pmod{13}$. Therefore $k \equiv 131 \pmod{429}$, whose least positive representative is 131.

The number 131 has no prime divisor at most $\sqrt{131} < 12$ (test $2, 3, 5, 7, 11$), so it is prime. The distinct prime divisors of $N = 1680 \cdot 131$ are $2, 3, 5, 7, 131$, with sum 148.

**Answer:** 148.

**Methods:** least common multiple, Chinese remainder theorem, congruence cancellation.

**Verification:** Direct search over $1 \le k \le 429$ verifies that only $k=131$ meets all three reduced congruences.

**Source comparison:** All six stored entries inspected. Several reach the correct minimum through different substitutions. The final entry incorrectly keeps modulus nine after division by three; its claimed full congruence class is false although the least candidate happens to agree. Another entry incorrectly excludes all noncoprime divisors. The present proof excludes actual divisors of 1680 and uses the correct reduced modulus.

**Archived entries inspected:** 6 of 6. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. The least integer is determined by divisibility, legal congruence cancellation, and CRT.

## 2013 AIME I Problem 15: Two arithmetic progressions with reordered residues

Count $0 \le A < B < C \le 99$ forming an arithmetic progression for which a prime $p$ and residues $0 \le b < a < c < p$ satisfy $A \equiv a$, $B \equiv b$, $C \equiv c \pmod p$, with $(b, a, c)$ also an arithmetic progression.

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### Solution

Write $(A, B, C) = (B - D, B, B + D)$ and $(b, a, c) = (a - d, a, a + d)$. Subtracting the congruences gives $D \equiv -d \equiv 2d \pmod p$, hence $p \mid 3d$. Since $0 < d < p$, $p = 3$, and the residues are $(a, b, c) = (1, 0, 2)$. Thus $B = 3k$ and $D \equiv 2 \pmod 3$, with $D \le \min(B, 99 - B)$. For $k = 1, \ldots, 16$ there are $k$ choices of

$D$; for $k = 17, \ldots, 32$ there are $33 - k$. These conditions are also sufficient using $p = 3$. The count is $2(1 + \cdots + 16) = 272$.

**Answer:** 272.

**Methods:** residue constraints, arithmetic progressions, case counting.

**Verification:** Independent exact computation passed in tools/verify_middle.py; matches archived answer.

**Source comparison:** Reviewed all available source arguments. The solution here gives an independently written proof, checked against the archived answer. Audit findings: Source 1: The listed ranges and counts for the progression difference disagree, and the final arithmetic sum is not 272. The middle-term count in the original solution avoids these errors.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Congruences force a prime and residue pattern, then bounded arithmetic progressions are counted.

## 2013 AIME II Problem 2: Nested logarithms impose a power-of-two divisor

For positive integers $a, b$, suppose $\log_2(\log_{2^a}(\log_{2^b}(2^{1000}))) = 0$. Sum the distinct possible values of $a+b$.

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### Solution

Undo the logarithms from outside inward. The outer argument must be one, its inner argument must be $2^a$, and therefore $1000/b = 2^a$. Since $1000 = 2^3 5^3$ and $a > 0$, the possible pairs are exactly $(1, 500)$, $(2, 250), (3, 125)$. All logarithms are defined for these pairs. Their sums $501, 252, 128$ are distinct and total 881.

**Answer:** 881.

**Methods:** logarithms, prime valuations.

**Verification:** The three possibilities are exhaustive by the valuation $\mathrm{v}_2(1000)$=3.

**Source comparison:** Both stored alternatives inspected. They differ only in the presentation of exponentiation and factorization. The new proof also checks domains and that summing pairs does not duplicate a value of a+b.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Undoing nested logarithms produces a prime-exponent restriction on integers.

## 2013 AIME II Problem 6: The first complete thousand-block between squares

Find the least positive integer $N$ for which none of $1000N, 1000N + 1, \ldots, 1000N + 999$ is a perfect square.

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**Solution**

Such a block lies strictly between consecutive squares $m^2, (m+1)^2$, so

$$m^2 + 1 \le 1000N \le (m+1)^2 - 1000.$$

The gap must be at least 1001, forcing $m \ge 500$. Write $m = 500 + k$, $k \ge 0$. The inequalities become

$$k + 250 + \frac{k^2+1}{1000} \le N \le k + 250 + \frac{(k+1)^2}{1000}.$$

For $0 \le k \le 30$, both bounds place $N$ strictly between the consecutive integers $k + 250$ and $k + 251$, impossible. For $k = 31$, the bounds permit $N = 282$: indeed $531^2 = 281961 < 282000$ and $532^2 = 283024 > 282999$. For $k \ge 32$, the lower bound gives $N > 282$, so no smaller solution is possible. Therefore the answer is 282.

**Answer:** 282.

**Methods:** consecutive squares, integer intervals, minimality.

**Verification:** Direct integer-square-root checks for every 1≤N≤282 confirm the first admissible N is 282.

**Source comparison:** All four stored entries inspected. The first three use inspection or a partial gap heuristic. The fourth supplies an endpoint argument; the new proof gives the complete finite exclusion and separates the gap size from its alignment with a thousand-block.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Consecutive-square gaps and strict endpoint inequalities certify a least integer.

## 2013 AIME II Problem 7: Staff reductions force a unique feasible integer solution

Each clerk sorts 30 files per hour. Starting with $x$ clerks, the same positive number $t$ leaves after each of the first three hours. The remaining group finishes 1775 files at time 3 hours 10 minutes. How many files were sorted during the first 90 minutes?

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**Solution**

The total work equation is

$$30x + 30(x-t) + 30(x-2t) + 5(x-3t) = 1775,$$

so $19x - 21t = 355$. Modulo nineteen this forces $t \equiv 3 \pmod{19}$. Hence $t = 3 + 19k$ and $x = 22 + 21k$, where positivity of $t$ requires $k \ge 0$. The final workforce must be positive because work continues for the last ten minutes; thus $x - 3t = 13 - 36k > 0$, forcing $k = 0$. This proves uniqueness rather than just selecting the least congruence solution. The first ninety minutes produce $30x + 15(x-t) = 660 + 285 = 945$ files.

**Answer:** 945.

**Methods:** rates, linear Diophantine equations, feasibility.

**Verification:** Exact enumeration of positive x,t with x>3t and 19x−21t=355 finds only (22,3).

**Source comparison:** Both stored solutions inspected. Each selects the smallest positive congruence solution without excluding larger ones. The new proof uses the positive workforce in the final interval to establish that the candidate is the only feasible solution.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. A rate equation and a linear congruence must be combined with a positive final workforce.

### 2013 AIME II Problem 14: Maximize a remainder by separating quotient ranges

For an integer $n > 1$, let $F(n)$ be the largest remainder obtained when $n$ is divided by an integer $k$ with $1 \le k \le n/2$. Find $\sum_{n=20}^{100} F(n)$ modulo 1000.

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#### Solution

Write $n = 3m + r$, where $0 \le r \le 2$; in the required range $m \ge 6$. If $k \ge m + 1$ is admissible, its quotient is two, so its remainder is at most $n - 2(m + 1) = m + r - 2$, with equality at $k = m + 1$. For $k \le m - 1$, the remainder is at most $k - 1 \le m - 2$, no larger. The remaining case $k = m$ has remainder $r \le m + r - 2$. Thus

$$F(n) = m + r - 2.$$

For $j = 0, \dots, 26$, the three inputs $20 + 3j, 21 + 3j, 22 + 3j$ contribute $(6 + j) + (5 + j) + (6 + j) = 17 + 3j$. Hence the total is

$$27 \cdot 17 + 3\frac{26 \cdot 27}{2} = 1512,$$

and its remainder modulo 1000 is 512.

**Answer:** 512.

**Methods:** remainders, quotient ranges, arithmetic series.

**Verification:** Brute-force maxima over every admissible k for all 20≤n≤100 agree with the formula and total 1512.

**Source comparison:** Both stored entries inspected. The first combines a pattern guess and a partial proof; the second asserts the same lemma and refers back. The new proof handles every divisor range explicitly and then sums complete residue blocks.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A universal bound on remainders proves a residue-class formula before summation.

### 2014 AIME I Problem 3: Reduced fractions with fixed numerator-plus-denominator

Count reduced fractions strictly between zero and one whose positive numerator and denominator sum to 1000.

Published problem and solutions · MAA AMC.

**Solution**

Write the fraction as $a/(1000-a)$ with $1 \le a \le 499$. It is reduced exactly when $\gcd(a, 1000) = 1$, so $a$ must be divisible by neither 2 nor 5. Inclusion-exclusion gives $499 - 249 - 99 + 49 = 200$. Equivalently, complementary residues pair the $\varphi(1000) = 400$ coprime values into reciprocal fractions.

**Answer:** 200.

**Methods:** Euclidean algorithm, totient, inclusion-exclusion.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All eight sources inspected. Totient pairing and exclusion of multiples of 2 and 5 agree. Some alternatives reverse divisibility notation or say a divides 1000 rather than shares a prime factor; the gcd equivalence is the correct criterion.

**Archived entries inspected:** 8 of 8. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, combinatorics. Inclusion-exclusion counts admissible coprime numerators; complementary residues also pair the possibilities. The finite coprime-set count is the objective.

## 2014 AIME I Problem 6: Two parabolas with a shared axis

The parabolas $y = 3(x-h)^2 + j$ and $y = 2(x-h)^2 + k$ have vertical intercepts 2013 and 2014. Each has two positive integer horizontal intercepts. Find $h$.

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**Solution**

The first pair of roots has product $2013/3 = 671 = 11 \cdot 61$ and sum $2h$. Its possible unordered pairs are $(1, 671)$ and $(11, 61)$, with sums $672, 72$. The second pair has product $2014/2 = 1007 = 19 \cdot 53$, giving pairs $(1, 1007)$ and $(19, 53)$ with sums $1008, 72$. The common sum must be 72, so $h = 36$; both indicated factor pairs verify existence.

**Answer:** 036.

**Methods:** Vieta formulas, divisor pairs.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All six sources inspected. Several omit the factor pair containing 1 or use an incorrect quadratic-formula denominator. Listing all positive factor pairs makes uniqueness immediate without assuming the answer is an integer.

**Archived entries inspected:** 6 of 6. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Two parabolas with a shared axis; Vieta formulas, divisor pairs supply the essential reduction.

## 2014 AIME I Problem 8: A four-digit idempotent

A positive integer and its square end in the same four digits, whose thousands digit is nonzero. Find the first three digits of that four-digit ending.

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**Solution**

If the ending is $r$, then $r(r-1) \equiv 0 \pmod{10000}$. Consecutive integers are coprime, so $r$ is independently 0 or 1 modulo 16 and modulo 625. The four Chinese-remainder solutions are $0, 1, 625, 9376$: the mixed cases follow from $625 \equiv 1 \pmod{16}$ and $15 \cdot 625 + 1 = 9376$. Only 9376 has a nonzero thousands digit. Thus the requested number is 937.

**Answer:** 937.

**Methods:** Chinese remainder theorem, idempotents.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All four sources inspected. The CRT alternative explicitly includes all four residues. Digit-lifting alternatives contain copied digit slips, including saying c=3 before writing 76 and a=3 before 9376; these are avoided.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A four-digit idempotent; Chinese remainder theorem, idempotents supply the essential reduction.

## 2014 AIME II Problem 4: Compatible repeating blocks

Digits $a, b, c$ satisfy $0.\overline{ab} + 0.\overline{abc} = 33/37$. Find the three-digit integer $100a + 10b + c$.

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**Solution**

Put $u = 10a + b$. Then $u/99 + (10u + c)/999 = 33/37$, which simplifies to $221u + 11c = 9801$. Reducing modulo 11 gives $u \equiv 0 \pmod{11}$, so $a = b$. Substitution yields $221a + c = 891$. Since $0 \le c \le 9$, necessarily $a = 4, c = 7$. The requested number is 447.

**Answer:** 447.

**Methods:** repeating decimals, denominator cancellation, digit bounds.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All four sources inspected. The denominator factor 11 forces equal first two digits; this avoids carrying through six-digit repeating blocks.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Compatible repeating blocks; repeating decimals, denominator cancellation, digit bounds supply the essential reduction.

## 2014 AIME II Problem 15: Prime multiplication as binary carrying

For positive $k$, let $p(k)$ be the smallest prime not dividing $k$, and let $X(k)$ be the product of smaller primes, with empty product 1. Define $x_0 = 1$ and $x_{n+1} = x_n p(x_n)/X(x_n)$. Find the first $t > 0$ with $x_t = 2090$.

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**Solution**

Every term is squarefree. Encode it by a binary integer whose bit $j-1$ records whether the $j$th prime divides it. The smallest missing prime identifies the first zero bit; dividing by all smaller primes clears exactly the preceding consecutive one bits, and multiplying by that prime sets the zero bit. This is precisely binary addition of 1. Starting from $x_0 = 1$, the encoded integer is therefore $n$. Since $2090 = 2 \cdot 5 \cdot 11 \cdot 19$ uses prime indices 1,3,5,8, its index is $2^0 + 2^2 + 2^4 + 2^7 = 149$. The encoding is bijective, proving this is the first occurrence.

**Answer:** 149.

**Methods:** binary carrying, squarefree integers, conjugate dynamics.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All three sources inspected. The binary encoding and its induction version explain the recurrence; the brute-force suggestion alone does not prove first occurrence.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, combinatorics. The prime-multiplication process is conjugate to a binary counter; a bijection to bit states proves the first occurrence index.

## 2015 AIME I Problem 3: A prime concealed in a cube

Find the prime $p$ such that $16p + 1$ is a positive perfect cube.

Published problem and solutions · MAA AMC.

**Solution**

Write $16p+1 = u^3$. Then $u$ is odd and $(u-1)(u^2+u+1) = 16p$. The second factor is odd and exceeds one, so it must equal $p$ and the first must equal 16. Thus $u = 17$ and $p = 17^2 + 17 + 1 = 307$. None of the primes $2, 3, 5, 7, 11, 13, 17$ divides 307, verifying primality and existence.

**Answer:** 307.

**Methods:** difference of cubes, prime factorization.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All ten alternatives inspected. Factorization proves uniqueness without an AIME answer bound or guessing cubes. One modular alternative incorrectly states the cube of 15 is 1 modulo 16; it is 15.

**Archived entries inspected:** 10 of 10. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A prime concealed in a cube; difference of cubes, prime factorization supply the essential reduction.

## 2015 AIME I Problem 8: Digit sums determine the carries

Find the least positive integer $n$ such that both $n$ and $n + 864$ have digit sum 20.

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### Solution

The value 695 works, since $695 + 864 = 1559$. It remains to rule out smaller values, which must be three-digit numbers $100a + 10b + c$ with $a + b + c = 20$. Adding 864 changes the sum from $20 + 18$ to 20, so exactly two carries occur. Since $a \geq 2$, the hundreds column carries. If the units column does not carry, then $c \leq 5$, so $a \geq 6$; for $a = 6$, the least choice is $b = 9, c = 5$, giving 695. If the units column carries and the tens column does not, then $c \geq 6, b \leq 2$, forcing $a \geq 9$. Thus no smaller number works.

**Answer:** 695.

**Methods:** digit sums, carry counting, lexicographic minimization.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All eight alternatives inspected. Many search by trial or invoke the answer range. Counting the two carries proves minimality independently of those assumptions.

**Archived entries inspected:** 8 of 8. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Digit sums determine the carries; digit sums, carry counting, lexicographic minimization supply the essential reduction.

## 2015 AIME I Problem 11: Integral isosceles triangles with a fixed incenter distance

An isosceles triangle has $AB = AC$, positive integer side lengths, and incenter $I$ satisfying $BI = 8$. Find its minimum perimeter.

Published problem and solutions · MAA AMC.

### Solution

Write $AB = AC = a$ and $BC = b$. If $\theta = \angle IBC$, the projection of $BI$ onto $BC$ gives $\cos\theta = b/16$, while $\cos 2\theta = b/(2a)$. The double-angle identity therefore gives $a = 64b/(b^2 - 128)$. Positivity requires $b^2 > 128$, and the triangle inequality $2a > b$ gives $b^2 < 256$. Thus $b = 12, 13, 14, 15$. The corresponding $a$ values are $48, 832/41, 224/17, 960/97$; only 48 is integral. The unique possible perimeter, and hence the minimum, is $2(48) + 12 = 108$.

**Answer:** 108.

**Methods:** incenter projection, double angles, integer bounds.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All five alternatives inspected. They derive equivalent rational formulas by half-angle identities, bisector lengths, or tangency. The four-case integer check establishes uniqueness as well as minimality.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory. Integral isosceles triangles with a fixed incenter distance; incenter projection, double angles, integer bounds supply the essential reduction.

## 2015 AIME II Problem 1: Two percentage relations force a least common multiple

A positive integer $N$ is 22 percent less than one integer and 16 percent greater than another. For the smallest such $N$, find its remainder modulo 1000.

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### Solution

The two comparison integers are $50N/39$ and $25N/29$. Since the fractions are reduced, they are integers exactly when $39 \mid N$ and $29 \mid N$. Thus the least positive value is $\operatorname{lcm}(39, 29) = 1131$, whose remainder is 131.

**Answer:** 131.

**Methods:** reduced denominators, least common multiples.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All three alternatives inspected. The denominator criterion is shortest. The third source initially writes 1.22 instead of 1.16 and 122 instead of 116 before returning to the correct numbers.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Two percentage relations force a least common multiple; reduced denominators, least common multiples supply the essential reduction.

## 2015 AIME II Problem 3: A multiple with a prescribed digit sum

Find the least positive multiple of 17 whose decimal digits sum to 17.

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### Solution

Write the number as $17k$. A decimal integer equals its digit sum modulo 9, so $17k \equiv 17 \pmod 9$ and $k \equiv 1 \pmod 9$. In increasing order, the first candidates are $17, 170, 323, 476$, with digit sums $8, 8, 8, 17$. Thus 476 is the first valid number.

**Answer:** 476.

**Methods:** digit sums modulo nine, arithmetic progressions.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All five alternatives inspected. Modulo 9 reduces the search to four candidates. The add 102 shortcut does not exclude intervening answers, as its own comment notes; the digit-case solution unnecessarily assumes the AIME answer bound.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A multiple with a prescribed digit sum; digit sums modulo nine, arithmetic progressions supply the essential reduction.

### 2015 AIME II Problem 6: Two possible cubics with positive integer roots

The polynomial $2x^3 - 2ax^2 + (a^2 - 81)x - c$ has three positive integer roots, counted with multiplicity. Knowing $a$ still leaves exactly two possible values of $c$. Find their sum.

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#### Solution

For roots $u \le v \le w$, Vieta gives $u + v + w = a$ and $2(uv + uw + vw) = a^2 - 81$. Subtraction yields $u^2 + v^2 + w^2 = 81$. Here $u \le 5$; for each $u = 1, 2, 3, 4, 5$, testing $u \le v \le \sqrt{(81 - u^2)/2}$ gives respectively $(v, w) = (4, 8)$, none, $(6, 6)$, $(4, 7)$, none. Thus the triples are $(1, 4, 8), (3, 6, 6), (4, 4, 7)$, with sums 13,15,15. The ambiguity forces $a = 15$, and $c = 2uvw$ is 216 or 224. Their sum is 440.

**Answer:** 440.

**Methods:** Vieta formulas, sums of three squares, bounded enumeration.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All three alternatives inspected. The sum-of-squares reduction is complete. The derivative alternative assumes the stationary point is an integer without justification and is not used.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Two possible cubics with positive integer roots; Vieta formulas, sums of three squares, bounded enumeration supply the essential reduction.

### 2015 AIME II Problem 8: An integer inequality bounds a cubic quotient

Positive integers $a, b$ satisfy $(ab + 1)/(a + b) < 3/2$. If the largest possible value of $(a^3b^3 + 1)/(a^3 + b^3)$ is $p/q$ in lowest terms, find $p + q$.

Published problem and solutions · MAA AMC.

#### Solution

Assume $a \le b$. The given inequality is equivalent to $(2a - 3)(2b - 3) < 5$. If $a = 1$, the objective equals 1. If $a \ge 2$, both factors are positive odd integers, so the only unordered possibilities are $(a, b) = (2, 2)$, $(2, 3)$. Their objective values are $65/16$ and $217/35 = 31/5$. The latter is greatest, giving 36.

**Answer:** 036.

**Methods:** shifted product bounds, finite integer optimization.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All four alternatives inspected. The factored inequality gives an exhaustive classification. The last two alternatives rely on unproved optimization claims or guesses; they are unnecessary.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. An integer inequality bounds a cubic quotient; shifted product bounds, finite integer optimization supply the essential reduction.

### 2015 AIME II Problem 13: Negative partial sums of successive sines

For $n \geq 1$, let $a_n = \sum_{k=1}^{n} \sin k$, with angles in radians. Find the index of the hundredth negative term.

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#### Solution

Telescoping $2\sin(1/2)\sin k = \cos(k-1/2) - \cos(k+1/2)$ gives $a_n = \sin(n/2)\sin((n+1)/2)/\sin(1/2)$. The denominator is positive. The two numerator factors have opposite signs exactly when a multiple $k\pi$ lies strictly between $n/2$ and $(n+1)/2$. Thus $2\pi k - 1 < n < 2\pi k$. Each such interval for positive integer $k$ contains exactly one integer, namely $\lfloor 2\pi k \rfloor$, since $\pi$ is irrational. The hundredth is $\lfloor 200\pi \rfloor = 628$.

**Answer:** 628.

**Methods:** trigonometric telescoping, irrational rotations, floor sequences.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All three alternatives inspected. Product-to-sum and geometric series yield equivalent sign conditions. The complex-argument alternative has loose wording about a complex number being negative; the real sine sign test is precise.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Negative partial sums of successive sines; trigonometric telescoping, irrational rotations, floor sequences supply the essential reduction.

### 2016 AIME I Problem 5: Two arithmetic totals share a day count

A reader starts with $n$ pages in $t$ minutes, then increases both quantities by one each day. She reads on at least two days, finishing 374 pages in 319 minutes; $n, t$ are positive integers. Find $n + t$.

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#### Solution

If there are $k$ days, the totals give $k(2n + k - 1) = 748$ and $k(2t + k - 1) = 638$. Subtraction gives $k(n - t) = 55$, so $k$ divides both 55 and $\gcd(748, 638) = 22$. Hence $k \mid 11$; since $k \geq 2$, $k = 11$. The two equations now give $n = 29, t = 24$, and $n + t = 53$.

**Answer:** 053.

**Methods:** arithmetic sums, divisibility, common factors.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All five alternatives inspected. The fully justified gcd argument also excludes even day counts; several alternatives simply infer eleven days from the common factor without this step.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Two arithmetic totals share a day count; arithmetic sums, divisibility, common factors supply the essential reduction.

## 2016 AIME I Problem 7: Real values of a complex radical expression

Using principal square roots, count ordered integer pairs $(a, b)$ for which $\sqrt{ab+2016}/(ab+100) - i\sqrt{|a+b|}/(ab+100)$ is defined and real.

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### Solution

If $ab \geq -2016$, the imaginary part vanishes exactly when $b = -a$. Then $|a| \leq 44$, excluding $a = \pm 10$ because the denominator would vanish:87 pairs. If $ab < -2016$, reality means $-ab - 2016 = |a+b|$. The signs are opposite; first take $a > 0, b = -c < 0$. For $c > a$, this becomes $(a-1)(c+1) = 2015 = 5 \cdot 13 \cdot 31$. The positive factor pairs with the first smaller are $(1, 2015), (5, 403), (13, 155), (31, 65)$, giving four solutions; each satisfies $ac > 2016$. Interchanging $a, c$ gives four more, and reversing both signs gives eight further pairs. Total: $87 + 16 = 103$.

**Answer:** 103.

**Methods:** principal complex square roots, shifted factorization, divisor pairs.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** The sole source correctly separates the real-radical and imaginary-radical cases. Counting only vanishing second radicals would miss sixteen pairs; the denominator exclusions are also essential.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Real values of a complex radical expression; principal complex square roots, shifted factorization, divisor pairs supply the essential reduction.

## 2016 AIME I Problem 10: Alternating geometric and arithmetic triples

A strictly increasing positive integer sequence has every triple $a_{2k-1}, a_{2k}, a_{2k+1}$ geometric and every triple $a_{2k}, a_{2k+1}, a_{2k+2}$ arithmetic. Given $a_{13} = 2016$, find $a_1$.

Published problem and solutions · MAA AMC.

### Solution

Let $b_k = \sqrt{a_{2k+1}}$ for $k \geq 0$. Geometric triples give $a_{2k+2} = b_k b_{k+1}$; the arithmetic condition then gives $b_k + b_{k+2} = 2b_{k+1}$. Thus $b_k = x + kd$ with $x, d > 0$. For every $k$, $b_k d = a_{2k+2} - a_{2k+1}$ is an integer, so subtracting consecutive such values shows $D = d^2$ is a positive integer. Since $b_6 = 12\sqrt{14}$, the integer $N = b_6 d$ satisfies $N^2 = 2016D$. Prime factorization forces $168 \mid N$, so $d = j\sqrt{14}$ for a positive integer $j$. Now $x = b_6 - 6d = (12-6j)\sqrt{14} > 0$ forces $j = 1$. Therefore $a_1 = x^2 = 504$. Conversely $b_k = (6+k)\sqrt{14}$ generates an increasing integer sequence with all required properties.

**Answer:** 504.

**Methods:** square-root linearization, integer differences, square divisibility.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All five alternatives inspected. Their square-pattern observations are useful, but several assume rational factors are integers or guess a smallest step. The integer-difference argument proves the step restriction and uniqueness.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Alternating geometric and arithmetic triples; square-root linearization, integer differences, square divisibility supply the essential reduction.

## 2016 AIME I Problem 12: The first value with four prime factors

Find the least positive integer $m$ for which $m^2 - m + 11$ has at least four prime factors, counted with multiplicity.

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### Solution

The value is odd. If an odd prime $p$ divides it, then $(2m-1)^2 \equiv -43 \pmod{p}$. For $p = 3, 5, 7$, the residues $2, 2, 6$ are not squares, so every prime factor is at least 11. A value with four or more prime factors is therefore either $11^4$ or at least $11^3 \cdot 13 = 17303$. The first possibility is impossible: $121^2 - 121 + 11 = 14531 < 14641 < 14773 = 122^2 - 122 + 11$, and the quadratic increases for positive integers. Since $132^2 - 132 + 11 = 17303 = 11^3 \cdot 13$, the least $m$ is 132.

**Answer:** 132.

**Methods:** quadratic residues, prime-factor lower bounds, monotonicity.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All four stored entries inspected. The first is a setup fragment; the third completes the sharp prime-factor bound. One factor-search alternative does not fully justify exhaustion, which the lower-bound proof supplies.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. The first value with four prime factors; quadratic residues, prime-factor lower bounds, monotonicity supply the essential reduction.

## 2016 AIME II Problem 5: An infinite chain of right-triangle altitudes

Right triangle $ABC_0$ has pairwise coprime positive integer sides and perimeter $p$. Let $C_1$ be the foot from $C_0$ to $AB$; for $n \geq 2$, let $C_n$ be the foot from $C_{n-1}$ to $C_{n-2}B$. If $C_0C_1 + C_1C_2 + \cdots = 6p$, find $p$.

Published problem and solutions · MAA AMC.

### Solution

Put $a = BC_0$, $b = AC_0$, $c = AB$. The first altitude is $ab/c$, and successive similar right triangles multiply it by $a/c$. Thus the total is $ab/(c-a) = 6(a+b+c)$. Using $c^2 = a^2 + b^2$ to simplify gives $7a = 6(b+c)$. Hence $6b = 7a - 6c$; squaring and substituting for $b^2$ yields $85a^2 = 84ac$, so $a/c = 84/85$. Coprimality forces $a = 84, c = 85$, and then $b = 13$. Therefore $p = 182$.

**Answer:** 182.

**Methods:** similarity chains, geometric series, primitive Pythagorean triples.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All six alternatives inspected. The geometric-series derivation is reliable; several finish by guessing a triple or choose the wrong parity orientation in Euclid's parametrization. Eliminating with the Pythagorean identity proves the unique primitive ratio.

**Archived entries inspected:** 6 of 6. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, algebra, number_theory. An infinite chain of right-triangle altitudes; similarity chains, geometric series, primitive Pythagorean triples supply the essential reduction.

## 2016 AIME II Problem 7: Three nested squares with two integer areas

Two concentric squares have parallel sides. The outer area is 2016 and the inner area is a smaller positive integer. A third square has one vertex on each outer side, while each inner vertex lies on one of its sides. Find the difference between the greatest and least possible positive integer areas of this third square.

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### Solution

If the third square is rotated through $\theta \in [0, \pi/4]$, its side length is the outer side divided by $\cos\theta + \sin\theta$. The inner side is divided by the same factor again. Consequently, if the middle and inner areas are $M$, $N$, then $M^2 = 2016N$ and $1008 \le M < 2016$. Since $2016 = 2^5 \cdot 3^2 \cdot 7$, integrality of $N$ requires $168 \mid M$. Thus the least and greatest possibilities are 1008 and 1848; their inner areas are 504 and 1694, both integers. The continuous rotation realizes both, and the difference is 840.

**Answer:** 840.

**Methods:** nested-square similarity, square divisibility, integer extrema.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** The single source states the area geometric progression and the extrema. The explicit divisibility condition supplies a complete proof that no larger integer middle area is possible.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory. Three nested squares with two integer areas; nested-square similarity, square divisibility, integer extrema supply the essential reduction.

## 2016 AIME II Problem 8: Three distinct factors of a product

Count unordered triples of distinct positive integers whose product is $11 \cdot 21 \cdot 31 \cdot 41 \cdot 51 \cdot 61$.

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### Solution

The product is $3^2$ times six distinct primes. Labeled triples receive the two factors 3 in $\binom{4}{2} = 6$ ways and each other prime in 3 ways, giving $6 \cdot 3^6$. An equal pair must both be 1 or both 3, because all

other prime exponents are 1. Each type has 3 choices for the unequal coordinate; all-equal is impossible. Remove 6 ordered triples and divide by $3!$ : $(6 \cdot 3^6 - 6)/6 = 728$.

**Answer:** 728.

**Methods:** prime-exponent distributions, stars and bars, symmetry.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** The first archived solution uses the same mathematical reduction; this argument is independently written.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Three distinct factors of a product; the proof uses prime-exponent distributions, stars and bars, symmetry.

## 2016 AIME II Problem 9: An arithmetic sequence plus a geometric sequence

Increasing positive integer sequences satisfy $a_1 = b_1 = 1$, with $a_n$ arithmetic and $b_n$ geometric. Put $c_n = a_n + b_n$. If $c_{k-1} = 100$ and $c_{k+1} = 1000$, find $c_k$.

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### Solution

Write $a_n = 1 + (n-1)d$ and $b_n = r^{n-1}$, where $d \geq 1, r \geq 2$ are integers. Since $c_1 = 2$, $k \geq 3$. The arithmetic contribution at $k+1$ is less than three times that at $k-1$, hence less than 300. Therefore $700 < r^k < 1000$. Also $r^3 < 1000$, so $2 \leq r \leq 9$. Checking these eight bases, the only powers in that interval with exponent at least 3 are $3^6 = 729$ and $9^3 = 729$. In the first case $k = 6$ and $d = (99 - 3^4)/4 = 9/2$, impossible. In the second $k = 3, r = 9, d = 90$. Thus $c_k = 1 + 2(90) + 9^2 = 262$, and the given neighboring values are verified.

**Answer:** 262.

**Methods:** mixed sequences, exponential bounds, divisibility.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All five alternatives inspected. One gives a valid power-interval search, while several merely guess k=3 or dismiss larger cases without details. The bounded power list proves uniqueness.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. An arithmetic sequence plus a geometric sequence; mixed sequences, exponential bounds, divisibility supply the essential reduction.

## 2016 AIME II Problem 11: Divisor counts of perfect powers

Call $N$ $k$-nice when some $k$th power has exactly $N$ positive divisors. Count $1 \leq N < 1000$ that are neither 7-nice nor 8-nice.

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**Solution**

The divisor formula shows $N$ is $k$-nice exactly when $N \equiv 1 \pmod{k}$: necessity follows by multiplying factors $ke_i + 1$; sufficiency uses $(2^{(N-1)/k})^k$. There are 143 integers $1 \pmod 7$, 125 integers $1 \pmod 8$, and 18 integers $1 \pmod{56}$ in $[1, 999]$. Inclusion-exclusion leaves $999 - (143 + 125 - 18) = 749$.

**Answer:** 749.

**Methods:** inclusion-exclusion, divisor counts.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** The archived count omits N=1 in two intermediate totals, though the errors cancel. This version counts every endpoint consistently.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Divisor counts of perfect powers; the proof uses inclusion-exclusion, divisor counts.

## 2017 AIME I Problem 2: Equal remainders determine two divisors

Dividing 702,787,855 by a positive integer $m$ gives the same positive remainder $r$. Dividing 412,722,815 by a positive integer $n$ gives the same positive remainder $ser$. Find $m + n + r + s$.

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**Solution**

The first divisor divides the differences 85 and 153, whose gcd is 17. Since the remainder is positive, $m \neq 1$, so $m = 17$ and $r = 5$. Similarly $n \mid \gcd(310, 403) = 31$ and $n \neq 1$, so $n = 31, s = 9$. The sum is $17 + 31 + 5 + 9 = 62$.

**Answer:** 062.

**Methods:** equal residues, gcd of differences.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All three alternatives inspected. The positive-remainder condition excludes modulus 1; one alternative accidentally labels the first remainder n rather than r.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Equal remainders determine two divisors; equal residues, gcd of differences supply the essential reduction.

## 2017 AIME I Problem 3: A periodic sequence of triangular-number digits

Let $d_n$ be the units digit of $1 + 2 + \cdots + n$. Find $\sum_{n=1}^{2017} d_n$ modulo 1000.

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**Solution**

For $T_n = n(n+1)/2$, $T_{n+20} - T_n = 20n + 210$ is divisible by 10, so the digits repeat with period 20. The first period is $1, 3, 6, 0, 5, 1, 8, 6, 5, 5, 6, 8, 1, 5, 0, 6, 3, 1, 0, 0$, with sum 70. There are 100 full periods and 17 remaining terms. The latter sum is 69 because the final three digits sum to 1. Thus the total is 7069 and the remainder is 69.

**Answer:** 069.

**Methods:** modular periodicity, triangular numbers.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** The single source lists the correct period and sum; the polynomial difference supplies a proof of periodicity.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A periodic sequence of triangular-number digits; modular periodicity, triangular numbers supply the essential reduction.

## 2017 AIME I Problem 5: One rational number in two bases

A rational number is $(ab.cd)_8$ and also $(bb.ba)_{12}$, with all four digits nonzero. Find the decimal integer $100a + 10b + c$.

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**Solution**

The integer parts agree: $8a + b = 13b$, so $2a = 3b$. Since $1 \le a, b \le 7$, $(a, b)$ is $(3, 2)$ or $(6, 4)$. Equality of fractional parts gives $c/8 + d/64 = b/12 + a/144 = 3b/32$, hence $8c + d = 6b$. For $(3, 2)$ this gives $(c, d) = (1, 4)$; for $(6, 4)$ it gives $(3, 0)$, excluded. Thus the requested decimal number is 321.

**Answer:** 321.

**Methods:** base expansions, integer and fractional parts, digit restrictions.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All four alternatives inspected. Separating integer and fractional parts avoids a large denominator equation. The approximation-based alternative does not establish that all digit choices were considered.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. One rational number in two bases; base expansions, integer and fractional parts, digit restrictions supply the essential reduction.

## 2017 AIME I Problem 9: A concatenation-type recurrence modulo99

Let $a_{10} = 10$ and $a_n = 100a_{n-1} + n$ for integers $n > 10$. Find the least $n > 10$ with $99 \mid a_n$.

Published problem and solutions · MAA AMC.

**Solution**

Modulo 99, $100 \equiv 1$, so $a_n \equiv 10 + 11 + \cdots + n = (n+10)(n-9)/2$. Since 2 is invertible modulo 99, we require $(n+10)(n-9) \equiv 0 \pmod{99}$. Modulo 9 this gives $n \equiv 0$ or 8; modulo 11 it gives $n \equiv 1$ or 9. The four Chinese-remainder classes modulo 99 are $9, 45, 53, 89$. The smallest representative greater than 10 is 45.

**Answer:** 045.

**Methods:** recurrence reduction, Chinese remainder theorem, polynomial congruences.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All eight alternatives inspected. Several shortcuts omit the n≡8 modulo 9 branch or assume a small search suffices. The four CRT classes give a complete minimality proof.

**Archived entries inspected:** 8 of 8. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A concatenation-type recurrence modulo99; recurrence reduction, Chinese remainder theorem, polynomial congruences supply the essential reduction.

## 2017 AIME I Problem 12: Product-free subsets

Count subsets $S$ of $\{1, \ldots, 10\}$ containing no $a, b, c$ (possibly equal) with $ab = c$.

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**Solution**

One cannot occur. If 2 occurs, 4 is excluded; on $\{3, 6, 9\}$ the allowed subsets are empty, $\{3\}, \{6\}, \{9\}$, $\{6, 9\}$, giving 5 choices; on $\{5, 10\}$ choose at most one, giving 3; 7 and 8 are free. This gives 60. If 2 is absent and 3 present, 9 is absent and six other entries are free, giving 64. If 2 and 3 are absent, every subset of $\{4, \ldots, 10\}$ works, giving 128. Total 252.

**Answer:** 252.

**Methods:** forbidden configurations, casework.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** The first archived solution uses the same mathematical reduction; this argument is independently written.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Product-free subsets; the proof uses forbidden configurations, casework.

## 2017 AIME I Problem 13: The last gap between consecutive cubes

For integer $m \geq 2$, let $Q(m)$ be the least positive integer such that every integer $n \geq Q(m)$ admits a cube with $n < k^3 \leq mn$. Find $\sum_{m=2}^{2017} Q(m)$ modulo 1000.

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**Solution**

For $j^3 \le n < (j+1)^3$, failure occurs exactly when $mn < (j+1)^3$. The ratio $((j+1)/j)^3$ decreases with $j$. For $m = 2$, only $j \le 3$ can fail, and the last failure is $n = \lfloor (64-1)/2 \rfloor = 31$, so $Q(2) = 32$. For $m = 3$, only $j \le 2$ can fail, with last failure $\lfloor 26/3 \rfloor = 8$, so $Q(3) = 9$. For $4 \le m \le 7$, only $j = 1$ fails, and only at $n = 1$, giving $Q(m) = 2$. For $m \ge 8$, no positive n fails, so $Q(m) = 1$. The sum is $32 + 9 + 4(2) + 2010 = 2059$, whose remainder is 59.

**Answer:** 059.

**Methods:** consecutive-power gaps, monotonic ratios, threshold analysis.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** Both alternatives inspected. Their numerical thresholds agree; explicitly locating the last failing interval proves the universal condition for every larger n, which a short sample of intervals alone would not establish.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. The last gap between consecutive cubes; consecutive-power gaps, monotonic ratios, threshold analysis supply the essential reduction.

## 2017 AIME I Problem 14: Nested logarithms become a monotone power equation

For $a > 1, x > 1$, suppose $\log_a(\log_a(\log_a 2) + \log_a 24 - 128) = 128$ and $\log_a(\log_a x) = 256$. Find $x$ modulo 1000.

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**Solution**

Put $b = a^{a^{128}} > 1$. Exponentiating the first equation gives $24 \log_a 2 = a^{128} b$. Multiplying by $\log a$ yields $24 \log 2 = b \log b$, so $b^b = 2^{24} = 8^8$. The function $u \log u$ is strictly increasing for $u > 1$, hence $b = 8$. Now put $t = a^{128}$; raising $b = a^t$ to 128 gives $t^t = 8^{128} = 64^{64}$, so $t = 64$ and $a = 2^{3/64}$. The second condition gives $x = a^{a^{256}} = a^{4096} = 2^{192}$. Repeated squaring modulo 1000 gives the successive residues for exponents $3, 6, 12, 24, 48, 96, 192$ as $8, 64, 96, 216, 656, 336, 896$. The answer is 896.

**Answer:** 896.

**Methods:** nested logarithms, monotonic power functions, repeated squaring.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All three alternatives inspected. The first has the useful power-tower substitution but does not state its uniqueness justification; the second guesses a parameter from its apparent factorization. Monotonicity makes both substitutions rigorous.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Nested logarithms become a monotone power equation; nested logarithms, monotonic power functions, repeated squaring supply the essential reduction.

## 2017 AIME II Problem 4: Ternary digit avoidance

Count positive integers at most 2017 with no zero digit in base 3.

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### Solution

Since $2017 = 2202201_3$, every valid number of at most 6 digits is allowed, contributing $\sum_{k=1}^{6} 2^k = 126$. A valid 7-digit numeral must begin 1 or 21, giving $2^6 + 2^5 = 96$ more. A prefix 22 forces the next positive digit above the bound. Total 222.

**Answer:** 222.

**Methods:** digit counting, prefix decomposition.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** The first archived solution uses the same mathematical reduction; this argument is independently written.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Ternary digit avoidance; the proof uses digit counting, prefix decomposition.

## 2017 AIME II Problem 6: A square squeezed between nearby shifted squares

Find the sum of positive integers $n$ for which $\sqrt{n^2 + 85n + 2017}$ is an integer.

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### Solution

For $n > 0$, $(n + 42)^2 < n^2 + 85n + 2017 < (n + 45)^2$. Thus an integer square root must be $n + 43$ or $n + 44$. Equating squares gives respectively $n = 168$ and $n = 27$, both positive and valid. Their sum is 195.

**Answer:** 195.

**Methods:** near-square bounds, finite Diophantine reduction.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All seven alternatives inspected. Completing a square and factoring 843 also works. The two nearby-square bounds justify the two-case inspection used without proof in some alternatives.

**Archived entries inspected:** 7 of 7. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A square squeezed between nearby shifted squares; near-square bounds, finite Diophantine reduction supply the essential reduction.

## 2017 AIME II Problem 8: Integrality of a truncated exponential series

Count positive integers $n < 2017$ for which $\sum_{j=0}^{6} n^j/j!$ is an integer.

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### Solution

After subtracting $1+n$ and multiplying by 720, the numerator is $F = n^6 + 6n^5 + 30n^4 + 120n^3 + 360n^2$. Modulo 2 and 3, integrality forces $n$ divisible by both, hence $n = 6u$. The terms through degree 4 are then integers, and the last two sum to $n^5(n+6)/720 = (324/5)u^5(u+1)$. This is integral exactly when $u \equiv 0$ or 4 modulo 5. Thus $n \equiv 0$ or 24 modulo 30. Below 2017 there are 67 positive values in each class, for a total of 134.

**Answer:** 134.

**Methods:** denominator valuations, necessary congruences, residue classes.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All four alternatives inspected. The complete numerator shows why 6 must divide n before the last two terms can be considered alone. The first alternative assumes that last-pair integrality is necessary without first excluding cancellation with earlier terms.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Integrality of a truncated exponential series; denominator valuations, necessary congruences, residue classes supply the essential reduction.

## 2018 AIME I Problem 2: A digit permutation in three bases

An integer is $(abc)_{14} = (acb)_{15} = (acac)_6$, with $a > 0$. Find its decimal value.

Published problem and solutions · MAA AMC.

### Solution

Expanding gives $196a + 14b + c = 225a + b + 15c = 222a + 37c$. The last equality yields $b = 22c - 3a$, and substitution into the first gives $a = 4c$. Since $1 \le a \le 5$ and $0 \le c \le 5$, necessarily $a = 4, c = 1$; then $b = 10$, an allowed base 14 digit. Thus $n = 222(4) + 37 = 925$.

**Answer:** 925.

**Methods:** base expansions, linear elimination, digit bounds.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All four alternatives inspected. Eliminating b first leaves a=4 c, making the base 6 bound decisive without casework.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A digit permutation in three bases; base expansions, linear elimination, digit bounds supply the essential reduction.

## 2018 AIME I Problem 6: Two disjoint root-of-unity families

Count complex numbers $z$ satisfying $|z| = 1$ and $z^{720} - z^{120} \in \mathbb{R}$. Find the count modulo 1000.

Published problem and solutions · MAA AMC.

### Solution

For $|z| = 1$, $\overline{z} = 1/z$. Equating the expression to its conjugate and multiplying by $z^{720}$ gives $(z^{840} + 1)(z^{600} - 1) = 0$. These factors have 840 and 600 distinct roots respectively, all on the unit circle. They share none: a common root would give $z^{4200} = (-1)^5 = -1$ from the first equation and $z^{4200} = 1^7 = 1$ from the second. Thus the count is 1440 and the remainder 440.

**Answer:** 440.

**Methods:** conjugation on the unit circle, roots of unity, disjoint root families.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All seven alternatives inspected. Trigonometric families give the same count, but several omit checking overlap. The polynomial degree alone is insufficient; the explicit factorization and incompatible powers prove distinctness.

**Archived entries inspected:** 7 of 7. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Two disjoint root-of-unity families; conjugation on the unit circle, roots of unity, disjoint root families supply the essential reduction.

## 2018 AIME I Problem 11: A two-digit ending requires prime-square orders

Find the least positive $n$ such that the final two digits of $3^n$ in base 143 are 01.

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### Solution

The condition is $3^n \equiv 1 \pmod{143^2}$, equivalently modulo 121 and 169. Modulo 121, the successive powers $3, 9, 27, 81, 243 \equiv 1$ show order 5. Modulo 13, the order is 3, so write $n = 3k$. The binomial theorem gives $3^{3k} = (1+26)^k \equiv 1 + 26k \pmod{169}$, which equals 1 exactly when $13 \mid k$. Thus the order modulo 169 is 39. The least exponent satisfying both is $\operatorname{lcm}(5, 39) = 195$.

**Answer:** 195.

**Methods:** multiplicative orders, prime-square lifting, Chinese remainder theorem.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All nine alternatives inspected. The binomial calculation proves the prime-square order directly. A few sources write congruences for n where they mean 3^n or invoke LTE before establishing the required divisibility of the exponent.

**Archived entries inspected:** 9 of 9. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A two-digit ending requires prime-square orders; multiplicative orders, prime-square lifting, Chinese remainder theorem supply the essential reduction.

### 2018 AIME I Problem 12: Subset sums modulo three

Choose a subset of [18] uniformly. Find the numerator of its reduced probability of having sum divisible by 3.

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#### Solution

The six multiples of 3 do not affect the probability. Pair the six residue 1 entries with the six residue 2 entries. A pair contributes residues $0, 1, 2$ with multiplicities $2, 1, 1$. If $A_k$ counts residue 0 choices among $k$ pairs and $B_k$ either nonzero residue, then $A_k+2B_k = 4^k$ and $A_k-B_k = 1$. Thus $A_6 = (4^6+2)/3 = 1366$. The probability is $1366/4^6 = 683/2048$; requested numerator 683.

**Answer:** 683.

**Methods:** residue states, generating functions.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** All thirteen sources inspected. Roots-of-unity filtering, residue convolution, Vandermonde sums, and block recurrences all prove the count. Indices 9 and 10 are empirical pattern/answer guesses, not proofs. Source index 6 pairs wrong residue-count classes, but their equal cardinalities conceal the error. The independent root-of-unity calculation is valid.

**Archived entries inspected:** 13 of 13. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Subset sums modulo three; the proof uses residue states, generating functions.

### 2018 AIME II Problem 2: A recurring triple proves periodicity

Let $a_0 = 2, a_1 = 5, a_2 = 8$, and for $n > 2$ let $a_n$ be the least nonnegative residue of $4(a_{n-1}+a_{n-2}+a_{n-3})$ modulo 11. Find $a_{2018}a_{2020}a_{2022}$.

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#### Solution

Applying the recurrence gives $a_0, \ldots, a_{12} = 2, 5, 8, 5, 6, 10, 7, 4, 7, 6, 2, 5, 8$. The triple at indices 10,11,12 equals the initial triple. Because each subsequent value is determined by its preceding triple, induction gives $a_{n+10} = a_n$ for every $n \geq 0$. Thus the product is $a_8a_0a_2 = 7 \cdot 2 \cdot 8 = 112$.

**Answer:** 112.

**Methods:** finite-state recurrence, modular periodicity.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** Both alternatives inspected. One computes the period; the other solves the characteristic polynomial over the field modulo 11. Returning to the entire three-term state, rather than merely one value, justifies the elementary periodicity proof.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A recurring triple proves periodicity; finite-state recurrence, modular periodicity supply the essential reduction.

### 2018 AIME II Problem 3: A square and a cube in the same base

Find the sum of integer bases $8 \leq b < 1000$ for which $(36)_b$ is a square and $(27)_b$ is a cube.

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#### Solution

Write $3b + 6 = u^2$ and $2b + 7 = v^3$. The first equation implies $3 \mid u$ and hence $b \equiv 1 \pmod 3$. Thus $3 \mid v$; also v is odd because $2b + 7$ is odd. The bound $2b + 7 < 2007 < 13^3$ leaves only $v = 3$ or 9. These yield $b = 10$ or 361, and the square values are 36 and 1089 respectively. Both bases are valid, so the sum is 371.

**Answer:** 371.

**Methods:** base notation, prime divisibility, bounded cubes.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All three alternatives inspected. The parity and modulo 3 restrictions reduce to two cubes. The third alternative misapplies the rational-root theorem to an equation containing another variable; that reasoning is not used.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A square and a cube in the same base; base notation, prime divisibility, bounded cubes supply the essential reduction.

### 2018 AIME II Problem 7: Rational lengths among equal-area parallel slices

A triangle has sides $AB = 9$, $AC = 12$, $BC = 5\sqrt{3}$. Lines parallel to BC divide it into 2450 equal-area regions. Let the kth parallel segment, including BC as the 2450 th, bound the first k regions nearest A. How many of these 2450 segments have rational length?

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#### Solution

The kth similar triangle has k/2450 of the total area, so its parallel side has length $5\sqrt{3}\sqrt{k/2450} = \sqrt{6k}/14$. A rational square root of an integer is an integer; thus the length is rational exactly when $6k$ is a square, equivalently $k = 6t^2$ for a positive integer t. The bound $6t^2 \leq 2450$ gives $1 \leq t \leq 20$. Therefore 20 segments have rational length.

**Answer:** 020.

**Methods:** similarity area ratios, squarefree kernels, rational square roots.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All three alternatives inspected. The squarefree kernel 6 controls the rational indices. The problem includes the base segment; it is irrational and does not alter the count, but the inclusive indexing is retained.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory. Rational lengths among equal-area parallel slices; similarity area ratios, squarefree kernels, rational square roots supply the essential reduction.

### 2019 AIME I Problem 1: A sum of strings of nines

Let $N = 9 + 99 + 999 + \cdots + (10^{321} - 1)$. Find the sum of its decimal digits.

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#### Solution

We have $N = \sum_{k=1}^{321} 10^k - 321$. The sum before subtraction ends in 1110 and has 318 further digits, all 1. Subtracting 321 changes only those last four positions, to 0789. The digit sum is therefore $318 + 0 + 7 + 8 + 9 = 342$.

**Answer:** 342.

**Methods:** decimal place value, subtraction with borrowing.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All four alternatives inspected. Place-value subtraction or the equivalent repunit identity gives the same digit pattern; direct long addition obscures why the pattern continues.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A sum of strings of nines; decimal place value, subtraction with borrowing supply the essential reduction.

### 2019 AIME I Problem 7: Prime exponents hidden in gcd and lcm logarithms

Positive integers x,y satisfy $\log_{10} x + 2\log_{10}\gcd(x,y) = 60$ and $\log_{10} y + 2\log_{10}\operatorname{lcm}(x,y) = 570$. If m,n count the prime factors of x,y with multiplicity, find $3m + 2n$.

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#### Solution

Adding and using $\gcd(x,y)\operatorname{lcm}(x,y) = xy$ gives $xy = 10^{210}$. Thus only 2 and 5 divide either number. From $x\gcd(x,y)^2 = 10^{60}$, each prime exponent in x is at most 60; the corresponding exponent in y is at least 150. Hence $x \mid y$ and $\gcd(x,y) = x$. Therefore $x^3 = 10^{60}$, giving $x = 10^{20}$ and $y = 10^{190}$. Thus $m = 40, n = 380$ and $3m + 2n = 880$.

**Answer:** 880.

**Methods:** gcd-lcm product, prime valuations, logarithms.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All seven alternatives inspected. Valuations rigorously establish x divides y. Several alternatives assume both numbers are powers of 10, or infer coordinatewise exponent inequalities merely from x<y; neither shortcut alone is justified.

**Archived entries inspected:** 7 of 7. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Prime exponents hidden in gcd and lcm logarithms; gcd-lcm product, prime valuations, logarithms supply the essential reduction.

## 2019 AIME I Problem 9: Consecutive integers with seven divisors in total

Let $\tau(n)$ count positive divisors. Find the sum of the six least positive integers n such that $\tau(n)+\tau(n+1)=7$.

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### Solution

Neither number can have one divisor: n=1 gives total 3. Thus the divisor counts are 2 and 5, or 3 and 4, in either order. A number with five divisors is $p^4$; for odd p both neighbors are even and exceed 2, so cannot be prime. The case p=2 gives n=16. A number with three divisors is $p^2$. For p≥5, $p^2-1$ has at least eight divisors: it is divisible by 8 and an odd prime. For p=2 neither neighbor has four divisors; p=3 gives n=8. To find all remaining answers up to 361, test primes $p=3,5,7,11,13,17,19$. Since $p^2+1=2((p^2+1)/2)$, it has four divisors precisely when the odd second factor is prime. These factors are 5,13,25,61,85,145,181, giving n=9,25,121,361. Consequently the first six are 8,9,16,25,121,361, with sum 540.

**Answer:** 540.

**Methods:** divisor-count patterns, prime powers, bounded prime search.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All three alternatives inspected. Classifying odd divisor counts reduces the search to prime squares and fourth powers; the bound 361 makes the short prime list exhaustive.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Consecutive integers with seven divisors in total; divisor-count patterns, prime powers, bounded prime search supply the essential reduction.

## 2019 AIME I Problem 11: A circle tangent to all three excircles

An integer-sided triangle has $AB=AC$. A circle centered at its incenter is internally tangent to the A-excircle and externally tangent to the other two excircles. Find the smallest possible perimeter.

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### Solution

Normalize $BC=2$ and $AB=AC=t>1$, with altitude $h=\sqrt{t^2-1}$. The inradius is $r=h/(t+1)$; the A-exradius is $h/(t-1)$, and the other exradii equal h. Put the base midpoint at the origin and A at $(0,h)$. The incenter is $(0,r)$ and a lateral excenter is $(t,h)$. Internal tangency forces the required circle radius to be $R=r+2h/(t-1)$; the A-excircle cannot contain its center, which is outside that excircle. External tangency then gives $\sqrt{t^2+(h-r)^2}=R+h$. Simplifying both sides yields $t\sqrt{2t/(t+1)}=t(t+3)/h$, hence $(t+3)^2=2t(t-1)$ and $(t-9)(t+1)=0$. Thus the side ratio is

9:9:2. An integral scaling has integer scale because gcd(9,2)=1. The smallest perimeter is 20, and these lengths satisfy all positive-radius equations.

**Answer:** 020.

**Methods:** inradius and exradii, circle tangency, primitive integer ratios.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All five alternatives inspected. Normalizing the base makes the three radii and center distance explicit. Positivity excludes the extraneous negative quadratic root, and coprimality proves the minimum integral scale.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory. A circle tangent to all three excircles; inradius and exradii, circle tangency, primitive integer ratios supply the essential reduction.

## 2019 AIME I Problem 14: An eighth power forces order sixteen

Find the least odd prime divisor of $2019^8 + 1$.

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### Solution

For any odd prime divisor p, $2019^8 \equiv -1 \pmod{p}$ and $2019^{16} \equiv 1 \pmod{p}$. Its multiplicative order is therefore exactly 16, so $16 \mid p - 1$. The only candidate primes below or equal to 97 are 17 and 97. Modulo 17, $2019 \equiv 13$ and $13^2 \equiv -1$, so the eighth power is 1 and 17 fails. Modulo 97, successive squaring gives $2019^2 \equiv 33$, $2019^4 \equiv 22$, $2019^8 \equiv -1$. Thus 97 is the least.

**Answer:** 097.

**Methods:** multiplicative order, prime divisor restrictions, repeated squaring.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All five stored entries inspected, including a totient-definition note. The exact-order argument is necessary: Fermat alone does not imply the order is 16.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. An eighth power forces order sixteen; multiplicative order, prime divisor restrictions, repeated squaring supply the essential reduction.

## 2019 AIME II Problem 4: A square product of four die rolls

Find numerator plus denominator of the probability that four fair die outcomes have square product.

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### Solution

The number of fives must be $0, 2$, or 4. On the other five faces, parity of prime exponents gives four states: the zero state has 2 labels $(1, 4)$, each other state one. If $a_k$ counts square products, $a_0 = 1$

and $a_{k+1} = a_k + 5^k$, hence $a_2 = 7, a_4 = 157$. The favorable count is $157 + \binom{4}{2}7 + 1 = 200$. Probability $200/6^4 = 25/162$; answer 187.

**Answer:** 187.

**Methods:** parity vectors, recurrence, probability.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** Parity recurrence and exhaustive casework agree. The final two stored alternatives were read in the separate 2023 audit supplement. Indices 5 and 6 reviewed. The minimal square-product blocks are single squares, equal nonsquare pairs, and the triple 2,3,6; their length-four casework gives 200. The parity-generating-function alternative is multiplication in the group algebra with x^2=y^2=z^2=1, and its constant coefficient is also 200. Both support 25/162.

**Archived entries inspected:** 8 of 8. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. A square product of four die rolls; the proof uses parity vectors, recurrence, probability.

## 2019 AIME II Problem 9: Multiples of twenty having twenty divisors

Call a positive integer n k-pretty when it has exactly k positive divisors and is divisible by k. Let S be the sum of all 20-pretty integers below 2019. Find $S/20$.

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### Solution

Write $n = 2^a 5^b m$ with $a \geq 2, b \geq 1$ and $\gcd(m, 10) = 1$. Then $(a+1)(b+1)\tau(m) = 20$. The possible a values are 3,4,9; a=3 forces b=4 and gives n≥5000, while a=9 gives n≥2560. For a=4, either b=3,m=1, giving 2000, or b=1 and m is a prime other than 2,5. The bound $80m < 2019$ leaves $m = 3, 7, 11, 13, 17, 19, 23$. Therefore $S/20 = 100 + 4(3 + 7 + 11 + 13 + 17 + 19 + 23) = 472$.

**Answer:** 472.

**Methods:** divisor-count factorization, prime valuations, bounded primes.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All three alternatives inspected. Exponents of 2 and 5 give a complete short classification; a general list of divisor patterns must still enforce the factor 20 and the upper bound.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Multiples of twenty having twenty divisors; divisor-count factorization, prime valuations, bounded primes supply the essential reduction.

## 2019 AIME II Problem 10: A tangent sign itinerary fixes every binary digit

An angle $0 < \theta < 90^\circ$ has $\tan(2^n\theta) > 0$ precisely when $3 \mid n$, and negative otherwise, for every integer $n \geq 0$. Write $\theta = p/q$ degrees in lowest terms and find $p + q$.

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**Solution**

Put $t = \theta/180 \in (0, 1/2)$. The sign of $\tan(180 \cdot 2^n t)$ is positive when the fractional part of $2^n t$ lies below $1/2$, and negative when it lies above $1/2$. In a binary expansion, this reads the next digit:0 for positive,1 for negative. No dyadic endpoint is allowed, because tangent there is zero or undefined. Thus the unique expansion is $t = 0.\overline{011}_2 = 3/(2^3 - 1) = 3/7$. Hence $\theta = 540/7$ degrees and the answer is 547.

**Answer:** 547.

**Methods:** binary expansion, doubling dynamics, trigonometric signs.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All five alternatives inspected. The binary-digit argument proves uniqueness. Several sources infer periodic angles directly from periodic signs without proof; that implication requires the complete itinerary, supplied here.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory, combinatorics. An infinite tangent-sign itinerary is encoded by the binary digits of a doubling orbit, and the periodic word determines the permitted angle.

## 2019 AIME II Problem 12: Divisibility chains with prescribed sum

Count finite strictly increasing positive integer sequences, each term dividing the next, whose sum is 360.

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**Solution**

Let $h(N)$ count such chains summing $N$ with first term at least 2; set $h(0) = 1, h(1) = 0$. Removing first term $d$ and dividing the suffix by $d$ is a bijection, so $h(N) = \sum_{d|N, d\geq 2} h(N/d - 1)$. Allowing first term 1 gives total $\sum_{d|360} h(360/d - 1)$. Its nontrivial values are $h(119) = 6, h(44) = 8, h(39) = 6, h(35) = 4$, $h(14) = 3, h(9) = 2, h(8) = 2, h(4) = 1$; all remaining divisor terms sum 15. Total 47.

**Answer:** 047.

**Methods:** divisor recurrences, canonical normalization.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** The source recurrence is valid with its initial values and the summation of partial chains; the new exposition makes those values explicit.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Divisibility chains with prescribed sum; the proof uses divisor recurrences, canonical normalization.

## 2019 AIME II Problem 14: A three-denomination Frobenius problem

With unlimited stamps of denominations $5, n, n + 1$, the largest unattainable value is 91. Find the sum of possible $n$.

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**Solution**

If $n \leq 23$, one of $n, n+1$ is coprime to 5 with value at most 23(or at most 21 when 5 divides $n$); the two-denomination Frobenius bound is then at most 87. Hence $n \geq 24$. Since 96 is attainable but 91 is not, a representation of 96 uses no 5-stamp. Using at most 4 other stamps forces $n \in \{24, 31, 32, 47, 48, 95, 96\}$. For 31 and 32, 91 is attainable; 48 cannot make 92; 95 and 96 cannot make 92. Values 24 and 47 fail at 91 but make every value 92 through 96, after which adding 5 covers all larger values. Sum 71.

**Answer:** 071.

**Methods:** numerical semigroups, residue classes.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** Several alternatives skip coprimality or finite-case justification. The original proof makes the choice of a denomination coprime to 5 explicit and checks the remaining candidates.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. A three-denomination Frobenius problem; the proof uses numerical semigroups, residue classes.

## 2020 AIME I Problem 3: A shared set of digits in bases eleven and eight

A positive integer N has base-eleven digits abc and base-eight digits 1 bca. Find the least possible decimal N.

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**Solution**

All three digits lie between 0 and 7, with a positive. Equating place values gives $121a + 11b + c = 512 + 64b + 8c + a$, hence $120a = 512 + 53b + 7c$. Thus $a \geq 5$. For a=5, $53b + 7c = 88$ forces b=0 or 1; only b=1 gives an integer digit c=5. Hence $N = 5 \cdot 121 + 11 + 5 = 621$. Any greater a makes the base-eleven number larger, so this is the minimum.

**Answer:** 621.

**Methods:** base conversion, digit bounds, linear congruences.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** Both alternatives inspected. Zero is allowed for b,c; the first source unnecessarily calls every digit positive. Explicitly testing b=0 retains the correct digit domain.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A shared set of digits in bases eleven and eight; base conversion, digit bounds, linear congruences supply the essential reduction.

## 2020 AIME I Problem 4: A truncated prefix must divide the suffix

Let S consist of positive integers ending in 2020 such that deleting those four digits leaves a positive divisor of the original integer. Find the sum of all decimal digits of all members of S.

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**Solution**

Write $N = 10000d + 2020$. The condition is equivalent to $d \mid 2020$. Since $2020 = 2^2 \cdot 5 \cdot 101$, its twelve divisors are 1,2,4,5,10,20,101,202,404,505,1010,2020. Their digit sums total 45. Each corresponding N contributes another 4 from its suffix, so the requested sum is $45 + 12 \cdot 4 = 93$.

**Answer:** 093.

**Methods:** prefix-suffix divisibility, divisor enumeration, digit sums.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All three alternatives inspected. Subtracting 10000 d gives the exact divisor criterion; the third source introduces an unnecessary factorization with an unjustified intermediate integrality implication.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A truncated prefix must divide the suffix; prefix-suffix divisibility, divisor enumeration, digit sums supply the essential reduction.

## 2020 AIME I Problem 9: A random chain of divisors

Independently sample three divisors $a, b, c$ of $20^9$ uniformly with replacement. Find the reduced numerator of $P(a \mid b \text{ and } b \mid c)$.

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**Solution**

Write divisors as $2^u 5^v$, with $0 \le u \le 18, 0 \le v \le 9$. Divisibility requires weakly increasing triples separately in both coordinates. The numbers of such triples are $\binom{21}{3}$ and $\binom{12}{3}$ by stars and bars. Probability $\binom{21}{3}\binom{12}{3}/190^3 = 77/1805$; numerator 77.

**Answer:** 077.

**Methods:** order-preserving tuples, stars and bars, probability.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** The shifted strictly increasing triples, stars and bars, and equality-case split give the same probability.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. A random chain of divisors; the proof uses order-preserving tuples, stars and bars, probability.

## 2020 AIME I Problem 10: A missing prime power constrains exponential divisibility

Positive integers m,n satisfy $\gcd(m + n, 210) = 1$, $n^n \mid m^m$, and $n \nmid m$. Find the least possible $m + n$.

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**Solution**

Every prime divisor of n divides m, hence divides m+n, so none of 2,3,5,7 divides n. Because n does not divide m, n has a repeated prime factor; thus n≥121. Also m>n, because $t^t$ increases on positive integers and equality would violate the last condition. The pair $(m, n) = (286, 121)$ works: m has one factor 11 and 286≥242, so $11^{242} \mid m^m$, and $407 = 11 \cdot 37$ is coprime to 210. Any better pair has n<407/2. The only possible n in this range with no prime below 11 and a repeated prime factor are 121 and 169. For n=169, m must have exactly one factor 13 and m≥338, giving sum≥507. For n=121, write m=11 k with $k \geq 22$ and $11 \nmid k$. The values k=22,23,24,25 fail either this restriction or coprimality of $11(k+11)$ with 210. The first valid k is 26, proving the minimum 407.

**Answer:** 407.

**Methods:** prime valuations, missing exponents, finite minimality bounds.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All three alternatives inspected. The explicit candidate supplies an upper bound, which restricts n to 121 or 169; this closes the minimality gap in the first short source.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A missing prime power constrains exponential divisibility; prime valuations, missing exponents, finite minimality bounds supply the essential reduction.

## 2020 AIME I Problem 12: Prime-power divisibility after selecting an exponent class

Let n be the least positive integer such that $3^3 5^5 7^7 \mid 149^n - 2^n$. Find the number of positive divisors of n.

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**Solution**

For an odd prime p dividing $a - b$ but not ab, the lifting formula is $v_p(a^n - b^n) = v_p(a - b) + v_p(n)$. Since $149 - 2 = 3 \cdot 7^2$, the 3 and 7 conditions are exactly $3^2 \mid n$ and $7^5 \mid n$. Modulo 5, the ratio 149/2 has order 4, so $4 \mid n$. Write n=4 k. Since $149^4 - 2^4 \equiv 1 - 16 \equiv 10 \pmod{25}$, its 5-adic valuation is 1; lifting now gives $v_5(149^{4k} - 2^{4k}) = 1 + v_5(k)$. Thus $5^4 \mid k$. The least n is $2^2 3^2 5^4 7^5$, and its divisor count is $3 \cdot 3 \cdot 5 \cdot 6 = 270$.

**Answer:** 270.

**Methods:** lifting the exponent, multiplicative order, least common multiples.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All four alternatives inspected. The order 4 condition must precede lifting at 5. Some source displays use equality for a divisibility threshold; the proof here states the necessary and sufficient inequalities.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Prime-power divisibility after selecting an exponent class; lifting the exponent, multiplicative order, least common multiples supply the essential reduction.

## 2020 AIME II Problem 1: Square divisors of a prime-power product

Count ordered pairs of positive integers m,n with $m^2n = 20^{20}$.

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### Solution

Factor $20^{20} = 2^{40}5^{20}$. Necessarily $m = 2^a5^b$ with $0 \le a \le 20$ and $0 \le b \le 10$; then $n = 2^{40-2a}5^{20-2b}$ is uniquely determined and positive. There are $21 \cdot 11 = 231$ pairs.

**Answer:** 231.

**Methods:** prime-factor exponents, square divisors.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** Both alternatives inspected. The one-to-one correspondence with square divisors gives independent exponent ranges; this is the standard arithmetic divisor-count problem rather than an arrangement problem.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Square divisors of a prime-power product; prime-factor exponents, square divisors supply the essential reduction.

## 2020 AIME II Problem 5: The smallest integer with nested digit-sum restrictions

Let f(n) be the base-four digit sum of n, and g(n) the base-eight digit sum of f(n). Let N be the least positive n for which the hexadecimal representation of g(n) uses a digit above 9. Find N modulo 1000.

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### Solution

For base b, the least integer with digit sum s is obtained by packing the least significant digits with b−1 and placing the remainder as the leading digit. Moving any available unit from a higher position to an unfilled lower position decreases the integer, proving minimality. These least integers increase strictly with s. A hexadecimal digit above 9 requires g(n)≥10. The least integer whose base-eight digit sum is at least 10 is $(37)_8 = 31$; therefore f(n)≥31. The least n with base-four digit sum at least 31 is $(1\underbrace{33\cdots3}_{10})_4 = 2 \cdot 4^{10} - 1$. It has g(n)=10, so the bound is attained. Modulo 1000 this is $2 \cdot 24^2 - 1 \equiv 151$.

**Answer:** 151.

**Methods:** minimal digit-sum representation, greedy place-value exchange, nested bases.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** Both alternatives inspected. The monotonicity of the least integer for a prescribed digit sum is essential: minimizing the final digit sum alone would otherwise not justify minimizing n.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. The smallest integer with nested digit-sum restrictions; minimal digit-sum representation, greedy place-value exchange, nested bases supply the essential reduction.

### 2020 AIME II Problem 10: Clear denominators before reducing a cube sum

Find the sum of all positive integers n for which $1^3 + \cdots + n^3$ leaves remainder 17 upon division by n+5.

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#### Solution

The cube sum is $n^2(n+1)^2/4$. Multiplying the required congruence by 4 and setting n≡−5 modulo n+5 gives $400 \equiv 68$, so $n+5 \mid 332 = 4 \cdot 83$. A remainder 17 requires n+5>17, leaving n=78,161,327. Substitution into the original integer cube-sum formula gives remainders 17,17,100 respectively. Thus the valid n sum to 239.

**Answer:** 239.

**Methods:** power-sum identity, modular denominator care, divisor candidates.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All five alternatives inspected. Multiplication by 4 gives a necessary condition, so candidates must be checked in the original congruence. One source illegally divides by 2 modulo an even modulus; another changes a cubic middle term into a square.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Clear denominators before reducing a cube sum; power-sum identity, modular denominator care, divisor candidates supply the essential reduction.

### 2021 AIME I Problem 3: Differences of powers of two

Count positive integers below 1000 expressible as a difference of integral powers of 2.

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#### Solution

For an integral positive difference the exponents must be nonnegative; a negative smaller exponent would leave an odd numerator over a power of 2. Write $2^b(2^{a-b}-1)$ with $a > b$. The 2-adic valuation uniquely determines $b$, and then a, so no duplicates arise. Necessarily $a \le 10$. Of the $\binom{11}{2} = 55$ exponent pairs, only $a = 10, b = 0, 1, 2, 3, 4$ exceed 999. Thus 50.

**Answer:** 050.

**Methods:** unique representations, valuation counting.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** Source 3 claims geometric sequences always have distinct differences, which is not a valid general principle. Binary valuation gives the required uniqueness for this sequence; source 0 also supplies it.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Differences of powers of two; the proof uses unique representations, valuation counting.

### 2021 AIME I Problem 5: Integer arithmetic progressions satisfying a square identity

A strictly increasing three-term integer arithmetic progression is special when the sum of its squared terms equals its middle term times the square of its common difference. Find the sum of the final terms of all special progressions.

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#### Solution

Write the terms as a−d,a,a+d with integer d>0. The condition gives $3a^2 + 2d^2 = ad^2$, so a>2 and $d^2 = 3a+6+12/(a-2)$. Thus a−2 is a positive divisor of 12, leaving a=3,4,5,6,8,14. The corresponding squared differences are 27,24,25,27,32,49. Only a=5,d=5 and a=14,d=7 work. Their final terms are 10 and 21, whose sum is 31.

**Answer:** 031.

**Methods:** arithmetic progression symmetry, rational divisibility, square testing.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All five alternatives inspected. Dividing the polynomial expression reduces to six possible middle terms and keeps the allowed zero first term. Several squarefree-kernel alternatives suppress the rational-square justification, which the divisor list avoids.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Integer arithmetic progressions satisfying a square identity; arithmetic progression symmetry, rational divisibility, square testing supply the essential reduction.

### 2021 AIME I Problem 10: A rational recurrence reduces at its first shared prime

Start with $a_1 = 2020/2021$. When $a_k = m/n$ is reduced with positive integers m,n, define $a_{k+1} = (m+18)/(n+19)$. Find the sum of all j for which $a_j = t/(t+1)$ for a positive integer t.

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#### Solution

Start at a term $m/(m+1)$. Before any reduction, k further steps give $(m+18k)/(m+1+19k)$, whose numerator-denominator gcd is $\gcd(m-18, k+1)$. If m−18>1, the first reduction occurs at k=p−1, where p is its least prime divisor. The denominator difference k+1 then equals p, so after reduction the next special numerator is $18+(m-18)/p$. Between these reductions the denominator difference exceeds 1 and the fractions are reduced, so no other term is special. Beginning with m=2020, the successive m−18 values are 2002,1001,143,13,1; the primes removed are 2,7,11,13. The special indices are therefore 1,2,8,18,30. Once m−18=1, the gcd is always 1 and no later special term occurs. Their sum is 59.

**Answer:** 059.

**Methods:** Euclidean algorithm, prime-driven reductions, rational recurrence.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** Both alternatives inspected. The first source states reducibility as full divisibility of k+1, which is not true in general; the first-event argument works because the first common factor is the least prime divisor. The second source supplies this distinction.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, combinatorics. The proof enumerates distinguished reduced-fraction states via a prime-removal dynamical process and proves no intermediate state is omitted.

## 2021 AIME I Problem 14: A universal divisor-sum congruence without advanced prime distribution

Let sigma(a) be the sum of positive divisors of a. Let n be the least positive integer for which 2021 | $\sigma(a^n) - 1$ for every positive integer a. Find the sum of the prime factors of n, counted with multiplicity.

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### Solution

Factor $2021 = 43 \cdot 47$. Taking the primes a=173 and 283, which are 1 modulo 43 and 47 respectively, shows $43 \mid n$ and $47 \mid n$. Taking prime a=3 modulo 43 requires $3^n = 1$; its order is 42, since $3^{21}, 3^{14}$, $3^6 \equiv 42, 36, 41 \pmod{43}$. Taking prime a=5 modulo 47 similarly requires $46 \mid n$, because $5^{23} \equiv -1$ and $5^2 \not\equiv 1 \pmod{47}$. Therefore $L = \operatorname{lcm}(43, 47, 42, 46)$ divides n. Conversely let n=L. For either p=43 or 47 and every prime q, the sum $1 + q + \cdots + q^{en}$ is 1 modulo p: this is immediate when q≡0; when q≡1 it follows from p|n; otherwise it follows from $(p-1) \mid n$ and the geometric-sum formula. Multiplicativity of sigma proves the condition for every a. Thus $n = 2 \cdot 3 \cdot 7 \cdot 23 \cdot 43 \cdot 47$, and the prime-factor sum is 125.

**Answer:** 125.

**Methods:** multiplicative divisor sums, explicit primitive roots, universal congruences.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All five alternatives inspected. Dirichlet is unnecessary: explicit primes 173,283 and prime primitive roots 3,5 prove necessity. Two sources incorrectly infer orders 42,46 from base 2, whose actual orders are 14,23. The residue-wise geometric sums prove sufficiency for all composite a as well.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A universal divisor-sum congruence without advanced prime distribution; multiplicative divisor sums, explicit primitive roots, universal congruences supply the essential reduction.

## 2021 AIME II Problem 3: Cyclic products modulo three

Permute $1, \ldots, 5$ around five labeled positions. Count permutations for which the sum of the five products of three cyclically consecutive entries is divisible by 3.

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**Solution**

Choose the position of 3 in 5 ways. The three terms containing it vanish modulo 3; the other two factor as a nonzero product times the sum of the two entries neighboring 3. These neighbors must have opposite nonzero residues: there are $2 \cdot 2 \cdot 2 = 8$ ordered choices from $\{1, 4\}$ and $\{2, 5\}$. Arrange the remaining entries in 2 ways. Total $5 \cdot 8 \cdot 2 = 80$.

**Answer:** 080.

**Methods:** residue classes, cyclic counting.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** All five residue counts agree; the position of 3 is chosen before counting the two nonzero residue classes.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Cyclic products modulo three; the proof uses residue classes, cyclic counting.

## 2021 AIME II Problem 9: Counting exponent pairs by valuations

Count ordered $1 \le m, n \le 30$ with $\gcd(2^m + 1, 2^n - 1) > 1$.

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**Solution**

Since $2^m - 1$ and $2^m + 1$ are coprime, the gcd condition is equivalent to $\gcd(2m, n) > \gcd(m, n)$, using $\gcd(2^u - 1, 2^v - 1) = 2^{\gcd(u,v)} - 1$. Thus $v_2(n) > v_2(m)$. The valuation class sizes are $15, 8, 4, 2, 1$. Count $15(15) + 8(7) + 4(3) + 2(1) = 295$.

**Answer:** 295.

**Methods:** valuation classes, number theory hybrid.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** Source 1 establishes necessity through multiplicative order but does not prove sufficiency. Its auxiliary GCD proof falsely says gcd(d 1,k)=gcd(d 2,k)=1; the valid assertion needed is gcd(xy,k)=1. The prime-exponent GCD factorization in the new proof proves both directions.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Counting exponent pairs by valuations; the proof uses valuation classes, number theory hybrid.

## 2021 AIME II Problem 13: Lift an exponential fixed point through powers of five

Find the least positive integer n with $1000 \mid 2^n + 5^n - n$.

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**Solution**

The cases n=1,2 fail. For n≥3, reduction modulo 8 requires n odd and n≡5 modulo 8. Write n=8 r+5. Modulo 125, $2^{8r+5} = 32(1+5)^r \equiv 32+160r+400r(r-1)$, so the remaining condition is $25r^2+2r+27 \equiv 0$ (mod 125). Modulo 5 this forces r=5 s+4. Substituting gives $10s + 60 \equiv 0 \pmod{125}$, equivalent to s≡19 modulo 25. The least nonnegative s is 19, giving r=99 and n=797. Every step is an equivalence after the stated substitutions, so this verifies both divisibility and minimality.

**Answer:** 797.

**Methods:** Chinese remainder theorem, truncated binomial expansion, congruence lifting.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All five alternatives inspected. The binomial expansion gives a short exact lifting calculation. Other sources contain parity slips, an incorrect exponent identity, or a very large quotient that does not help establish minimality.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Lift an exponential fixed point through powers of five; Chinese remainder theorem, truncated binomial expansion, congruence lifting supply the essential reduction.

## 2021 AIME II Problem 15: Walking to the next square with two step sizes

For positive integers n, define f(n)=$\sqrt{n}$ at squares and f(n)=1+f(n+1) otherwise. Define g(n)=$\sqrt{n}$ at squares and g(n)=2+g(n+2) otherwise. Find the least n for which f(n)/g(n)=4/7.

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**Solution**

A square gives ratio 1, so write n=$m^2$+k with integers m≥1 and 0<k<2 m+1. If k is odd, n has the same parity as $(m+1)^2$, and both recursions stop there, giving equal values. Thus k is even, and g skips that square to $(m+2)^2$. Directly summing the increments gives $f(n) = 3m+2-k$ and $g(n) = 5m+6-k$. The ratio equation reduces to m=3 k+10. Positive even k has minimum 2, giving m=16,n=258, and n increases with k. This candidate satisfies k<2 m+1 and gives f=48,g=84, verifying the ratio.

**Answer:** 258.

**Methods:** next-square stopping times, parity, linear Diophantine reduction.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All six alternatives inspected. Using distance above the preceding square turns the ratio condition directly into m=3 k+10. The longer quadratic-formula approaches require additional interval checks and do not shorten the proof.

**Archived entries inspected:** 6 of 6. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Walking to the next square with two step sizes; next-square stopping times, parity, linear Diophantine reduction supply the essential reduction.

## 2022 AIME I Problem 2: A cyclic digit shift in two bases

The decimal integer with digits abc equals the base-nine integer with digits bca. Find the decimal integer.

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### Solution

The equality gives $100a + 10b + c = 81b + 9c + a$, or $99a = 71b + 8c$. Here a,b are nonzero and all three digits are at most 8. Modulo 9, b+c is divisible by 9; its range 1 through 16 forces b+c=9. Substitution gives $11a = 7b + 8$. Thus a≡2 modulo 7, and 1≤a≤8 forces a=2, then b=2,c=7. The integer is 227.

**Answer:** 227.

**Methods:** positional notation, digit bounds, modular elimination.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All six alternatives inspected. Modular equations together with exact digit ranges establish uniqueness. Several trial-based alternatives find 227 without exhausting the digit possibilities; one inequality argument does not justify its asserted lower bound.

**Archived entries inspected:** 6 of 6. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A cyclic digit shift in two bases; positional notation, digit bounds, modular elimination supply the essential reduction.

## 2022 AIME I Problem 7: A minimal positive ratio of digit products

Partition digits 1 through 9 into three ordered triples. Find numerator plus denominator of the least positive value $(abc - def)/(ghi)$.

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### Solution

If the numerator is at least 2, the ratio is at least $2/(7 \cdot 8 \cdot 9) > 1/288$. For numerator 1, write $u = v + 1$ for the two triple products. Their six distinct factors have product at least $6! = 720$, so $v \geq 27$. The pair $27, 28$ would require the triples $\{1, 3, 9\}$ and $\{1, 4, 7\}$, which overlap. Every consecutive pair with lower member $28, \ldots, 34$ contains one of $29, 31, 33, 34$, impossible as a product of available digits because it has a prime factor greater than 7. Products $35, 36$ are possible using $\{1, 5, 7\}$ and $\{2, 3, 6\}$, leaving denominator $4 \cdot 8 \cdot 9 = 288$. Larger consecutive products have smaller remaining denominator $9!/(uv)$. Hence the minimum is $1/288$, and the answer is 289.

**Answer:** 289.

**Methods:** extremal counting, product partitions.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** Source 0 omits the impossible overlapping 27/28 factor triples; sources 1–2 rely on search or skip candidates and do not fully prove the minimum. The original proof explicitly eliminates 27/28 and proves that all numerator values at least 2 are worse.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. A minimal positive ratio of digit products; the proof uses extremal counting, product partitions.

## 2022 AIME I Problem 13: Distinct reduced numerators

Consider all nonzero repeating four-digit decimals. Count their distinct reduced numerators modulo 1000.

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### Solution

Each fraction has denominator dividing $D = 9999 = 3^2 \cdot 11 \cdot 101$. A positive numerator $u$ occurs iff it is at most the largest divisor of $D$ coprime to $u$. If 101 divides $u$, this bound is at most 99, impossible. If neither 3 nor 11 divides $u$, there are $\varphi(D) = 6000$ choices. If 3 alone divides $u$, the bound 1111 gives $\lfloor 1111/3 \rfloor - \lfloor 1111/33 \rfloor - \lfloor 1111/303 \rfloor = 334$. If 11 alone divides $u$, the bound 909 gives $82 - 27 = 55$. If both divide $u$, the bound 101 allows $33, 66, 99$. Total 6392, remainder 392.

**Answer:** 392.

**Methods:** inclusion-exclusion, divisor lattice.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** Source 1 reverses divisibility signs and contains false logical equivalences. Source 0 has imprecise wording about cancellation but its case counts are valid. The new maximal-coprime-divisor criterion states the exact necessary-and-sufficient condition.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Distinct reduced numerators; the proof uses inclusion-exclusion, divisor lattice.

## 2022 AIME I Problem 14: Perimeter-bisecting cleavers determine a120-degree angle

A triangle has BC=219 and positive integer sides AB,AC. The lines through the midpoints of AB and AC that divide the triangle into two regions of equal perimeter meet at 30 degrees. Find its perimeter.

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### Solution

A perimeter-bisecting line through a side midpoint is parallel to the opposite internal angle bisector. To verify this, put C=0,A=b u,B=a v for unit vectors u,v and M=(b u+a v)/2. If a≥b, the other boundary point X satisfies CX=(a−b)/2, so $X - M = -b(u + v)/2$; the case b≥a is analogous. Consequently the acute angle between the two given lines is 90−A/2, forcing A=120 degrees. Put p=AB,q=AC. The cosine law gives $p^2 + pq + q^2 = 219^2$. Modulo 3, p≡q; substituting this into the equation modulo 9 forces 3|p and 3|q. Write p=3 x,q=3 y, and u=x+y,v=x−y. Then $3u^2 + v^2 = 4 \cdot 73^2 = 21316$. Positivity gives 73<u≤84. For u=74 through 84, the possible v squared values are 4888,4441,3988,3529,3064,2593,2116,1633,1144,649,148. Only 2116=46 squared is a square. Thus u=80, {x,y}={63,17}, and the perimeter is 219+3(80)=459.

**Answer:** 459.

**Methods:** cleaver theorem, cosine law, quadratic-form bounds.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All five alternatives inspected. The cleaver direction is proved by a vector calculation. A short bounded search after extracting the factor 3 replaces unsupported uniqueness claims or a much longer Eisenstein-integer argument.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory. Perimeter-bisecting cleavers determine a120-degree angle; cleaver theorem, cosine law, quadratic-form bounds supply the essential reduction.

## 2022 AIME II Problem 1: A population ratio constrains the total size

Adults were 5/12 of a crowd before a bus brought 50 people. They were 11/25 afterward. Find the least possible number of adults after the bus arrived.

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### Solution

If N is the original crowd size, the reduced fractions force 12|N and 25|(N+50), hence 300|N. The least positive possibility is N=300. It gives 125 adults before and 154 after, so the bus contains 29 adults and 21 others, which is feasible. Every larger N gives more final adults. The minimum is 154.

**Answer:** 154.

**Methods:** divisibility of population totals, least common multiple, feasibility.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All three alternatives inspected. Divisibility gives the candidate; explicitly checking the 50 bus occupants establishes attainability.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. A population ratio constrains the total size; divisibility of population totals, least common multiple, feasibility supply the essential reduction.

## 2022 AIME II Problem 5: Prime-difference triangles

Twenty cyclic vertices are labeled $1, \ldots, 20$; connect labels whose ordinary difference is prime. Count triangles with original vertices.

Published problem and solutions · MAA AMC.

### Solution

For increasing labels $x < y < z$, all three gaps must be prime and the largest is the sum of the other two. Thus one smaller gap is 2, and the other is a twin-prime lower member $p = 3, 5, 11, 17$. Each $p$ has two gap orders and $20 - (p + 2)$ starting labels. Count $2(15 + 13 + 7 + 1) = 72$.

**Answer:** 072.

**Methods:** gap encoding, prime parity.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** Source 0 falsely treats 18 starting positions as available for every prime pair and obtains 72 coincidentally. Entry 1 warns about this; entry 2 gives the valid separate spans 15,13,7,1 and both gap orders.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Prime-difference triangles; the proof uses gap encoding, prime parity.

### 2022 AIME II Problem 8: Identifiability from three floor values

Count positive integers $n \le 600$ uniquely determined among positive integers by $\lfloor n/4 \rfloor, \lfloor n/5 \rfloor, \lfloor n/6 \rfloor$.

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#### Solution

The common fiber is a consecutive integer interval. It is a singleton exactly when a floor jumps at $n$ and another jumps at $n+1$; equivalently $n$ and $n+1$ are each divisible by at least one of $4, 5, 6$. This repeats modulo 60. The qualifying residues are $4, 5, 15, 24, 35, 44, 54, 55$. Ten periods give 80. The upper bound restricts which $n$ are counted, not the possible comparison integers. If the comparison integers are instead restricted to be at most 600, the value 600 becomes one additional singleton, giving 81.

**Answer:** 080.

**Methods:** floor fibers, periodicity.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** Source 1 wrongly lists divisors 3,4,5 instead of 4,5,6. Sources 2–3 use 24 instead of lcm(4,6)=12 in overlap formulas, with compensating arithmetic errors. All-positive comparison gives 80; bounded comparison adds 600 and gives 81. Archived notes report both accepted, but this audit has not obtained a primary MAA acceptance notice.

**Archived entries inspected:** 7 of 7. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Identifiability from three floor values; the proof uses floor fibers, periodicity.

### 2022 AIME II Problem 13: A coefficient reduced to stars and bars

Find coefficient $x^{2022}$ in $(x^{2310}-1)^6/[(x^{105}-1)(x^{70}-1)(x^{42}-1)(x^{30}-1)]$.

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#### Solution

Below degree 2310, the numerator contributes only 1. Geometric-series expansion counts $105a + 70b + 42c + 30d = 2022$ in nonnegative integers. Modular reductions force $a = 2A, b = 3B, c = 5C + 1$, $d = 7D + 3$. Substitution gives $A + B + C + D = 9$. This bijection yields $\binom{12}{3} = 220$.

**Answer:** 220.

**Methods:** generating functions, modular reduction, stars and bars.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** Source 0 drops binomial coefficients from a full expansion, harmless only after restricting to degree 2022. Source 1 skips the modular exhaustion argument. Sources 2 and 4 write infinite series after polynomial divisions; that equality needs truncation. The original proof uses formal coefficients below 2310 and explicit residues.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, algebra, number_theory. A coefficient reduced to stars and bars; the proof uses generating functions, modular reduction, stars and bars.

## 2022 AIME II Problem 14: Complete coverage by a finite stamp collection

For denominations $a < b < c$, let $f$ be the minimum number of stamps, using every denomination, whose subcollections represent every value from 1 through 1000. Find and sum the three least values of $c$ permitting $f = 97$.

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### Solution

Necessarily $a = 1$. A sorted stamp collection covers every value up to its total exactly when each next stamp is at most one plus the preceding total. Choosing the largest currently admissible denomination maximizes the total after each step, so is optimal. Put $q = \lceil c/b \rceil$. Use $b-1$ ones, then $q-1$ stamps of value $b$, reaching $L = bq-1$, then $t = \max(1, \lceil (1000-L)/c \rceil)$ stamps of value $c$. Thus $f = b+q-2+t$. For $c \le 10$, 97 stamps have insufficient total. Let $T = \lceil 1001/c \rceil$. Since $bq \ge c$, we have $t \le T-1$ for $c < 1001$; and $b+q \le c+1$, giving $f \le c-2+T$. For $12 \le c \le 86$, the inequality $c(98-c) \ge 1001$ makes this bound at most 96. For $c = 87$ and $b \le 85$, $b + \lceil 87/b \rceil \le 87$ gives $f \le 96$; the remaining $b = 86$ also gives 96 directly. Finally $(c,b) = (11,7), (88,87), (89,87)$ give $f = 97$. These are therefore the three least possible $c$, whose sum is 188.

**Answer:** 188.

**Methods:** complete sequences, greedy dominance, coverage.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** The sole stored entry contains its own corrections but still uses an invalid general minimum formula and wrong witnesses b=86 for c=88,89. The new coverage-interval greedy proof gives the exact minimum and witnesses(11,7),(88,87),(89,87).

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Complete coverage by a finite stamp collection; the proof uses complete sequences, greedy dominance, coverage.

### 2023 AIME I Problem 4: Sum all divisors with prescribed exponent parity

Sum all positive integers m for which 13!/m is a perfect square. If the sum is $2^a3^b5^c7^d11^e13^f$, find a+b+c+d+e+f.

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#### Solution

Legendre exponents give $13! = 2^{10}3^5 5^2 7 \cdot 11 \cdot 13$. Thus the admissible exponents in m are 0,2,…,10 for 2;1,3,5 for 3;0,2 for 5; and 1 for each remaining prime. The sum factors as $(1 + 4 + \cdots + 4^5)(3 + 27 + 243)(1 + 25)7 \cdot 11 \cdot 13 = 1365 \cdot 273 \cdot 26 \cdot 7 \cdot 11 \cdot 13 = 2 \cdot 3^2 \cdot 5 \cdot 7^3 \cdot 11 \cdot 13^4$. Its exponent sum is 12.

**Answer:** 012.

**Methods:** prime valuations, parity-restricted divisor sums, geometric series.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All four stored entries inspected. The first two give valid primewise sums. The third explicitly labels its observed pattern as false, and the final entry is mathematically invalid despite ending with the reported answer; neither is used.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Sum all divisors with prescribed exponent parity; prime valuations, parity-restricted divisor sums, geometric series supply the essential reduction.

### 2023 AIME I Problem 7: Pairwise-distinct remainders

Count positive $n < 1000$ whose remainders modulo $2, 3, 4, 5, 6$ are all different.

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#### Solution

The condition repeats modulo 60. If $n$ is even, the residues modulo $2, 4$ must be $0, 2$; distinctness then forces residues $1, 4$ modulo $3, 6$, and 3 modulo 5, so $n \equiv 58 \pmod{60}$. If $n$ is odd, the $2, 4$ residues are $1, 3$; the $3, 6$ residues must be $2, 5$, and the 5 residue 0 or 4, giving $n = 35$ or $59 \pmod{60}$. The three counts are $16, 17, 16$, total 49.

**Answer:** 049.

**Methods:** Chinese remainder theorem, residue patterns.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** All four stored solutions inspected. They reduce to a period 60 and residues 35,58,59. The complete residue casework proves necessity; empirical listing alone needs exhaustive coverage. The independent argument derives the three classes.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Pairwise-distinct remainders; the proof uses Chinese remainder theorem, residue patterns.

## 2023 AIME I Problem 10: Bound the floor error before summing residues

For a positive integer a, put $U = \sum_{n=1}^{2023} \lfloor (n^2 - na)/5 \rfloor$. Exactly one a makes −1000<U<1000. Find a+U.

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### Solution

The sum without floors is $T = \frac{2023 \cdot 1012}{5}(1349 - a)$. Since 0≤T−U<2023, the prescribed range forces −1000<T<3023. The coefficient 2023 times 1012/5 exceeds 400000, so the only integral a is 1349, giving T=0. For n modulo 5 equal to 0,1,2,3,4, the residues of n(n−1349) are 0,2,1,2,0. Each of residues 1,2,3 occurs 405 times in 1 through 2023. The total fractional-part correction is 405(2+1+2)/5=405. Thus U=−405 and a+U=944.

**Answer:** 944.

**Methods:** floor-error bounds, residue cycles, polynomial sums.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All four alternatives inspected. A rigorous error bound identifies a before the residue calculation. The integral approximation in the final alternative does not by itself prove uniqueness and also gives the fractional-part sum the wrong sign.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Bound the floor error before summing residues; floor-error bounds, residue cycles, polynomial sums supply the essential reduction.

## 2023 AIME I Problem 15: A bounded Gaussian-integer search with all sign cases

Find the largest prime p<1000 for which a Gaussian integer z has absolute value $\sqrt{p}$ and p,Re(z cubed),Im(z cubed) are positive side lengths of a triangle.

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### Solution

Gaussian units and conjugation permute and change signs of the two cubic components. Thus existence is equivalent to testing their absolute values for z=a+bi with integers a≥b≥0 and a squared+b squared=p. The case b=0 cannot give a prime norm, while a=b gives only p=2. For a>b, if a squared≤3 b squared, the absolute component difference is $(a-b)(a^2+4ab+b^2) > p$, impossible. Otherwise both components are positive, their sum exceeds their Euclidean norm $p^{3/2} > p$, and the remaining triangle condition is $(a+b)|a^2-4ab+b^2| < a^2+b^2$. The bounds are 1≤b≤15 and $\sqrt{3}$ times b<a≤floor($\sqrt{999 - bsquared}$). Testing these explicit integer intervals gives the following possible norm lists for b=1 through 8: (10,17),(53,68),(130,153),(241),(349,386),(520,565),(725,778),(905,964); for b=9 through 15 there are none. The prime norms are 17,53,241,349. Finally z=18+5 i has cube 4482+4735 i, whose component difference 253 is less than 349. Hence the largest prime is 349.

**Answer:** 349.

**Methods:** Gaussian integer norms, triangle inequalities, exhaustive bounded search.

**Verification:** Exhaustive integer pairs in the stated bounds reproduce the displayed table; primality and the witness cube are independently checked.

**Source comparison:** Both alternatives inspected. Their reductions of the signs are not fully justified: mixed signs can give positive cubic components. Gaussian-unit symmetry followed by the separate a squared≤3 b squared branch covers every sign case. The explicit bounded table makes the finite exclusion reproducible.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. A bounded Gaussian-integer search with all sign cases; Gaussian integer norms, triangle inequalities, exhaustive bounded search supply the essential reduction.

## 2023 AIME II Problem 2: Enumerate the four-digit octal palindromes

Find the greatest positive integer below 1000 that is palindromic in both decimal and base 8.

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### Solution

The value 585=1111 in base 8 is a candidate, so only values above 511 need be considered. A base-eight palindrome in this range has four digits abba, with value 513 a+72 b. Since the value is below 1000, a=1. For b=0 through 6 the values are 513,585,657,729,801,873,945; b=7 exceeds the bound. Of these only 585 is a decimal palindrome. Therefore 585 is the greatest.

**Answer:** 585.

**Methods:** base representation, palindrome digit constraints, bounded enumeration.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** The single alternative inspected. Establishing 585 first justifies ignoring all shorter octal representations.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Enumerate the four-digit octal palindromes; base representation, palindrome digit constraints, bounded enumeration supply the essential reduction.

## 2023 AIME II Problem 5: The cancellation divisor determines a rational number

Let S consist of positive rationals r for which r and 55 r, each reduced, have the same numerator-plus-denominator sum. Write the sum of S as p/q reduced; find p+q.

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### Solution

Write r=a/b reduced and g=gcd(55,b). Then 55 r has reduced numerator 55 a/g and denominator b/g, so equality of sums gives $(55-g)a = (g-1)b$. Since g divides 55, it is 1,5,11,or 55. The endpoint choices force a=0 or b=0 and are impossible. For g=5 the ratio is a/b=2/25; for g=11 it is 5/22. Both have exactly the claimed cancellation divisor, so both work and exhaust S. Their sum is 169/550, giving 719.

**Answer:** 719.

**Methods:** fraction reduction, gcd parameter, finite divisor cases.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** The single alternative inspected. The general cancellation-divisor equation replaces three separately named denominator cases and automatically verifies completeness.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. The cancellation divisor determines a rational number; fraction reduction, gcd parameter, finite divisor cases supply the essential reduction.

## 2023 AIME II Problem 10: Balanced proper colorings of a ladder

Place $1, \ldots, 12$ in a 2 by 6 grid so adjacent cells never have equal residues modulo 3. Let $N$ count placements. Find the number of positive divisors of $N$.

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### Solution

Each residue occurs 4 times. Each column contains two distinct residues; counting appearances forces exactly two columns of each unordered type $01, 02, 12$. Order the types in $6!/(2!)^3 = 90$ ways. The first column has two orientations; each subsequent column has exactly one orientation avoiding a horizontal match. Thus 180 residue patterns. Assign actual numbers in $(4!)^3$ ways, so $N = 180(24)^3 = 2^{11}3^5 5$. Its divisor count is $12 \cdot 6 \cdot 2 = 144$.

**Answer:** 144.

**Methods:** graph coloring, multiset permutations.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** Both stored solutions inspected. The first uses residue-column types correctly. The second initially counts all column orientations without enforcing horizontal restrictions and compensates with an unjustified value-assignment count; its appended shorter formula has the right total. The independent proof explains why only two global orientations exist for each type order.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Balanced proper colorings of a ladder; the proof uses graph coloring, multiset permutations.

## 2023 AIME II Problem 13: Prove the integrality class before counting last digits

For acute A, tan A=2 cos A. Count positive n≤1000 for which sec to the nth power A plus tan to the nth power A is an integer ending in 9.

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**Solution**

Put alpha=sec A,beta=tan A, so alpha beta=2 and alpha squared−beta squared=1. Thus $u = \alpha^2 = (\sqrt{17}+1)/2$ and $v = \beta^2 = (\sqrt{17}-1)/2$. Conjugation in the field of $\sqrt{17}$ sends u to−v and v to−u. Therefore u to an odd power plus v to that power is nonzero and changes sign under conjugation, so is irrational. If n is odd, $(\alpha^n+\beta^n)^2 = u^n + v^n + 2^{n+1}$ is irrational; if n=2 k with k odd, the original sum is irrational. For n=4 m, let c_m=u to the power 2 m plus v to the power 2 m. Since u squared+v squared=9 and u squared times v squared=16, $c_0 = 2, c_1 = 9, c_m = 9c_{m-1} - 16c_{m-2}$, proving integrality. For m≥1 the terms are odd. Modulo 5 their residues repeat 4,4,2, so the units digit is 9 exactly when 3 does not divide m. There are 250−floor(250/3)=167 such m.

**Answer:** 167.

**Methods:** quadratic conjugation, irrationality, power-sum recurrence, residue periods.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** Both alternatives inspected. Each asserts without proof that only multiples of 4 yield integers. The conjugation argument supplies that missing exclusion, including all odd exponents. The recurrence then rigorously gives the last-digit cycle.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Prove the integrality class before counting last digits; quadratic conjugation, irrationality, power-sum recurrence, residue periods supply the essential reduction.

## 2023 AIME II Problem 15: A finite carry state detects repeated modular inverses

For each n≥1, let a_n be the smallest positive multiple of 23 congruent to 1 modulo 2 to the nth power. Count n≤1000 for which a_n=a_(n+1).

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**Solution**

Put $b_n = (a_n - 1)/2^n$. To pass to modulus $2^{n+1}$, the least admissible multiple is either a_n or a_n+23 times 2 to the nth power: all candidates for the old modulus differ by that increment. Consequently $b_{n+1} = b_n/2$ when b_n is even, and $(b_n + 23)/2$ when it is odd. Starting at b_1=11, the successive states are 11,17,20,10,5,14,7,15,19,21,22, followed again by 11. Determinism proves the 11-term cycle repeats forever. Equality of consecutive a values occurs exactly at the even states, or n≡0,3,4,6 modulo 11. The 90 complete cycles in 1 through 990 contribute 360, and 991 through 1000 contribute three more. The answer is 363.

**Answer:** 363.

**Methods:** modular inverse lifting, finite-state recurrence, periodic carries.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All three alternatives inspected. The carry-state recurrence proves periodicity exactly. The first source incorrectly makes the multipliers a_n/23 themselves periodic, and the binary discussion gives an unsupported 16-state bound; only the bounded carry state repeats.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, combinatorics. A finite carry automaton counts the indices where consecutive least modular inverses agree; the state cycle and the final partial cycle are essential.

## 2024 AIME I Problem 13: Lift the four roots of minus one

Let p be the least prime for which some positive n satisfies p squared divides n to the fourth power plus 1. Find the least positive m satisfying the same divisibility for that p.

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### Solution

The prime 2 is impossible modulo 4. For an odd p, a fourth root of−1 has multiplicative order 8, so 8 divides p−1; the least possible prime is 17. Modulo 17 the four roots are 2,8,9,15. For each such a, put m=a+17 t. Modulo 289 the equation becomes $(a^4+1)/17+4a^3t\equiv 0 \pmod{17}$. The derivative coefficient is nonzero, so each root has exactly one lift. For a=2,8,9,15, the least lifted representatives are 155,110,179,134, respectively. Thus 17 really works, and the least m is 110.

**Answer:** 110.

**Methods:** multiplicative order, prime-power root lifting, least residue.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All three alternatives inspected. Listing all four unique lifts establishes minimality. The primitive-root and modular-square-root alternatives require additional root-enumeration facts that the linear lift avoids.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Lift the four roots of minus one; multiplicative order, prime-power root lifting, least residue supply the essential reduction.

## 2024 AIME II Problem 2: A mode and a missing median constrain an integer partition

A finite list of positive integers sums to 30, has unique mode 9, and has an integral median that is absent from the list. Find the sum of squared entries.

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### Solution

The list length is even, and 9 occurs at least twice but at most three times. If it occurs three times, the remaining sum is 3: an even list has either one extra entry 3, giving median 9, or three entries 1, giving a tied mode. Thus exactly two 9 s occur and every other entry is distinct. Their sum is 12, so there are either two or four other entries; six distinct positive entries already sum to 21. Four distinct entries summing to 12 are only (1,2,3,6) or(1,2,4,5); each gives median 9/2 after the two 9 s are appended. For two entries, the distinct positive pairs summing to 12 are (1,11),(2,10),(3,9),(4,8),(5,7). Only(5,7) meets both the mode and median conditions. Thus the list is 5,7,9,9, and its square sum is 25+49+81+81=236.

**Answer:** 236.

**Methods:** integer partitions, multiplicity constraints, exhaustive case split.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All four alternatives inspected. The first three analyze only four-element lists and leave longer lists unexcluded. The final source supplies the missing length analysis; the authored proof explicitly exhausts both the frequency and length possibilities.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Mode multiplicities and integer partitions classify every permitted list length and eliminate all but one multiset.

### 2024 AIME II Problem 6: Encoding a set by maxima of subsets

A finite set $A$ of positive integers has the property that exactly 2024 finite nonempty sets of positive integers have maximum in $A$. Find the sum of the elements of $A$.

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#### Solution

Exactly $2^{k-1}$ sets have maximum $k$: include $k$ and choose any subset of the smaller positive integers. Thus $\sum_{k\in A} 2^{k-1} = 2024$. Its unique binary expansion is $2^{10} + 2^9 + 2^8 + 2^7 + 2^6 + 2^5 + 2^3$. Therefore $A = \{11, 10, 9, 8, 7, 6, 4\}$ and its sum is 55.

**Answer:** 055.

**Methods:** binary encoding, subset maxima.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** Both sources use unique binary expansion, once at 2024 and once at 4048. Both recover the same seven elements.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Encoding a set by maxima of subsets; the proof uses binary encoding, subset maxima.

### 2024 AIME II Problem 7: Sum digit replacements before solving the digits

Let N be the greatest four-digit integer such that replacing any one digit by 1 produces a multiple of 7. If N=1000 Q+R with 0≤R<1000, find Q+R.

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#### Solution

The sum of the four replaced numbers is 3 N+1111, hence 3 N+5≡0 modulo 7 and N≡3. For digits a,b,c,d, replacing each position therefore requires $6(a-1) \equiv 2(b-1) \equiv 3(c-1) \equiv d-1 \equiv 3 \pmod 7$. The digit ranges give a=5,b=6,d=4, and c=2 or 9. Taking the larger gives N=5694. Its

four replacements 1694,5194,5614,5691 are respectively 7 times 242,742,802,813, verifying all conditions. Thus Q+R=5+694=699.

**Answer:** 699.

**Methods:** place-value congruences, summing constraints, lexicographic maximum.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All four alternatives inspected. Summing the replacement equations determines N modulo 7 without trial over leading digits. The possible tens digits 2 and 9 are distinguished by maximality.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Sum digit replacements before solving the digits; place-value congruences, summing constraints, lexicographic maximum supply the essential reduction.

## 2024 AIME II Problem 13: Evaluate a root product at two conjugate constants

Let omega≠1 satisfy omega to the 13 th power=1. Find the product of $2-2\omega^k+\omega^{2k}$ for k=0 through 12, modulo 1000.

Published problem and solutions · MAA AMC.

### Solution

Since 13 is prime, the powers of omega run through all 13 th roots. Factor each term as $(1+i-\omega^k)(1-i-\omega^k)$. Using $\prod_{k=0}^{12}(z-\omega^k)=z^{13}-1$, the product is $|(1+i)^{13}-1|^2$. Now $(1+i)^{13}=(1+i)(2i)^6=-64-64i$. Thus the product is 65 squared+64 squared=8321, whose remainder is 321.

**Answer:** 321.

**Methods:** roots-of-unity resultant, conjugate norm, modular remainder.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All four alternatives inspected. Factoring first turns the 13-factor product into one exact Gaussian-integer norm; the primality of 13 ensures every nontrivial 13 th root is primitive.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Evaluate a root product at two conjugate constants; roots-of-unity resultant, conjugate norm, modular remainder supply the essential reduction.

## 2024 AIME II Problem 14: Digit-square conditions count CRT idempotents

For integer b≥2, call a positive integer special if its base-b representation has exactly two digits whose sum equals its square root. Find the least b with more than ten special integers.

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**Solution**

Write the digits a,c and put k=a+c=$\sqrt{n}$, B=b−1. Since n<b squared, k≤B; the leading digit is positive, so k≥2. The equation k squared=ba+c becomes $k(k-1) = Ba$. Conversely every k in 2 through B satisfying B divides k(k−1) gives a=k(k−1)/B and c=k−a, with 1≤a≤k−1≤B−1 and 1≤c≤B−1, so it gives a valid distinct integer. For each prime power dividing B, coprimality of k,k−1 forces k≡0 or 1 modulo that power. The Chinese remainder theorem gives exactly $2^{\omega(B)}$ residues modulo B; representing them in 1 through B and excluding k=1 leaves $2^{\omega(B)} - 1$ special integers. More than ten requires at least four distinct prime factors. The least B with four is 2 times 3 times 5 times 7=210, which gives 15 integers. Hence b=211.

**Answer:** 211.

**Methods:** Chinese remainder theorem, prime-power assignments, idempotent counting, digit bounds.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** The single alternative inspected. The authored proof supplies the full converse digit bounds and uses CRT for existence, not merely uniqueness, of every prime-power assignment.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, combinatorics. Each prime-power factor is assigned independently to one of two consecutive integers. CRT gives the bijective count2^omega(B)−1.

## 2025 AIME I Problem 1: Reduce a base-divisibility condition to one fixed integer

Find the sum of integer bases b>9 for which the base-b number 17 divides the base-b number 97.

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**Solution**

The two values are b+7 and 9 b+7. Their divisibility is equivalent to b+7 dividing $(9b+7)-9(b+7) = -56$. Since b+7>16, the possible positive divisors are 28 and 56. Thus b=21 or 49, and the sum is 70.

**Answer:** 070.

**Methods:** base notation, fixed remainder, bounded divisors.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All five alternatives inspected. The fixed remainder−56 gives the complete base list; several duplicated versions contain minor digit-subtraction typographical errors.

**Archived entries inspected:** 5 of 5. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Reduce a base-divisibility condition to one fixed integer; base notation, fixed remainder, bounded divisors supply the essential reduction.

## 2025 AIME I Problem 4: Count a union of two primitive lattice lines

Count ordered integer pairs (x,y) with −100≤x,y≤100 satisfying $12x^2 - xy - 6y^2 = 0$.

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**Solution**

Factor the expression as $(3x+2y)(4x-3y)$. On the first line, coprimality gives $(x,y)=(2t,-3t)$, with $-33\le t\le 33$, yielding 67 pairs. On the second, $(x,y)=(3t,4t)$, with $-25\le t\le 25$, yielding 51 pairs. The lines share only(0,0), so their union has 67+51−1=117 pairs.

**Answer:** 117.

**Methods:** factorization, primitive lattice directions, inclusion-exclusion.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All three alternatives inspected. Factoring without division retains the origin automatically; subtracting it once corrects the overlap of the two cases.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, combinatorics. The integer solutions are the union of two bounded primitive lattice lines; inclusion-exclusion removes their common origin.

## 2025 AIME I Problem 5: Digit permutations divisible by twenty-two

Let $N$ count permutations of digits 1 through 8 that form an eight-digit integer divisible by 22. Find $N-2025$.

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**Solution**

Divisibility by 11 requires both alternating-position digit sums to be 18: their difference is even and has absolute value at most 16. The four-subsets summing to 18 are $1278, 1368, 1458, 1467, 2358, 2367, 2457, 3456$; the total number of even digits across this list is 16. If such a subset occupies the even positions, choose its even final digit, then arrange the remaining positions in 3!4! ways. Thus the favorable count is $16\cdot 3!4! = 2304$. The requested difference is $2304-2025=279$.

**Answer:** 279.

**Methods:** divisibility tests, subset sums.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** All four sources inspected. They split digits into two four-subsets of sum 18. Enumeration confirms eight subsets, each with two even digits, justifying the factor one-half for an even final digit. The independent solution uses an explicit subset list.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Digit permutations divisible by twenty-two; the proof uses divisibility tests, subset sums.

## 2025 AIME I Problem 15: Cubic residues and three-variable counting

Count ordered triples $1\le a,b,c\le 729$ with $a^3+b^3+c^3$ divisible by 2187, and give the remainder modulo 1000.

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**Solution**

Split inputs into category $A$ (multiples of 9, size 81), category $B$ (multiples of 3 but not 9, size 162), and category $C$ (units, size 486). Their cube residues are respectively $729j$ with multiplicity 27; $\pm 27 + 243j$ with multiplicity 9; and $\pm 1 + 9j$ with multiplicity 1. These multiplicities follow by expanding a cube, or by $v_3(u^3 - v^3) = v_3(u - v) + 1$ for units of the same residue modulo 3. Reduction modulo 9 and then 243 permits only patterns AAA, ABB, ACC, BCC. For any first two entries, the required third residue has the stated multiplicity. The ordered counts are $81^2 \cdot 27$, $3 \cdot 81 \cdot 162 \cdot 9$, $3 \cdot 81 \cdot 486$, and $3 \cdot 162 \cdot 486$. Their sum is $15 \cdot 3^{10} = 885735$, giving 735.

**Answer:** 735.

**Methods:** valuation stratification, residue distributions.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** All four sources inspected. Source index 1 explicitly contains a false vanishing exponential-sum claim and then inserts an unjustified factor 5; it is rejected. Source index 2 has lengthy omitted and inconsistent modular subcounts and is not used as a proof. Source index 3 offers a useful valuation recursion, but its appeal to ordinary Hensel lifting needs justification because the cube derivative vanishes modulo 3. The independent cube-residue distribution was checked by exact convolution.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Cubic residues and three-variable counting; the proof uses valuation stratification, residue distributions.

## 2025 AIME II Problem 2: Evaluate a polynomial at the modulus root

Find the sum of positive integers n for which n+2 divides $3(n+3)(n^2+9)$.

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**Solution**

Modulo n+2, substitute n≡−2. The expression becomes 3(1)(13)=39. Thus n+2 must be a positive divisor of 39 exceeding 2, namely 3,13,39. The corresponding n values 1,11,37 all work and sum to 49.

**Answer:** 049.

**Methods:** polynomial remainder, finite divisor reduction.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** All three alternatives inspected. The direct substitution modulo n+2 proves equivalence without long division.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Evaluate a polynomial at the modulus root; polynomial remainder, finite divisor reduction supply the essential reduction.

## 2025 AIME II Problem 7: The least common multiple of a random divisor subset

Choose uniformly a subset of the positive divisors of 2025. Find numerator plus denominator of the probability that its least common multiple is 2025.

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### Solution

Since $2025 = 3^4 5^2$, there are 15 divisors. The subset must contain a divisor of 3-exponent 4 and one of 5-exponent 2. There are respectively 3 and 5 such divisors, with one in common. Inclusion-exclusion gives $2^{15} - 2^{12} - 2^{10} + 2^8$ favorable subsets. Dividing by $2^{15}$ yields 109/128, so the answer is 237.

**Answer:** 237.

**Methods:** divisor lattice, inclusion-exclusion.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** All four source solutions inspected. Inclusion-exclusion and splitting on inclusion of 2025 are equivalent. Missing displayed expressions in two scraped alternatives do not obscure the underlying count.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. The least common multiple of a random divisor subset; the proof uses divisor lattice, inclusion-exclusion.

## 2025 AIME II Problem 8: When greedy coin change succeeds

With coin values $1, 10$, and 25, count amounts from 1 through 1000 for which repeatedly taking the largest affordable coin uses the fewest coins.

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### Solution

Write $N = 25q + r$, $0 \le r < 25$, and put $h(s) = \lfloor s/10 \rfloor + (s \bmod 10)$, the optimal count using ones and tens. Using two fewer quarters changes the count by $-2 + h(s + 50) - h(s) = 3$, so only $q$ and $q - 1$ quarters need comparison. For $q \ge 1$, using $q-1$ improves the count exactly when $r \bmod 10 \ge 5$, namely $r = 5, \dots, 9, 15, \dots, 19$. There are 39 complete positive-quarter blocks and thus 390 failing amounts. The remaining $1000 - 390 = 610$ succeed.

**Answer:** 610.

**Methods:** greedy algorithms, periodic optimization.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** The single stored source uses coin-case analysis. The independent two-quarter exchange explains why only the largest allowable quarter count and one less need comparison.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. When greedy coin change succeeds; the proof uses greedy algorithms, periodic optimization.

## 2025 AIME II Problem 13: Track reduction before simplifying a rational recurrence

Define $x_1 = 25/11$ and $x_{k+1} = (x_k + 1/x_k - 1)/3$. If x 2025=m/n reduced with positive integers m,n, find (m+n) modulo 1000.

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### Solution

Write x=a/b reduced, with S=a+b divisible by 3; this holds initially. Then neither a nor b is divisible by 3, and the next unreduced fraction is $(a^2 - ab + b^2)/(3ab)$. Its numerator is coprime to ab and divisible by 3, so exactly the common factor 3 cancels. The next reduced numerator and denominator are $(a^2 - ab + b^2)/3$ and ab, whose sum is S squared/3, again divisible by 3. Thus $S_k = 3 \cdot 12^{2^{k-1}}$. To reduce S 2025 modulo 1000, it is 0 modulo 8. Modulo 125, Euler's theorem reduces the exponent modulo 100. Since 2 to the 2024 th power is 0 modulo 4 and 16 modulo 25, it is 16 modulo 100. Repeated squaring gives 12 to the 16 th power≡41 modulo 125, so S 2025≡123 modulo 125. The unique residue below 1000 satisfying this and divisibility by 8 is 248.

**Answer:** 248.

**Methods:** fraction reduction invariant, numerator-denominator sum, exponent reduction, Chinese remainder theorem.

**Verification:** The exact derivation yields the archived answer.

**Source comparison:** Both alternatives inspected. The first leaves its reduction invariant to an assertion about modulus 9; the second omits the cancellation proof. Coprimality plus the invariant 3 divides a+b proves the reduced recurrence directly.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. Track reduction before simplifying a rational recurrence; fraction reduction invariant, numerator-denominator sum, exponent reduction, Chinese remainder theorem supply the essential reduction.

## 2026 AIME I Problem 4: Products of two unequal factors

How many integers at most 100 have the form $a + b + ab$ for distinct positive integers $a, b$?

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### Solution

Adding one converts the expression to $(a + 1)(b + 1)$. We therefore count integers $N$ from 2 through 101 that have two unequal factors at least two. The exclusions are primes and squares of primes: if $p$ is the least prime factor of a composite $N$, then $p$ and $N/p$ are unequal unless $N = p^2$. There are 26 primes through 101 and four prime squares, $4, 9, 25, 49$. Thus the count is $100 - 26 - 4 = 70$.

**Answer:** 070.

**Methods:** factorization, least prime divisor, prime-square exceptions.

**Verification:** Independent enumeration of all a<b with a+b+ab≤100 gives exactly 70 distinct values; direct primality checking gives 26 primes from 2 through 101.

**Source comparison:** All three alternatives read, including the full eight-stage sieve table. The product reduction is sound; counting representations instead of distinct products would overcount. The least-prime-factor argument proves that the listed exceptional composites are exhaustive.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory. Completing (a+1)(b+1) reduces the possible values to composites other than prime squares.

### 2026 AIME I Problem 6: A logarithmic quadratic determines a product

Let $P$ be the product of all positive real solutions of $\sqrt[20]{x^{\log_{2026} x}} = 26x$. Find the number of positive divisors of $P$.

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#### Solution

Set $u = \log_{2026} x$. Taking logarithms gives $u^2 - 20u - 20 \log_{2026} 26 = 0$. Its discriminant is positive, so its two real roots correspond bijectively to two positive solutions. Their sum is 20; hence their corresponding $x$ values have product $2026^{20}$. Since $2026 = 2 \cdot 1013$ and 1013 is prime (none of the primes through 31 divides it), the divisor count is $(20 + 1)^2 = 441$.

**Answer:** 441.

**Methods:** logarithmic substitution, Vieta formulas, divisor count.

**Verification:** The discriminant is strictly positive because log base 2026 of 26 is positive; primality of 1013 checked by trial division through its square root.

**Source comparison:** All six alternatives read. Sources 0, 1, 4, and 5 use a logarithmic quadratic, though several displays are missing. Source 2 is answer guessing rather than a proof. Source 3 inserts an invalid exponent quotient before arriving at the correct quadratic. Taking logarithms directly repairs that step.

**Archived entries inspected:** 6 of 6. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. A logarithmic quadratic determines the product by Vieta, after which prime exponents determine its divisor count.

### 2026 AIME I Problem 8: Divisor residues determined by exponent parity

Count divisors of $17017^{17}$ congruent to 5 modulo 12, and give the count modulo 1000.

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#### Solution

Factor $17017 = 7 \cdot 11 \cdot 13 \cdot 17$. A divisor is $7^a 11^b 13^c 17^d$, with exponents from 0 through 17. Its residue is 5 precisely when $a + b$ is even and $b + d$ is odd, by reduction modulo 4 and 3. Choose $b, c$ in $18^2$ ways; each determines the required parity of $a, d$, with 9 choices each. Thus there are $18^2 \cdot 9^2 = 26244$ divisors, giving remainder 244.

**Answer:** 244.

**Methods:** residue classes, exponent parity.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** All four sources inspected. They reduce exponents to parity using residues 7,11,1,5 modulo 12. Both two-pattern enumeration and simultaneous modulo 3/modulo 4 conditions give 26244.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Divisors correspond to exponent vectors, whose parities determine residue five modulo twelve.

### 2026 AIME I Problem 13: Binomial residue sums over a finite field

For $0 \le r < 502$, let $S_r$ sum $\binom{10000}{k}$ over $0 \le k \le 10000$ congruent to $r$ modulo 502. How many $S_r$ are divisible by 503?

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#### Solution

Over $\mathbb{F}_{503}$, put $P(x) = \sum_{r=0}^{501} S_r x^r$. For every nonzero $x$, Fermat gives $P(x) = (1+x)^{10000} = (1+x)^{462}$, since $10000 \equiv 462 \pmod{502}$; the equality also holds at $x = -1$. The difference has degree at most 501 and 502 roots, so it is identically zero. Thus $S_r = \binom{462}{r}$ modulo 503 for $r \le 462$, and $S_r = 0$ for larger $r$. No binomial coefficient in row 462 vanishes modulo the larger prime 503, because all relevant factorials are invertible. Exactly $501 - 462 = 39$ sums qualify.

**Answer:** 039.

**Methods:** finite-field interpolation, binomial coefficients.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** Both sources inspected. The finite-field root argument and quotient-ring interpretation are valid; they must handle x=-1 separately before reducing the exponent. Direct modular coefficient calculation confirms 39 zeros.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, algebra, number_theory. Binomial coefficients grouped by congruence class are analyzed using polynomial identities over a finite field.

### 2026 AIME II Problem 1: Arithmetic progressions indexed by common divisors

Find the sum of the eleventh terms of all integer arithmetic sequences beginning with 4 and containing both 24 and 34. An alternate archived version asks for the tenth terms instead.

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#### Solution

The common difference $d$ is a positive integer dividing both $24 - 4 = 20$ and $34 - 4 = 30$. Conversely every positive divisor of their gcd 10 gives such a sequence. Thus $d = 1, 2, 5, 10$. The eleventh terms

are $4+10d$, and their sum is $4\cdot 4+10(1+2+5+10)=196$. For the alternate tenth-term version the same calculation is $4\cdot 4+9\cdot 18=178$.

**Answer:** 196.

**Methods:** arithmetic sequences, greatest common divisor, divisor sums.

**Verification:** Enumerating the four positive divisors of 10 gives eleventh-term sum 196 and tenth-term sum 178.

**Source comparison:** The sole stored alternative read. It solves the main eleventh-term statement and obtains 196. The clean alternate statement asks for tenth terms and reports 178. These are genuinely different requests, and both are solved explicitly here.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** algebra, number_theory. Allowed integer common differences are the positive divisors of the gcd of two term offsets.

## 2026 AIME II Problem 3: An indented pentagon gives one residue class

A simple nonconvex pentagon $ABCDE$ has interior angles $A=E=90^\circ$, $B=D=45^\circ$, with $DE<AB$, $AE=20$, $BC=14\sqrt{2}$. Vertices $B,C,D$ lie on one side of $AE$. How many integer lengths $AB<2026$ make its area a multiple of 16?

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### Solution

Put $A=(0,0)$, $E=(0,20)$ and $B=(b,0)$. The angle conditions force $C=(b-14,14)$ and $D=(b-8,20)$: the other possible direction of $BC$ would give $C=(b-14,-14)$ and $DE=b+20$, contradicting $DE<AB$. The same-side condition is exactly $b>14$. These coordinates give a simple pentagon with the required reflex angle at $C$ for every $b>14$. Its area by the shoelace formula is $20b-164$. This is divisible by 16 exactly when $b\equiv 1 \pmod 4$. The permitted integers are $17,21,\dots,2025$, numbering $(2025-17)/4+1=503$.

**Answer:** 503.

**Methods:** angle-constrained coordinates, polygon area, congruence counting.

**Verification:** Exact enumeration of integers 15 through 2025 confirms 503 with (20 b−164) divisible by 16; the coordinate determinants verify the stated side lengths and turns.

**Source comparison:** The sole stored alternative read. Its rectangle-and-two-triangles calculation is correct, but it leaves the geometric branch and the positivity of the horizontal offset implicit. Coordinates justify both before counting the residue class.

**Archived entries inspected:** 1 of 1. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory. The angle data fix an indented polygon shape up to one integer length, and its area imposes a residue class.

## 2026 AIME II Problem 4: A numeral equal to its value in a smaller base

For positive decimal $n$, reinterpret its digit string in the base one larger than its greatest digit. Count $n < 1000$ unchanged by this operation.

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### Solution

Every one-digit number is unchanged. For a longer numeral the value is strictly increasing in its base, so equality requires base 10, or equivalently a digit 9. There are $90 - 8 \cdot 9 = 18$ qualifying two-digit numerals and $900 - 8 \cdot 9^2 = 252$ three-digit numerals. Including the nine one-digit numbers gives $9 + 18 + 252 = 279$.

**Answer:** 279.

**Methods:** complement counting, digit restrictions.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** Both sources inspected. Monotonicity in the base gives the clean criterion; the digit-by-digit alternative reaches the same count by inclusion-exclusion.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Monotonicity of positional values reduces fixed numerals to a digit-pattern count including at least one nine.

## 2026 AIME II Problem 5: Equal hypergeometric probabilities

An urn has at least seven marbles of each of two colors. Drawing seven without replacement gives equal probabilities of exactly four red marbles and exactly five red marbles. Sum the five smallest possible total populations.

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### Solution

If the color counts are $r, b$, cancellation in $\binom{r}{4}\binom{b}{3} = \binom{r}{5}\binom{b}{2}$ gives $(b-2)/3 = (r-4)/5$, or $3r - 5b = 2$. All solutions have $r = 4 + 5k$, $b = 2 + 3k$. Both counts are at least seven exactly when $k \geq 2$. The five smallest totals are $22, 30, 38, 46, 54$, whose sum is 190.

**Answer:** 190.

**Methods:** hypergeometric ratios, Diophantine counting.

**Verification:** The displayed mathematical count agrees with the archived answer.

**Source comparison:** Both sources inspected. Cancellation of adjacent hypergeometric coefficients gives 3 r-5 b=2 and the five totals 22,30,38,46,54.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Equal neighboring hypergeometric probabilities cancel to a linear Diophantine condition on color populations.

## 2026 AIME II Problem 8: An integer isosceles triangle with a tiny incenter triangle

In an isosceles triangle $ABC$, $AB = BC$ and $I$ is the incenter. The perimeters of $ABC$ and $AIC$ have ratio $125 : 6$. All side lengths of both triangles are integers. Find the least possible $AB$.

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### Solution

Write $AC = 2x$, $AB = L$, and $AI = CI = u$. The altitude from $B$ contains $I$, so $\cos A = x/L$ and $\cos(A/2) = x/u$. With $t = u/x > 1$, the double-angle formula gives $L/x = t^2/(2 - t^2)$, where $t < \sqrt{2}$. The perimeter ratio becomes $2/((2 - t^2)(t + 1)) = 125/6$, or $(t - 7/5)(t^2 + 12t/5 + 34/25) = 0$. The quadratic factor is positive for $t > 0$, so $t = 7/5$ and $L/x = 49$. If $AC = b$ is an integer, then $AI = 7b/10$ forces $10 \mid b$; also $L = 49b/2$. Hence $L \geq 245$, attained at side lengths $245, 245, 10$ and $AI = CI = 7$. These lengths satisfy the ratio and the angle identity, verifying attainment.

**Answer:** 245.

**Methods:** half-angle identity, unique positive root, integrality scaling.

**Verification:** The cubic factors exactly as shown; for base 10 the altitude is $100\sqrt{6}$, the inradius is $2\sqrt{6}$, and AI$^2$=$5^2$+$(2\sqrt{6})^2$=49. Perimeters 500 and 24 have ratio 125:6.

**Source comparison:** All three alternatives read. Source 0 introduces an integer substitution after normalization without proving it and then guesses the root. Source 1 mislabels the base as BC but contains the intended half-angle reduction. Source 2 finds a working example without proving minimality. The factored cubic proves the unique shape, and divisibility by 10 then proves the minimum.

**Archived entries inspected:** 3 of 3. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, algebra, number_theory. A half-angle relation determines a unique isosceles shape, and denominator divisibility gives its smallest integral scale.

## 2026 AIME II Problem 9: A divisor-generating decimal with controlled carries

Let $S = \sum_{k \geq 1} 1/(10^k - 1)$. Find $\lfloor 10^{100} S \rfloor$ modulo 1000.

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### Solution

Expand each summand as $\sum_{j \geq 1} 10^{-kj}$. Since all terms are positive, regrouping gives $S = \sum_{m \geq 1} \tau(m) 10^{-m}$. After multiplying by $10^{100}$, the terms through $m = 100$ are integers. Their last three places contribute $100\tau(98) + 10\tau(99) + \tau(100) = 600 + 60 + 9 = 669$. It remains to rule out a carry from the infinite tail $T$. Here $\tau(101) = 2$, and for $j \geq 2$ the divisor-pair bound gives $\tau(100 + j) \leq 2\sqrt{100 + j} \leq 20 + j$. Therefore $0 < T \leq 2/10 + \sum_{j \geq 2} (20 + j) 10^{-j} = 361/810 < 1$. Thus the tail contributes zero to the floor and the remainder is 669.

**Answer:** 669.

**Methods:** Lambert series, divisor function, rigorous tail bound.

**Verification:** Exact divisor counts are $\tau(98)$=6, $\tau(99)$=6, $\tau(100)$=9; the geometric tail majorant is 361/810<1. Rational truncation with a rigorous remainder bound independently confirms the floor residue 669.

**Source comparison:** Both alternatives read. They correctly identify divisor counts as the uncarried decimal coefficients but dismiss all later carrying solely because 101 is prime. That observation alone is insufficient; the explicit bound on the entire infinite tail completes the proof.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** number_theory, algebra. The repeating-decimal series has divisor-count coefficients, and a rigorous tail bound controls the final carry.

## 2026 AIME II Problem 10: An angle bisector and tangent secants

In triangle $ABC$, $AB = 200$, $AC = 225$, and $AD$ bisects angle $A$, with $D$ on $BC$. A circle through $A$ is tangent to $BC$ at $D$ and meets $AB, AC$ again at $E, F$ on those segments. If $AE, AF, BD, CD$ are positive integers, find the greatest possible $BC$. An alternate version asks for the sum of all possible values.

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### Solution

The angle-bisector theorem and integrality give $BD = 8m$, $CD = 9m$ for a positive integer $m$. Tangent-secant power gives $BE = BD^2/200 = 8m^2/25$ and $CF = CD^2/225 = 9m^2/25$. Integrality forces $5 \mid m$; write $m = 5k$. Then $AE = 200 - 8k^2$, $AF = 225 - 9k^2$, and $BC = 85k$. Positivity gives $k = 1, 2, 3, 4$. Conversely each of these gives triangle sides $200, 225, 85k$, satisfying the strict triangle inequalities. The unique circle through $A$ tangent to $BC$ at its bisector foot $D$ exists: its center lies on the perpendicular at $D$, and the equal-distance equation to $A, D$ determines a positive radius. Its second secant intersections have the computed positive lengths less than 200 and 225, so they lie on the required segments. Hence the possible bases are $85, 170, 255, 340$. Their maximum is 340; their sum, requested in the alternate version, is 850.

**Answer:** 340.

**Methods:** angle bisector theorem, tangent-secant power, divisibility, existence converse.

**Verification:** Exact enumeration of the angle-bisector factor yields k=1,2,3,4; the corresponding bases are 85,170,255,340 and their sum is 850.

**Source comparison:** All four alternatives read. Source 0 explicitly notes an international sum variant. Source 1 initially lets the common factor be real and then infers integer divisibility without stating why; BD,CD integral and coprime ratio 8:9 supply that step. The completed converse checks triangle existence and segment intersections for every candidate. The main maximum answer 340 and alternate sum answer 850 are retained separately.

**Archived entries inspected:** 4 of 4. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** geometry, number_theory. The angle-bisector ratio and tangent-secant powers turn four integrality conditions into one bounded integer parameter.

## 2026 AIME II Problem 15: A Fano-plane residue count

Count ordered seven-tuples with entries in $\{1, 2, 3\}$ for which both $\sum_i a_i$ and $\sum_i a_i a_{i+1} a_{i+3}$ are divisible by three; indices are cyclic modulo seven.

**Solution**

Replace entries by their residues $0, \pm 1$. The seven triples $\{i, i+1, i+3\}$ are the lines of the Fano plane: each pair lies on one line. Classify by support size $w$. For $w = 0, 1, 2, 3$, counts are $1, 0, 42, 56$: two signs must oppose, while three signs must agree and their support must avoid the seven lines. For $w = 4$, signs split two and two; the support must contain no line, hence be the complement of a line. Count $7\binom{4}{2} = 42$. For $w = 5$, its two contained lines meet at one point. Four signs agree and one differs; their two products cancel exactly when the minority sign is at one of the other four points. Count $\binom{7}{2} \cdot 8 = 168$. For $w = 6$, all equal signs fail; among the $\binom{6}{3} = 20$ balanced assignments, exactly eight fail because the plus or minus triple is one of the four contained lines. Count $7(20 - 8) = 84$. For $w = 7$, signs split five and two; the seven products comprise three of one sign and four of the other, so none qualify. Total $1 + 42 + 56 + 42 + 168 + 84 = 393$.

**Answer:** 393.

**Methods:** finite geometry, Fano plane, residue counting.

**Verification:** Independently enumerated all $3^7 = 2187$ residue assignments; the support-count distribution is $1, 0, 42, 56, 42, 168, 84, 0$.

**Source comparison:** Both sources inspected. Source index 0 incorrectly replaces entries by a_i-2 rather than their actual residues; its later support analysis corresponds to the correct residue mapping. Source index 1 uses the correct mapping but has mistaken line-containment wording and omits all-equal signs at weight 6, so it is not a complete proof. The independent Fano support count was exhaustively checked on all 2187 tuples. Clean statement includes all seven cyclic terms.

**Archived entries inspected:** 2 of 2. Zero-based indices are recorded in the data; linked media are not implicitly claimed reviewed.

**Subject classification:** combinatorics, number_theory. Residues on the seven Fano-plane points turn cyclic triple products into a finite incidence and sign count.